

Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Glossary of acronyms

OMIE	Operador del Mercado Ibérico de Energía	PDBC	Programa Diario Base de Casación
REE	Red Eléctrica de España	PDBF	PDBC plus bilateral contracts
CNMC	Comisión Nacional de los Mercados y la Competencia	PIBCI	Programa Intradiario Base de Casación Incremental
ENTSO-E	European Network of TSOs for Electricity	TR	Tiempo Real (real-time technical restrictions)
ESIOS	Sistema de Información del Operador del Sistema	Fase I/II	pre-/post-IDA technical-restriction redispatch
EU	European Union	PO	Procedimientos de Operación
DA	day-ahead auction	aFRR	automatic Frequency Restoration Reserve
IDA	intraday auctions	mFRR	manual Frequency Restoration Reserve
SIDC	Single Intraday Coupling	RR	Replacement Reserve
XBID	cross-border intraday continuous market	BRP	Balance Responsible Party
EPEX	European Power Exchange	BSP	Balance Service Provider
MTU	market time unit (MTU60 hourly; MTU15 15-min)	CCGT	combined-cycle gas turbine
ISP15	imbalance settlement at 15 min (2024-12-11)	RES	renewable energy sources
ID15	intraday at 15 min (2025-03-19)	PV	photovoltaic
DA15	day-ahead at 15 min (2025-10-01)	MIC	minimum-income condition (offer type)
DA60	pre-DA15 hourly day-ahead	HHI	Herfindahl–Hirschman Index
MCP	market clearing price	MW / MWh / GWh	megawatt / megawatt-hour / gigawatt-hour

1 Introduction

[To be written. Concepts to cover, mirroring the “Question and headline” slide:]

- **Setting.** Spanish wholesale electricity is a sequence of auctions cleared from one day before delivery down to real time. In 2025 the trading time unit moved from 60 minutes to 15 minutes in two staggered reforms — intraday in March (ID15), day-ahead in October (DA15).
- **Question.** Does finer time resolution reshape strategic bidding? Does market power go up or down? What is the effect on prices, on the day-ahead vs intraday wedge, and on imbalance penalties?
- **Headline answer.** Granularity made the market more competitive without moving aggregate quantities: prices fell substantially at ID15 and were unchanged at DA15; per-firm residual demand thickened on every Big-4 dominant firm at every reform; dispatchables grade their critical-hour ladders more finely while renewables compress; imbalance penalty per MWh dropped by a third at ID15 with aggregate imbalance volume essentially unchanged.
- **Identification challenge.** An April 2025 blackout triggered a separate “operación reforzada” regulatory regime that inflated ancillary-services costs. Granularity and that regime are calendar-overlapping but mechanistically distinct; the empirical windows are constructed to keep them separable.
- **Contribution.** Bid-level identification on the new MTU15 sequence; a sequential-markets model in which the optimal step offer is solved exactly — the bid ladder places equal-MW tranches on an even price grid along the bidder’s monopoly locus, so the measured bid-shape statistics $(\sigma_p, \text{HHI}_p, \hat{\beta})$ have closed-form model counterparts; per-firm Ito-Reguant residual-demand recovery for Spanish data; convex-renewable robustness on the price effect; a strategic-markup proxy that nets out infra-marginal own production from the Cournot first-order condition.
- **Roadmap.** Institutional setting (§2); theory (§3); data (§4); empirical strategy (§5); results (§6); conclusion (§7).

2 Institutional setting: sequential markets, three reforms, and a blackout

2.1 Spanish wholesale architecture: sequential auctions and balancing

Takeaway. Spain runs a sequential auction stack — day-ahead → Fase I restrictions → three intraday auctions → continuous intraday → Fase II → balancing settlement — under uniform-price multi-unit clearing. OMIE auction-cleared programmes are pre-restriction by construction (REE redispatch enters through separate offers), so we can isolate competitive clearing from system-operator interventions. Wholesale concentration is moderate ($\text{HHI} \sim 1100$); ancillary-service concentration is high ($\text{HHI} > 4000$).

[To be written. Concepts to cover:]

- OMIE clears the day-ahead market (PDBC), three European intraday auctions (PIBCI), and the continuous intraday market; REE runs the technical-restrictions process before IDA (*Fase I*) and after IDA (*Fase II*), real-time balancing, and the imbalance settlement.

- Day-ahead and intraday auctions both use *uniform-price multi-unit clearing*; within-hour discrimination is possible only through the bid stack, not through differential clearing rules.
- OMIE auction-cleared programmes are *pre-restriction by construction*: REE’s redispatch enters through separate restriction-offer markets, so OMIE-cleared volumes reflect competitive clearing rather than the system operator’s restriction calls.
- Wholesale HHI $\sim 1\,100$ (moderate) vs ancillary-services HHI $> 4\,000$ (high): the strategically interesting margin is per-product within each unit’s competing tier, and concentration migrates from wholesale to ancillary services.

2.2 The three 15-minute reforms: ISP15, ID15, DA15

Takeaway. Three layered reforms in seven months. ISP15 (Dec 2024): REE’s imbalance settlement period moves from 60 to 15 minutes; OMIE contracts stay hourly. ID15 (Mar 2025): intraday auctions and continuous intraday market move from 60 to 15 minutes. DA15 (Oct 2025): day-ahead market follows. Wherever an effect could plausibly attach to more than one reform, we identify the channel separately by window.

[To be written. Concepts to cover:]

- **ISP15** (2024-12-11): REE’s imbalance settlement reference moves from hour to quarter-hour; OMIE contract grid stays hourly. Within-hour deviations from schedule now resolve at 15-min resolution.
- **ID15** (2025-03-19): intraday auctions and continuous intraday market move to 15 minutes. Each clock-hour of delivery now hosts four 15-min products in IDA where it previously hosted one.
- **DA15** (2025-10-01): day-ahead market follows the same logic, completing the granularity rollout.
- Two earlier 2024 reforms set the stage: the European IDA reform of 14 June 2024 (six MIBEL auctions $\rightarrow$ three SIDC auctions), and the CNMC PO 3/14 modification (March 2024) that moved balancing services and BSP habilitation to 15-min.
- The reforms are *layered* rather than independent — ISP15 changes the settlement reference for every subsequent auction; ID15 lands on the new reference; DA15 lands on both.

2.3 The April 2025 blackout and the reinforced-operation regime

Takeaway. The blackout (28 April 2025) cascaded from voltage- and reactive-power-control failures, not from market microstructure (ENTSO-E Expert Panel). REE’s continuous reinforced-operation regime (*operación reforzada*) that followed is a separate regulatory layer: it tightens voltage-control requirements and expands Fase I up-redispatch payments by EUR ~ 5 M/day, almost entirely on combined-cycle gas. The DA15 BSTS pre-window starts on the blackout date, holding *reforzada* constant across the cutover by window construction; the ID15 windows sit entirely before the blackout.

[To be written. Concepts to cover:]

- ENTSO-E Expert Panel report (ENTSO-E Expert Panel, 2026) attributes the cascade to voltage- and reactive-power-control failures: fixed-power-factor renewable plants, manually-operated shunt reactors, divergent overvoltage protection, converter-driven instability, missing power-system stabilisers. Market microstructure is *not* in the panel’s root-cause tree.

- REE’s response was *operación reforzada* (CNMC, 2025b), continuous from 28 April 2025: tightened voltage-control requirements and expanded Fase I up-redispatch to keep CCGT synchronised for voltage support even when not day-ahead competitive. Fase I cost balloons by EUR ~ 5 M/day, almost entirely on CCGT.
- **Identification consequence.** Reforzada overlaps DA15 calendar-wise but *not* ID15 (clean 40-day pre-blackout post-window). DA15 windows start on the blackout date so reforzada is held constant across the DA15 cutover by window construction. The headline ajuste-cost rise across 2025 is reforzada-driven, not MTU15-driven.

3 Theory: a sequential-market model of bid ladders with granularity selectors

3.1 Setup: three sequential stages, three agents, and the within-hour state

Takeaway. A sequential-market model in the Ito and Reguant (2016) tradition — a strategic bidder, a forecast-driven renewable aggregator, and a capacity-bounded arbitrageur — augmented on two dimensions: a granularity selector $Q^m \in \{1, Q\}$ at each stage, and an explicit within-hour state (a deterministic ramp plus redistributive news) that hourly products mechanically average away. The strategic bidder’s step offer is solved exactly: the optimal n -tranche ladder places equal-MW tranches on an even price grid along its monopoly locus, with closed forms for the bid-shape statistics $(\sigma_p, \text{HHI}_p, \hat{\beta})$ measured in §4.3. The reform path $(1, 1) \rightarrow (1, Q) \rightarrow (Q, Q)$ delivers seven sign predictions for prices, bid shapes, wedge dispersion, cleared MWh, and imbalance costs.

We extend Ito and Reguant (2016)’s sequential-markets framework on two dimensions — an explicit operational renewable aggregator and a granularity variable $Q^m \in \{1, Q\}$ at each stage — and trace the Spanish backward reform path $(1, 1) \rightarrow (1, Q) \rightarrow (Q, Q)$. The mechanism in one line: granularity determines which within-hour states are *contractible*; the strategic bidder’s step offer tracks its monopoly locus over the states its product is exposed to, so when a venue becomes per-quarter its competing tier widens and regrades precisely in the hours where the within-hour state varies; equilibrium price levels move only through residual-demand slopes and venue supply, because ladder geometry is level-neutral; and the within-hour wedge dispersion does not collapse at (Q, Q) because per-product arbitrage capacity binds at K_a/Q . Full derivations live in Appendix D; the main body collects the closed forms and the comparative statics that §6 takes to the data.

A representative clock-hour contains Q equal-length subperiods. Three sequential stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own *granularity* $Q^m \in \{1, Q\}$ — it clears either once per hour or once per subperiod — defining a $2^3 = 8$ -state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1, 1) \rightarrow (1, Q) \rightarrow (Q, Q)$. The forward and spot markets map empirically to the electricity day-ahead and intraday auctions respectively; we use the more general terminology throughout. Figure 1 summarises.

Bidders and primitives. The **strategic bidder** is a residual monopolist with constant marginal cost c ; it delivers *exactly* (no production shock) and therefore does not internalise the imbalance penalty γ . Its offer per delivery product is a non-decreasing step function — a ladder of tranches $\{(p_k, \Delta q_k)\}$ — submitted before the product’s clearing state is known and cleared at uniform price (§3.2). The **operational aggregator** is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$ (forecast-error range Δ_η); it allocates total hourly MWh $\bar{Y}_h$ across forward and spot and *does*

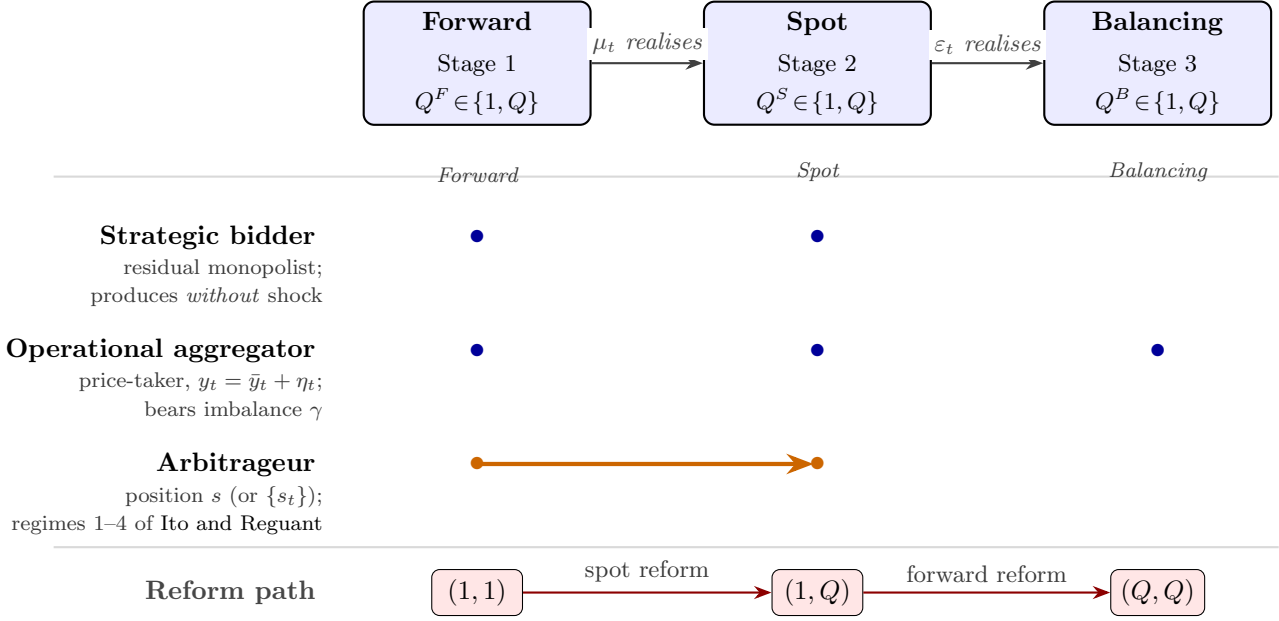


Figure 1: Sequential markets, granularity, and agents. *Top:* a forward market, a spot market, and a balancing settlement on a timeline; each has its own granularity selector $Q^m \in \{1, Q\}$ and information resolves between them. The forward market maps empirically to the electricity day-ahead auction, the spot market to the intraday auction. *Middle:* which agents act at which stages (blue dots), and the arbitrageur position s linking forward to spot (orange arrow). The strategic bidder produces without uncertainty and does not internalise the imbalance penalty; only the operational aggregator does, since it bears η_t . *Bottom:* the auction-side reform path $(Q^F, Q^S) = (1, 1) \rightarrow (1, Q) \rightarrow (Q, Q)$ traversed in the main analysis.

internalise the imbalance penalty, at the rate $\gamma(Q^B)$ set by the balancing granularity. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§3.3); arbitrage transmits price impact but not imbalance risk.

The within-hour state. Write the strategic bidder’s residual demand for delivery quarter t in market m as $q_{mt} = A_{mt} - b_{mt} p_{mt}$, net of arbitrage positions and aggregator supply (Appendix D.1 states the full Ito and Reguant specification). The intercept decomposes into three pieces,

$$A_{mt} = \bar{A} + \zeta_t + v_{mt}.$$

(i) ζ_t is the *deterministic within-hour profile* (the ramp): $\sum_t \zeta_t = 0$, dispersion $\sigma_\zeta^2 = Q^{-1} \sum_t \zeta_t^2$, large in critical hours and ≈ 0 in flat hours (§4.5), known to everyone at all times. (ii) v_{mt} is the *clearing state* the bidder screens at stage m : unknown at submission, embedded in the residual demand it faces at clearing. (iii) Between the forward and spot gates the within-hour *update* ω_t realises and becomes the spot’s traded object: $\omega_t = \zeta_t + \mu_t$ while the forward is hourly (the profile is not yet contractible upstream), and $\omega_t = \mu_t$ once the forward is per-quarter, where μ_t is mean-zero redistributive news ($\sum_t \mu_t = 0$: ramp-timing shifts that move MW between adjacent quarters — a sunset fifteen minutes earlier moves the solar drop from one quarter to its neighbour with no hour-level trace). We write $\sigma_\omega^2 = \sigma_\zeta^2 + \sigma_\mu^2$ and $\sigma_\omega^2 = \sigma_\mu^2$ for the two cases.

Granularity as exposure to states. Decompose the stage- m clearing state as $v_{mt} = \theta_m + \mu_{mt}$: an hour-level component and a redistributive component with $\sum_t \mu_{mt} = 0$. An hourly product clears against the within-hour average, $Q^{-1} \sum_t v_{mt} = \theta_m$ — redistributive components cancel mechanically — while a per-quarter product clears against its own v_{mt} . We assume the screened state is uniform at each granularity: $\theta_m \sim U[-\Delta_m, \Delta_m]$ for an hourly product, and $v_{mt} \sim$

$U[-\rho\chi\Delta_m, +\rho\chi\Delta_m]$ for a quarter product, with an amplification $\rho\chi \geq 1$ specific to the hour class: $\rho \approx 1$ in flat hours and $\rho > 1$ in ramp hours, where ramp-timing uncertainty concentrates.¹ Granularity thus changes *which* states a product is exposed to: hourly products are blind to the within-hour redistributive component at every stage; per-quarter products price it.

Solution concept. We solve the game by **backward induction**: at Stage 2 the strategic bidder best-responds per spot product taking the forward outcome as given; the arbitrageur takes the implied price spread and chooses s within its regime; at Stage 1 the strategic bidder commits its forward ladder internalising the spot continuation value (the forward price anchors spot demand), which matters both for the forward price level and — through the slope of the stage-1 locus — for the forward ladder (§3.5). Full step-by-step backward induction in Appendices D.4–D.6.

Model abstractions: residual monopolist, supply functions, and the passive balancing stage. Three simplifications deserve flagging. First, the strategic bidder is a single residual monopolist per product, even though Spanish wholesale concentration is moderate (HHI ~ 1100) and several firms compete on the bid stack. The residual-monopolist is a stylised device that delivers qualitative comparative statics on within-band conduct; its empirical counterpart is the per-firm residual demand of Ito and Reguant (2016) recovered in Appendix A.10. Second, within a product the bid *is* a supply function, and with a single strategic bidder facing one-dimensional residual-demand uncertainty the optimal schedule is exact (§3.2); what we do not model is strategic interaction in supply functions across several large firms — the Klemperer and Meyer (1989) equilibrium concept for uniform-price multi-unit auctions — which gets intractable in a sequential-market setting. Third, the balancing stage in the model is a passive settlement of the residual imbalance, *not* an active auction with bidding. The aggregator internalises its own deviation through the penalty γ ; the strategic bidder produces with certainty and so carries no individual imbalance; the unpriced within-hour residual collects in C_{sys}^I for the welfare account (Appendix D.7, D.8). In reality REE runs aFRR/mFRR auctions where firms bid actively; we abstract from that to keep the focus on how forward/spot granularity reshapes the auction stages. The model’s prediction that finer auctions absorb the within-hour state upstream and shrink the balancing residual is what §6.3 reads against the empirical ajuste-cost decomposition.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant’s residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m : the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup delivers the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on two objects our data record directly: *why* the fringe’s willingness to supply changes with granularity, and *what shape* the strategic bidder’s offer takes. We make the fringe explicit (the operational aggregator with forecast errors and an imbalance penalty γ), so the hedging gain from per-period scheduling has the closed form $G(\Delta_\eta) = \gamma Q \Delta_\eta / 2$ (Appendix D.2); and we let the strategic bidder choose the step offer the OMIE files record, so the empirical bid-shape quintuple of §4.3 acquires exact model counterparts (§3.2) rather than serving as proxies.

¹The uniform family is exactly closed under this aggregation in a tiling construction: if $\theta_m \sim U[-\Delta_m, \Delta_m]$ and $\mu_{mt} \in \{-\Delta_m, +\Delta_m\}$ with equally many quarters up and down, then $v_{mt} = \theta_m + \mu_{mt} \sim U[-2\Delta_m, 2\Delta_m]$ exactly, i.e. $\rho = 2$ exactly; redistributive news supported on odd multiples of Δ_m delivers any even integer ρ . We carry a general $\rho\chi$.

3.2 Bid ladders within a product: step offers along the monopoly locus

Takeaway. With linear residual demand and one-dimensional clearing-state uncertainty, a continuous supply schedule replicates the full-information monopoly outcome state by state, so a finite tranche count is the only friction. The optimal n -tranche ladder splits the screened states into equal intervals: n equal-MW competing tranches on an even price grid spanning the conditional monopoly-price range, with closed forms $\text{HHI}_p = 1/n$, σ_p , and measured ladder slope $\hat{\beta} = 1/b$. The ladder's geometry is level-neutral — the expected clearing price is the same for every n — so bid shape and price levels are separate channels of the reform. The granularity–variation interaction itself has a one-line closed form: a product's conditional monopoly-price range is exactly Δ/b , and granularity multiplies the screened range Δ by $\rho_{\mathcal{X}}$ precisely where within-hour variation lives (Corollary 2).

Fix one delivery product with residual demand $q = A - bp$, $A = \bar{a} + v$, $v \sim U[-\Delta, \Delta]$, and constant marginal cost c . The bidder submits a non-decreasing step offer before v realises; the auction clears at the crossing of the realised residual demand with the offer. When the crossing lands on a horizontal segment, the marginal tranche is partially filled and sets the price; when it lands on a vertical riser, the cumulative quantity is fixed and the realised demand sets the price. Both regimes occur in equilibrium and both are visible in the OMIE clearing data.

Lemma 1 (Continuous benchmark). *The supply schedule $S^*(p) = b(p - c)$ clears at the full-information monopoly point*

$$p^m(A) = \frac{A + bc}{2b}, \quad q^m(A) = \frac{A - bc}{2}$$

in every state A , and is the unique non-decreasing schedule that does so on the contested range. Expected profit equals $\mathbb{E}[(A - bc)^2]/(4b)$, the unconditional monopoly value.

Under one-dimensional uncertainty the bidder loses nothing to not knowing the state: the supply function traces the monopoly locus — the single-firm benchmark of Klemperer and Meyer (1989). Real offers are step functions with few tranches (OMIE caps tranche counts, and bid management is costly), so the relevant problem is the best n -step approximation of the locus. It has an exact solution.

Assumption 1. $q^m(\bar{a} - \Delta) \geq \Delta/(2(2n + 1))$: the monopoly quantity is interior in every state, by a margin of a quarter cell.

Proposition 1 (Optimal n -tranche ladder). *Let $w = 2\Delta/(2n + 1)$ and let Assumption 1 hold. The optimal offer with n competing price levels consists of an inframarginal base of $q^m(\bar{a} - \Delta + w/2)$ MW priced at the floor, plus n competing tranches of exactly w MW each at prices*

$$p_k = p^m\left(\bar{a} - \Delta + \left(2k - \frac{1}{2}\right)w\right), \quad k = 1, \dots, n,$$

evenly spaced with grid step $p_{k+1} - p_k = w/b$ and spanning $(n - 1)w/b$. The ladder splits the state range into $2n + 1$ equal intervals, each served at its midpoint monopoly point — the clearing rule itself places the regime switches exactly midway between adjacent tranches — and expected profit is

$$\mathbb{E}[\pi_n] = \frac{\mathbb{E}[(A - bc)^2]}{4b} - \frac{\Delta^2}{12b(2n + 1)^2}.$$

(Proof: Appendix D.3.)

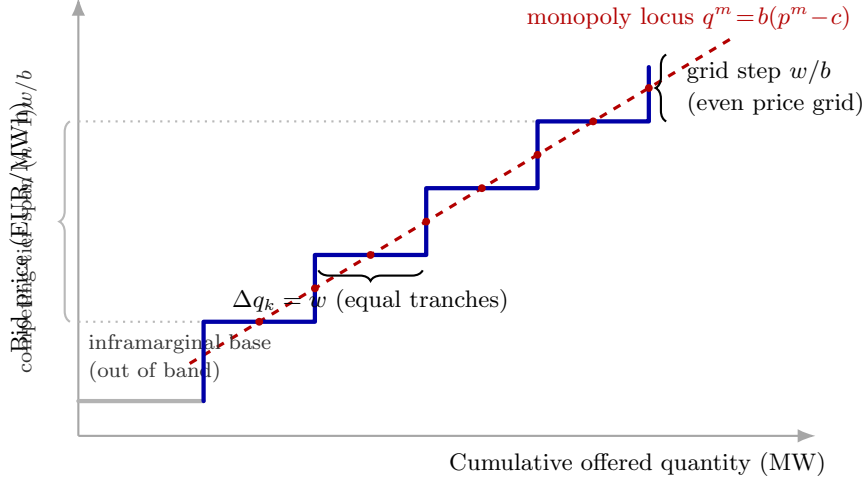


Figure 2: The optimal ladder and the monopoly locus ($n = 4$ competing tranches). The blue step offer weaves across the dashed monopoly locus: every horizontal segment crosses it at its midpoint (states where the tranche price is exactly the monopoly price) and every vertical riser crosses it at its midpoint (states where the cumulative quantity is exactly the monopoly quantity); red dots mark the implemented points. Competing tranches carry equal MW $w = 2\Delta/(2n + 1)$ on an even price grid of step w/b ; the inframarginal base sits below the contested range and outside the empirical band of §4.3.

Corollary 1 (Bid-shape statistics). *Computed over the competing tranches — the in-band construction of §4.3, which nets out the inframarginal base and parked tiers — the optimal ladder delivers in closed form*

$$HHI_p = \frac{1}{n}, \quad \sigma_p = \frac{2\Delta}{(2n + 1)b} \sqrt{\frac{n^2 - 1}{12}} \xrightarrow{n \rightarrow \infty} \frac{\sigma_v}{2b}, \quad \hat{\beta} = \frac{1}{b}, \quad \hat{\gamma} = 0,$$

with $\sigma_v = \Delta/\sqrt{3}$ the SD of the screened state. The OLS slope of price on cumulative in-band quantity identifies the inverse per-product residual-demand slope exactly; curvature is zero under linear residual demand, so a non-zero $\hat{\gamma}$ in the data picks up curvature in residual demand or marginal cost that the model omits — one more reason §4.3 treats $\hat{\beta}, \hat{\gamma}$ as supporting rather than headline objects.

Remark 1 (Level neutrality). $\mathbb{E}[\text{clearing price}] = p^m(\bar{a})$ for every n : within each interval of states the realised price averages to the monopoly price at the interval's midpoint, so ladder geometry reshapes dispersion and surplus but never the mean price (Appendix D.3). Bid-shape changes and price-level changes are therefore *separate channels*: a reform can regrade ladders strongly while leaving price levels untouched. §6 finds exactly this pattern at DA15.

Proposition 2 (Tranche budget). *With a linear cost $\kappa > 0$ per competing tranche (treating n as continuous; the integer optimum is one of the two adjacent integers), the optimal depth satisfies*

$$2n^* + 1 = \left(\frac{\Delta^2}{3b\kappa} \right)^{1/3} :$$

ladders deepen with the screened range, $n^* + \frac{1}{2} = \frac{1}{2} \Delta^{2/3} (3b\kappa)^{-1/3}$, and flatten with market thickness b and complexity cost κ . Jointly with Corollary 1, a wider screened range raises σ_p and lowers HHI_p together — the joint signature the bid-shape DiD of §5.2 tests.

Corollary 2 (Granularity $\times$ within-hour variation, in closed form). *Let a venue switch from hourly to per-quarter products, so that in hour class $\mathcal{X}$ the screened range moves from Δ to*

$\rho_X \Delta$ while the per-product thinness moves from b to b' . Write $\sigma_p(\Delta; b)$ for the fixed- n value in Corollary 1. Then, exactly:

- (i) the conditional monopoly-price range of the product — the range of clearing prices under the continuous benchmark of Lemma 1 — moves from $\frac{\Delta}{b}$ to $\frac{\rho_X \Delta}{b'}$;
- (ii) at fixed depth n , the tranche-price span moves from $\frac{2(n-1)\Delta}{(2n+1)b}$ to $\rho_X \frac{2(n-1)\Delta}{(2n+1)b'}$ and the dispersion from $\sigma_p(\Delta; b)$ to $\rho_X \sigma_p(\Delta; b')$, while $\text{HHI}_p = 1/n$ is unchanged;
- (iii) with the depth endogenous (Proposition 2), $2n^*+1$ moves from $(\Delta^2/(3b\kappa))^{1/3}$ to $\rho_X^{2/3} (\Delta^2/(3b'\kappa))^{1/3}$, so $\text{HHI}_p = 1/n^*$ falls in critical hours exactly when σ_p rises;
- (iv) the critical-minus-flat, post-minus-pre double difference nets out the thinness change, which is common to both hour classes: with $\rho_{\text{flat}} = 1$ and at fixed depth n ,

$$\underbrace{[\sigma_p(\rho_X \Delta; b') - \sigma_p(\Delta; b)]}_{\text{critical hours}} - \underbrace{[\sigma_p(\Delta; b') - \sigma_p(\Delta; b)]}_{\text{flat hours}} = (\rho_X - 1) \frac{2\Delta}{(2n+1)b'} \sqrt{\frac{n^2 - 1}{12}},$$

strictly increasing in the within-hour variation that ρ_X encodes (Lemma 2) and exactly zero when the hour is internally flat.

(Proof: Appendix D.3.)

This is the mechanism in one display: the reform changes bid shape exactly where, and exactly as much as, within-hour variation amplifies the state a product screens. Anything common to both hour classes — thinness changes, complexity costs, price-level seasonality — drops out of (iv) by construction.

Assumption 2 (Stage-1 second-order condition). $B_2 \equiv \sum_t b_{2t} < 2b_1$: the total continuation slope of the spot products hanging off the forward product is strictly less than twice the forward slope. The condition delivers an interior stage-1 monopoly locus and the closed form for the continuation slope in Corollary 3.

Corollary 3 (Continuation slope and forward locus). *Under Assumption 2, the slope of the stage-1 monopoly locus is*

$$\frac{dp_1^*}{dv} = \frac{1}{2b_1 - B_2/2},$$

the measured forward ladder slope is $\hat{\beta}_1 = 1/(b_1 - B_2/2)$, and the stage-1 clearing-price range over the screened states is exactly $2\Delta_1/(2b_1 - B_2/2)$. As the spot venue becomes per-quarter, B_2 jumps from a single hourly slope to the sum of Q per-quarter slopes; the forward locus, the measured slope, and the price range all steepen and widen correspondingly — the cross-market anticipation channel of §3.5(M4).

Remark 2 (Price-takers post blocks). A price-taking unit optimally offers its marginal-cost schedule; for a zero-marginal-cost renewable that is a single block at the floor — $\sigma_p = 0$, $\text{HHI}_p = 1$, out of band, for any granularity (panel (a) of Figure 5). The bid-shape margin is therefore dispatchable-specific; the renewable margin of the reform is the quantity re-scheduling of §3.5 (M1).

Numerical example. With $\Delta = 200$ MW, $b = 50$ MW per EUR/MWh, and $n = 5$: tranches of 36 MW, a grid step of 0.73 EUR/MWh, a competing-tier span of 2.9 EUR/MWh, $\sigma_p = 1.03$ EUR/MWh, and $\text{HHI}_p = 0.2$ — the order of magnitude of the CCGT critical-hour cells in

Table 6. In a ramp hour with $\rho_X = 2$ the same product as a quarter contract spans 5.8 EUR/MWh with $\sigma_p = 2.06$ EUR/MWh at the same depth, and the endogenous depth rises to $2n^* + 1 = 2^{2/3} \cdot 11 \approx 17.5$ (so $n^* \approx 8$, $\text{HHI}_p \approx 0.12$); in a flat hour ($\rho = 1$) nothing moves.

3.3 Arbitrage regimes: four benchmarks, one empirical

Takeaway. The arbitrageur's position s across the four Ito and Reguant regimes (no arbitrage, full, limited, strategic) determines whether the day-ahead/intraday wedge is closed or stays positive. Empirically, real arbitrageurs have bounded capacity and market power, so the limited regime (capacity ceiling binds) is the relevant benchmark; once granularity makes the wedge per-product at (Q, Q) , the per-product capacity constraint binds tightly at K_a/Q and delivers the non-collapse of within-hour wedge dispersion documented in §6.4.

Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s(p_1 - p_2)$. Four regimes pin s :

- (1) *No arbitrage*: $s = 0$, positive forward premium $p_1 > p_2 > c$.
- (2) *Full / competitive arbitrage*: s closes the spread, $p_1 = p_2$; the strategic bidder withholds more in response.
- (3) *Limited arbitrage*: a capacity ceiling $s \leq K_a$ binds, sustaining a positive wedge whose size depends on K_a .
- (4) *Strategic arbitrage*: the arbitrageur internalises price impact and stops short of full, giving the closed-form benchmark

$$\boxed{p_1 - p_2 = \frac{p_1 - c}{4} > 0.} \quad (1)$$

Comparative-statics signs are the same across (1), (3), (4) (Ito and Reguant Table 1).

3.4 Pre-reform (1, 1): scarcity wedge, no within-hour objects

Both auctions hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains under any non-full-arbitrage regime (Appendix D.4); the hourly spot trades the hour-level update $\bar{\omega}$, but neither venue can carry the within-hour profile or the redistributive news, so the balancing discrepancy is $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$. All within-hour dispersion objects (the $D^\bullet$ outcomes of §4.4, the within-hour wedge SD) are mechanically zero, and the competing tiers of both ladders screen the hour-level ranges Δ_1 and Δ_2 — the pre-reform baseline of the bid-shape statistics. Two features carry through every subsequent reform: the scarcity wedge survives under any non-full-arbitrage regime, and whatever the contracting grid cannot carry loads onto the balancing residual.

3.5 Spot reform (1, Q): contractibility arrives

Takeaway. The first granular venue makes the within-hour state contractible. Four mechanisms activate: (M1) the aggregator's hedging gain pulls supply to the granular leg; (M2) the spot price level re-clears against the migrated supply and per-product slopes, and transmits to the forward through arbitrage; (M3) within-hour spot dispersion and the wedge SD turn on at $\sigma_\omega/(2b_2)$, concentrated in ramp hours; (M4) the spot competing tier is exposed to ρ -amplified states (widening and regrading in critical hours), and the *forward* ladder steepens through the continuation slope — the cross-market anticipation channel. The balancing residual collapses to ε_t already at this state.

The spot market becomes per-period; the forward market stays hourly.

(M1) The aggregator gains from granular spot scheduling. The aggregator commits a forward schedule at Stage 1, observes the forecast update η_t , and then sets a spot schedule at Stage 2. The reform changes WHAT it can choose at Stage 2:

	Hourly spot ($Q^S = 1$)	Granular spot ($Q^S = Q$)
At Stage 2 the aggregator chooses	ONE number S_2 for the whole hour	Q numbers $\{S_{2,t}\}$ per period
Per-period schedule	uniform $(S_1 + S_2)/Q$	$S_1/Q + S_{2,t}^*$
Can respond to η_t per period?	no	yes, exactly
Per-period imbalance	η_t	0

Granular spot lets the aggregator wait until η_t resolves and submit a per-period schedule that absorbs the forecast update. Hourly spot forces a single schedule before η_t is known, so the forecast update mechanically loads onto the imbalance bill.

The closed-form hedging gain under uniform shocks. Take $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$, i.i.d. across t (PDF $1/(2\Delta_\eta)$), and linear penalty $\gamma|\mathcal{I}_t|$ per period. Then

$$\mathbb{E}|\eta_t| = \int_{-\Delta_\eta}^{\Delta_\eta} \frac{|x|}{2\Delta_\eta} dx = \frac{\Delta_\eta}{2}.$$

Expected hourly imbalance cost: $\mathbb{E}[\mathcal{C}_{\text{hour}}] = \gamma Q(\Delta_\eta/2)$. Granular expected cost: $\mathbb{E}[\mathcal{C}_{\text{gran}}] = 0$. The aggregator's expected hourly hedging gain is the avoided cost:

$$\boxed{G(\Delta_\eta) = \mathbb{E}[\mathcal{C}_{\text{hour}}] - \mathbb{E}[\mathcal{C}_{\text{gran}}] = \frac{\gamma Q \Delta_\eta}{2}, \quad \frac{\partial G}{\partial \Delta_\eta} = \frac{\gamma Q}{2} > 0.} \quad (2)$$

The gain is strictly positive and strictly increasing in the forecast-error range. *Numerical example.* With $Q = 4$ (quarter-hours), $\gamma = 1$ EUR/MWh, and $\Delta_\eta = 10$ MW, $G = 20$ EUR per clock-hour; a wind unit with twice the forecast range gets $G = 40$ EUR per clock-hour. Derivation in Appendix D.2.

Note on Δ^ .* The price-taker setup gives a corner solution for the individual aggregator's forward-vs-spot split (the imbalance cost is independent of the split once we impose no-average-bias, so the FOC reduces to $p_1 = p_2$). The migration onto the granular leg is an aggregate equilibrium statement: $G > 0$ generates a participation premium on the granular leg that equilibrium prices internalise, producing the sell-share migration of Result 6.4.

(M2) Price levels: venue re-clearing and cross-market transmission. The spot best response per quarter is $p_{2t}^* = (p_1 + c)/2 + (\omega_t + s_t - S_{2t}^{op})/(2b_{2t})$ (Appendix D.5). Averaging over the hour, the spot level falls with migrated aggregator supply ($\partial \bar{p}_2 / \partial S_2^{op} = -1/(2b_2) < 0$) and with any per-product thickening of the slope through the markup identity (6) below. On the forward leg two channels cut in opposite directions: arbitrage transmits the spot discount upstream (under full or strategic arbitrage the forward tracks the spot down one-for-one up to the regime wedge of §3.3), while the strengthened continuation value pushes toward more stage-1 withholding (Appendix D.5). The empirical DA-side drop of the same magnitude as the IDA drop (§6.1) is consistent with the arbitrage channel dominating; the model signs each channel and lets the data adjudicate the net.

(M3) Block parity reveals within-hour spot dispersion. The hourly forward can be arbitrated only against the *average* of the Q spot prices — block parity — so arbitrage disciplines

the mean wedge but not the within-hour dispersion of the granular leg. The spot prices spread across quarters with the update:

$$\boxed{\text{sd}_t(p_{2t}) = \frac{\sigma_\omega}{2b_2}, \quad \sigma_\omega^2 = \sigma_\zeta^2 + \sigma_\mu^2,} \quad (3)$$

plus a thinness-asymmetry component when b_{2t} varies across quarters (Appendix D.5). Since p_1 is one number per clock-hour, (3) is also the within-hour wedge SD — mechanically zero before the reform, and largest in ramp hours where σ_ζ is large.

(M4) Bid ladders: own-venue exposure and cross-market anticipation. Two exact ladder results. (a) *Spot tier, own venue.* Each quarter product screens the ρ_χ -amplified state (§3.1): by Corollary 2, at fixed depth the critical-hour spot dispersion multiplies by exactly ρ_χ and the conditional monopoly-price range moves from Δ_2/b_2 to $\rho_\chi\Delta_2/b_{2t}$, while the endogenous depth multiplies $2n^* + 1$ by exactly $\rho_\chi^{2/3}$, so $\text{HHI}_p = 1/n^*$ falls — all relative to flat hours, where $\rho_\chi = 1$ leaves the tier unchanged and only the class-common thinness shift remains, which the critical-vs-flat contrast nets out (Corollary 2(iv)) — exactly the contrast the bid-shape DiD of §5.2 is built on. (b) *Forward tier, cross-market.* The stage-1 problem internalises the spot continuation value, and the slope of the stage-1 monopoly locus is

$$\boxed{\frac{dp_1^*}{dv} = \frac{1}{2b_1 - B_2/2}, \quad B_2 \equiv \sum_t b_{2t},} \quad (4)$$

with B_2 the total slope of the spot products hanging off the forward product (Appendix D.3; we maintain $B_2 < 2b_1$, Assumption 2). At the spot reform the continuation coefficient jumps from the single hourly slope b_2 to the sum of Q per-quarter slopes — empirically the per-quarter spot slopes are of the same order as the hourly one (Table 17), so the jump is large — steepening the forward locus and hence the forward competing tier: the stage-1 clearing-price range is exactly $2\Delta_1/(2b_1 - B_2/2)$ and the measured ladder slope becomes $\hat{\beta}_1 = 1/(b_1 - B_2/2)$ (Corollary 3, Appendix D.3). This is the model’s account of the day-ahead bid-shape response at ID15 — anticipation through backward induction — and it matches the sign pattern of the ID15 DA cells in §6.5: σ_p and $\hat{\beta}$ up. The tranche-count response on the cross-market cell is *not* signed by this channel alone (the value of extra tranches falls as the locus flattens in quantity space, while any accompanying rise in the forward screen pushes the other way); we do not claim it.

Balancing: the quarterly spot absorbs $\zeta_t + \mu_t$ and the aggregator hedges η_t , so the balancing discrepancy is $\Delta_t = \varepsilon_t$ already at this state (Appendix D.7). Contractibility arrives with the *first* granular venue.

3.6 Forward reform (Q, Q): relocation and limited per-product arbitrage

Takeaway. The forward reform *relocates* the already-contractible within-hour objects from the spot to the forward rather than creating new contractibility: forward quarter prices separate by the ramp, forward ladders carry the amplified screen, the spot’s update demand shrinks to μ -news, and the within-hour wedge SD does NOT collapse because per-product arbitrage capacity binds at K_a/Q . Balancing is unchanged at ε_t , so the model predicts no further imbalance gain at DA15 — the empirical null is the prediction.

(i) The ramp becomes forward-contractible. Per-quarter forward products price ζ_t at the day-ahead margin: $\mathbb{E}[p_{1t}] = p_1^*(\bar{A} + \zeta_t)$ with pass-through $\lambda_1 \equiv (2b_1 - B_2/2)^{-1}$ per unit of ramp,

so the across-quarter dispersion of forward prices and curve levels turns on at

$$\text{sd}_t(\mathbb{E} p_{1t}) = \lambda_1 \sigma_\zeta$$

— the model counterpart of the jump in the within-hour dispersion outcomes $D^{\bar{p}}$ and D^{MCP} on the DA at DA15 (§4.5). The spot’s update demand correspondingly shrinks from $\zeta_t + \mu_t$ to μ_t .

(ii) Per-product limited arbitrage — and arbitrageur market power. The arbitrageur takes a position s_t from forward to spot per matched quarter, earns $s_t (p_{1t} - p_{2t})$, and faces a capacity cap on its bounded balance sheet. Two objects matter: the wedge MEAN (does forward sit above spot on average?) and the wedge within-day SD (does the gap fluctuate within the day?). The full-arbitrage benchmark gives $p_{1t} = p_{2t}$ per product (wedge SD zero); the strategic-arbitrage benchmark gives the Ito and Reguant wedge (1) per product, with wedge SD $\lambda_1 \sigma_\zeta / 4$. *Both benchmarks rely on the arbitrageur being able to build any per-product position, and both fail empirically.* Arbitrage capacity K_a is bounded; with Q matched products per clock-hour, per-product capacity is K_a / Q , which binds tightly. The symmetric state is therefore in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to

$$\boxed{\text{sd}_h(p_{1t} - p_{2t}) = \frac{\lambda_1 \sigma_\zeta}{2}} \quad (5)$$

(Appendix D.6.3), plus thinness-asymmetry corrections when b_{2t} varies across quarters. The wedge SD therefore does *not* collapse to zero at the symmetric state: arbitrageur market power generates the non-collapse documented empirically at DA15 (§6.4).

(iii) Bid ladders relocate, not duplicate. On the forward side the screen widens from the hour-level Δ_1 to the ρ_χ -amplified per-quarter range — the redistributive component becomes contractible at the forward — so by Corollary 2 the forward competing tier widens and deepens in critical hours exactly as the spot tier did at the spot reform (a critical-vs-flat contrast now on the DA): σ_p multiplies by ρ_χ at fixed depth, $2n^* + 1$ by $\rho_\chi^{2/3}$ at the endogenous depth. Symmetrically, the spot’s per-quarter object loses its ramp-timing component once the ramp is priced upstream. Writing $\rho_\chi^\mu \leq \rho_\chi$ for the amplification generated by the news-only redistributive support — Lemma 2 applied to the smaller support that remains when ramp-timing shifts no longer reach the spot; $\rho_\chi^\mu = \rho_\chi = 1$ in flat hours — the spot screen falls from $\rho_\chi \Delta_2$ to $\rho_\chi^\mu \Delta_2$, and the spot competing tier *tightens* by the same closed forms run in reverse: σ_p multiplies by $\rho_\chi^\mu / \rho_\chi \leq 1$ at fixed depth and $2n^* + 1$ by $(\rho_\chi^\mu / \rho_\chi)^{2/3}$. The venue of expression changes; the ladder mathematics is the same Corollary applied with each venue’s own screen.

(iv) Markup channel and the Cournot identity. The Cournot first-order condition for the strategic bidder against an inverse residual demand of slope b_{mt} is

$$p_t - c = \frac{q_m}{b_{mt}}, \quad \text{markup}_m = \frac{q_m}{b_{mt}}, \quad (6)$$

empirically tested via the strategic markup proxy q_f^{strat} / b_f of §6.6 (Appendix A.10). The per-firm residual-demand slope b_f thickens at every reform on every Big-4 dominant firm; equation (6) then delivers markup compression — the headline market-power result. Note that the price-level prediction operates through the markup identity, *not* through ladder geometry, by Remark 1 (level neutrality).

(v) **Balancing unchanged.** The spot already absorbs the residual update at $(1, Q)$, and the per-quarter forward absorbs the ramp at (Q, Q) ; nothing new loads onto the balancing residual. $\Delta_t = \varepsilon_t$ at both $(1, Q)$ and (Q, Q) . The model’s no-further-imbalance-gain prediction at DA15 is the empirical null on aggregate imbalance volume at DA15 (§6.2).

3.7 Comparative statics across the reform path

Takeaway. Seven sign predictions (P1–P7), pinned by the contractibility logic: the spot reform creates new contractible objects (imbalance gain, wedge SD, spot bid-shape regrading, cross-market forward-locus steepening); the forward reform *relocates* the already-contractible objects (forward bid-shape regrading, spot tier tightening, ramp-induced within-hour DA dispersion) without producing further imbalance gain. Markup compression via the Cournot identity operates wherever per-firm residual demand thickens.

The model delivers seven comparative-statics predictions across the three states, summarised in Table 1.

Table 1: Comparative statics across the reform path under limited per-product arbitrage (empirical regime). Notation: $\lambda_1(B) \equiv (2b_1 - B/2)^{-1}$ is the stage-1 locus slope at continuation coefficient B (Corollary 3; with the venue’s own forward slope, b_{1t} per product at (Q, Q)); $\rho_{\mathcal{X}}^{\mu} \leq \rho_{\mathcal{X}}$ is the news-only amplification of §3.6(iii), and $\rho_{\mathcal{X}}^{\mu} = \rho_{\mathcal{X}} = 1$ in flat hours. In the two ladder rows, twice the product of the entries is the product’s conditional monopoly-price range (Corollary 2(i)); σ_p , the tranche-price span, and HHI_p follow in closed form from Corollary 1 and Proposition 2 evaluated at the screen and slope shown. Imbalance row: quadratic penalty $\gamma \mathbb{E}[\Delta_t^2]$, omitting the unpriceable σ_{ε}^2 common to all states; for uniform shocks $\sigma_{\mu}^2 = \Delta_{\mu}^2/3$ and $\sigma_{\eta}^2 = \Delta_{\eta}^2/3$ (§3.1, §3.8).

Outcome	$(1, 1)$	$(1, Q)$	(Q, Q)
Aggregator hedging gain $G(\Delta_{\eta})$	0	$\frac{\gamma Q \Delta_{\eta}}{2}$	$\frac{\gamma Q \Delta_{\eta}}{2}$
Spot tier: screen $\times$ locus slope	$\Delta_2 \times \frac{1}{2b_2}$	$\rho_{\mathcal{X}} \Delta_2 \times \frac{1}{2b_{2t}}$	$\rho_{\mathcal{X}}^{\mu} \Delta_2 \times \frac{1}{2b_{2t}}$
Forward tier: screen $\times$ locus slope	$\Delta_1 \times \lambda_1(b_2)$	$\Delta_1 \times \lambda_1(\sum_t b_{2t})$	$\rho_{\mathcal{X}} \Delta_1 \times \lambda_1(b_{2t})$
Within-hour wedge SD	0	$\frac{\sigma_{\omega}}{2b_2}, \sigma_{\omega}^2 = \sigma_{\zeta}^2 + \sigma_{\mu}^2$	$\frac{\lambda_1 \sigma_{\zeta}}{2} + \text{thin. asym.}$
Within-hour DA dispersion $D^{\bar{p}}$	0	0	$\lambda_1 \sigma_{\zeta}$
Balancing Δ_t	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$	ε_t	ε_t
Imbalance cost quadratic	$C^I/(\gamma Q), \sigma_{\zeta}^2 + \sigma_{\mu}^2 + \sigma_{\eta}^2$	0	0

Arbitrage regime is empirical. Real arbitrageurs have market power and bounded balance sheets: the persistent forward premium observed across electricity markets rules out full arbitrage, and the per-product capacity constraint binds at (Q, Q) . The relevant Ito and Reguant column is Result 3 (limited), not Result 4 (strategic, unlimited); per-product limited arbitrage at (Q, Q) delivers the model’s account of the empirical asymmetric-persistence of wedge dispersion (§3.6(ii)).

Predictions to test empirically. The model delivers seven sign predictions across the reform path; §6 verifies them in turn: (P1) aggregator quantity shifts from forward to spot at the spot reform, larger in clock-hours with bigger forecast uncertainty; (P2) the granular-side equilibrium price drops at the spot reform via venue re-clearing of migrated aggregator supply and Cournot markup compression as per-firm b_f thickens — *not* through ladder geometry, by Remark 1; (P3) within-hour wedge dispersion turns on at the spot reform at $\sigma_{\omega}/(2b_2)$, largest in ramp hours; (P4) the spot competing tier widens and deepens in critical hours at the spot reform

(Corollary 2: σ_p multiplies by $\rho_{\mathcal{X}}$ at fixed depth, $2n^* + 1$ by $\rho_{\mathcal{X}}^{2/3}$), and the forward competing tier steepens at the spot reform through the continuation-slope channel (Corollary 3); (P5) the forward competing tier widens and deepens at the forward reform (the ramp is now forward-contractible: screen $\Delta_1 \rightarrow \rho_{\mathcal{X}}\Delta_1$), and the spot competing tier correspondingly tightens (screen $\rho_{\mathcal{X}}\Delta_2 \rightarrow \rho_{\mathcal{X}}^{\mu}\Delta_2$) — the relocation pattern; (P6) within-hour wedge SD does not collapse at (Q, Q) — the empirical asymmetric-persistence finding under per-product limited arbitrage; (P7) production-unit arbitrageurs exist and are capacity-constrained (we test this directly by checking whether production plants submit both buy and sell offers in OMIE data).

Asymmetry of activation across the two reform steps. The contractibility framing pins which reform delivers which channel. New contractibility arrives at the *spot* reform: imbalance gain (the residual collapses to ε_t once the per-quarter spot absorbs $\zeta + \mu$), within-hour wedge SD (mechanically zero before, $\sigma_{\omega}/(2b_2)$ after), spot bid-shape regrading (the spot tier screens the $\rho_{\mathcal{X}}$ -amplified state), and cross-market forward-ladder steepening through the continuation-slope channel. The *forward* reform *relocates* the bid-shape regrading from spot to forward and turns on within-hour DA price dispersion, but it does *not* produce a further imbalance gain — the spot already absorbed everything. This is the model basis for the empirical pattern of §6.2: -33% penalty per MWh at ID15, flat at DA15.

3.8 Welfare across the reform path

Takeaway. Three positive welfare channels — imbalance saving (delivered at ID15 only, by contractibility); markup compression via per-firm b_f thickening at every reform (Cournot identity); and the discreteness loss that bid-shape regrading reduces (§3.2, Proposition 1). One distributional channel — bid-ladder rent — is mostly $\text{CS} \rightarrow \pi^{SB}$ transfer.

- **Imbalance saving, ID15 only (dominant gain).** The per-quarter spot absorbs both the ramp ζ_t and the redistributive news μ_t , and the aggregator hedges η_t . Imbalance cost falls from $\gamma Q(\sigma_{\zeta}^2 + \sigma_{\mu}^2 + \sigma_{\eta}^2)$ at $(1, 1)$ to 0 at $(1, Q)$ and stays at 0 at (Q, Q) (quadratic penalty, Appendix D.8). Net of the small allocative cost of per-quarter monopoly pass-through (the discrimination term of Appendix D.8), the saving is positive whenever $\gamma > 1/(8b_2)$ — comfortably satisfied at observed balancing premia. The model predicts *no further imbalance gain at DA15* — consistent with the flat-DA15 imbalance penalty of §6.2.
- **Markup compression at every reform.** Per-firm b_f thickens at every reform on every Big-4 firm (Table 4); equation (6) delivers markup compression linearly. Operates through venue re-clearing and per-firm slope thickening, *not* through ladder geometry (Remark 1).
- **Discreteness loss.** The optimal-ladder expected profit loss relative to the continuous benchmark is $\Delta^2/(12b(2n^*+1)^2)$ (Proposition 1); a deeper n^* at the granular-leg critical hours (Proposition 2) reduces it. Mostly $\text{CS} \rightarrow \pi^{SB}$ transfer within a product.
- **Less strategic withholding under limited per-product arbitrage.** The spot markup falls from $3(p_1 - c)/4$ (strategic) to $(p_1 - c)/2$ (limited per product), per Ito-Reguant Result 2.

Per-component derivations under quadratic imbalance penalty $\gamma\mathcal{I}^2$ in Appendix D.8.

4 Data: OMIE bid-level offers, weather, redispatch

4.1 Sources

Takeaway. Bid-level OMIE data (every unit, every period, every offer tranche) covering 2024-06-14 to 2026-03-06, plus ENTSO-E load and renewable generation, plus ESIOs daily settlement of REE’s adjustment-cost channels. The bid-level granularity is what lets the bid-shape DiD identify within-day shape changes; the system-level data feeds the BSTS counterfactual and the cost decomposition.

[To be written. Concepts to cover:]

- **OMIE** — bid-level offers in DA (PDBC, files *cab/det*) and IDA (PIBCI, files *icab/idet*); one row per (unit, date, period, tranche); structural type (simple, minimum-income condition, block); OMIE spec v1.37. Daily resolution 2022-01-01 onwards; bid-level detail 2024-06-14 onwards (post-IDA-reform 3-session regime).
- **ENTSO-E Transparency Platform** — 15-min actual total load (code A65), per-technology actual generation (A75: B16 solar PV, B18 wind onshore, B19 wind offshore), installed capacity, cross-border flows. Builds the residual-demand series (load – wind – solar) used for the critical/flat partition and the wind/solar covariates for the BSTS and OLS price specs.
- **ESIOS (REE)** — daily settlement of REE’s adjustment-cost components (TR-up/dn, aFRR-up/dn, mFRR-up/dn, Fase I up/dn, Fase II), 139 curated indicators, per-BSP settlement detail (archives 17/203), aFRR offer curves, unit outages. Token-authenticated via `.env`.
- **Magnitude.** 420 GB raw, 200 GB processed parquet, ~10 billion rows on a dedicated external SSD. Full automated reproducible pipeline.

4.2 Bid offers, reform path, and notation

Indices. Throughout: u unit, d date, p market period (a clock-hour pre-MTU15, a 15-minute quarter post-MTU15), $h \in \{1, \dots, 24\}$ clock-hour, $q \in \{1, \dots, 4\}$ within-hour quarter (post-MTU15 only), s IDA session, t technology, $m \in \{\text{DA}, \text{IDA}\}$ market, k tranche of a step ladder, T_r reform cutover date, $\mathcal{X} \in \{\text{morning ramp}, \text{midday}, \text{evening ramp}, \text{flat}\}$ hour-class.

Bid offer. Each (u, d, p) sell offer is a step ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$ sorted by ascending price, with p_k the bid price (EUR/MWh) and Δq_k the incremental MW added to the unit’s step supply at p_k . Cumulative supply at any price p is $S_{u,d,p}(p) = \sum_{k: p_k \leq p} \Delta q_k$. Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods). Most technologies use simple bids; CCGT is the outlier with the bulk of its offers in MIC or block form.

Market clearing. $\text{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh) of the auction in which the offer was submitted; $Q_{d,p}^*$ is the system cleared quantity. Per-tech cleared MWh per clock-hour: $Q_{t,m,d,h}^{\text{cl}}$.

Reform path. The Spanish wholesale market underwent three granularity reforms in seven months: **ISP15** (2024-12-11), REE’s settlement period moved from 60 to 15 min while the OMIE contract grid stayed at 60 min; **ID15** (2025-03-19), intraday auctions and continuous market moved from 60 to 15 min; **DA15** (2025-10-01), day-ahead moved from 60 to 15 min. The April 2025 blackout (2025-04-28) triggered a continuous *operación reforzada* regime (CCGT-led

voltage support, post-DA redispatch) that overlaps DA15 calendar-wise but operates on a different mechanism; DA15 windows are constructed so this regime is held constant on both sides of the cutover. Figure 3 maps the windows that partition the sample.

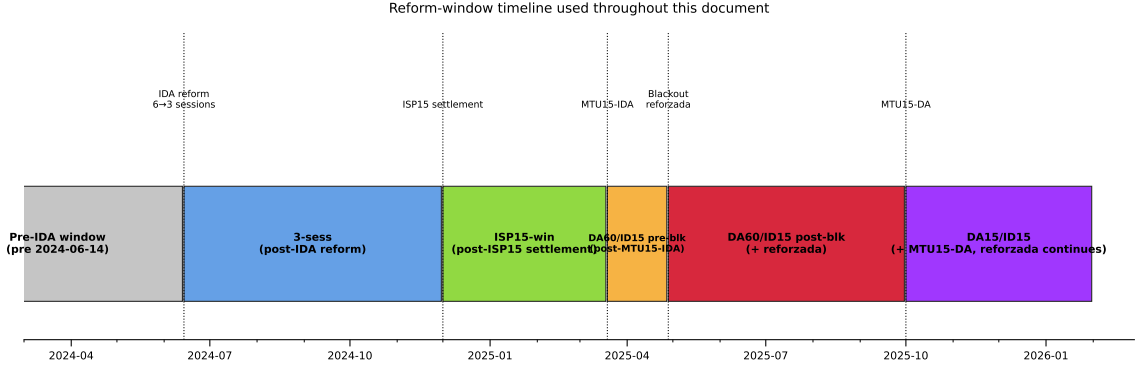


Figure 3: Reform-window timeline. Six named windows partition the 2022-2026 sample at the five reform dates. *Pre-IDA*: baseline 6-session IDA, hourly. *3-sess*: post-IDA-reform 3-session IDA, hourly. *ISP15-win*: REE settlement 15-min, OMIE grid still 60-min. *DA60/ID15 pre-blk*: 40-day pre-blackout window with quarter-hourly intraday. *DA60/ID15 post-blk*: post-blackout reforzada, quarter-hourly intraday. *DA15/ID15*: full 15-min grid post-2025-10-01.

4.3 In-band region and bid-shape outcomes

Takeaway. The bandwidth $\delta = p_{90} - p_{50}$ of the contemporary MCP distribution isolates the *competing tier* of each ladder — tranches close enough to clearing to be in real contention. Per in-band curve we compute five jointly informative bid-shape statistics: MW-weighted price dispersion σ_p , intercept $\hat{\alpha}$ (level), linear slope $\hat{\beta}$, curvature $\hat{\gamma}$ (convex > 0 / concave < 0), and bid-stack concentration HHI_p (MW-weighted Herfindahl of in-band tranches). σ_p is reported alongside the $(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ polynomial fit because it is well-defined on every curve — including single- and few-tranche curves where the OLS slope is noisy — and so serves as the robust price-side anchor of the bid-shape DiD.

Bandwidth. Let δ be a bandwidth that isolates the tranches in contention — close enough to the clearing price to compete in the *competing tier* of the offer (CCGT’s two-tier structure makes this distinction explicit, Appendix C). We anchor δ to the contemporary contested margin: $\delta = p_{90} - p_{50}$ of the MCP distribution in the analysis window, market by market. The eight bandwidths (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are reported below; wider- δ robustness in Appendix A.5.

Window	δ in DA (EUR/MWh)	δ in IDA (EUR/MWh)
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

Table 2: Bandwidths δ used to define the in-band region. δ is set to $p_{90} - p_{50}$ of the MCP distribution in each (window, market). Window-and-market-specific by construction. Bandwidth robustness in Appendix A.5.

In-band set. For each (u, d, p) curve:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \text{MCP}_{d,p}| \leq \delta\}, \quad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}_{u,d,p}} \Delta q_j}, \quad \bar{p}_{u,d,p} = \sum_{k \in \mathcal{B}} w_k p_k.$$

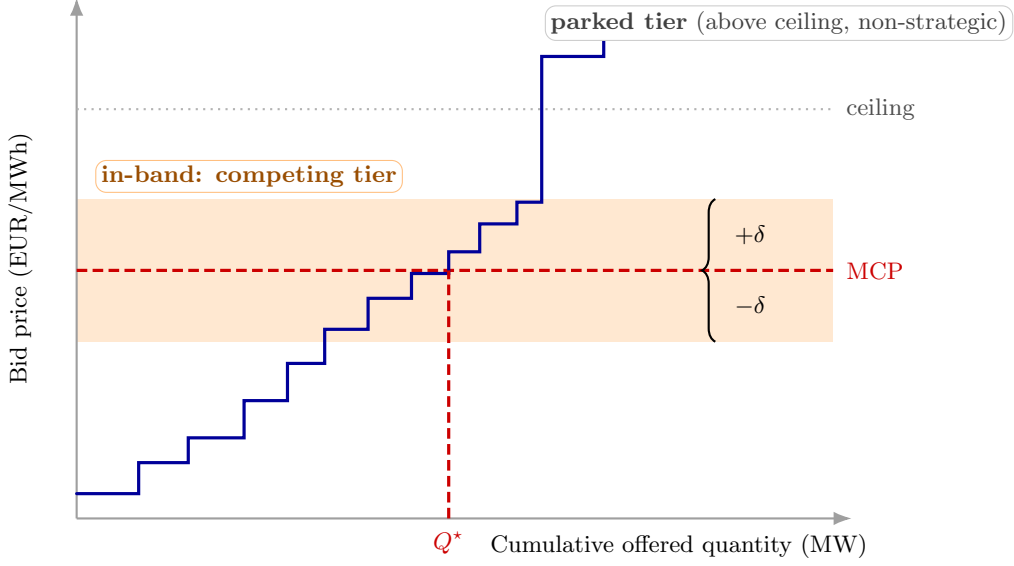


Figure 4: Bandwidth δ and the in-band region. The blue step function is the sorted sell-bid ladder for one quarter. Only tranches falling within $\pm\delta$ of the clearing price (orange band) enter the bid-shape outcomes; the parked tier above the clearing-price ceiling is excluded (Appendix C).

Representative bid curves per technology in critical and flat hours are in Appendix C.4 (Figures 13–14); they illustrate the dispatchable-ladder vs renewable-block contrast.

Five bid-shape outcomes. Per in-band curve we summarise the ladder with a quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p)$. Let $Q_k = \sum_{j \leq k} \Delta q_j$ be the cumulative in-band MW after the k th tranche.

Price-side dispersion (the robust price-side anchor, well-defined on every curve including single- and few-tranche ones where the OLS slope is noisy):

$$\sigma_{p,u,d,p} = \left[\sum_{k \in \mathcal{B}} w_k (p_k - \bar{p}_{u,d,p})^2 \right]^{1/2}.$$

Level and linear slope (unweighted OLS on in-band step endpoints):

$$p_k = \hat{\alpha}_{u,d,p} + \hat{\beta}_{u,d,p} Q_k + e_k.$$

$\hat{\alpha}$ is the geometric level of the ladder; $\hat{\beta}$ is the average slope (EUR/MWh per MW) and the direct empirical counterpart of the model’s per-product residual-demand slope b_{mt} (§3.1).

Curvature (separate quadratic fit; we keep only the quadratic coefficient):

$$p_k = \tilde{\alpha}_{u,d,p} + \tilde{\beta}_{u,d,p} Q_k + \hat{\gamma}_{u,d,p} Q_k^2 + \tilde{e}_k.$$

$\hat{\gamma} > 0$ means convex (cheap MW packed at the bottom, marginal MW priced high); $\hat{\gamma} < 0$ means concave.

Bid-stack concentration (MW-weighted Herfindahl of in-band tranches):

$$\text{HHI}_{p,u,d,p} = \sum_{k \in \mathcal{B}} w_k^2, \quad w_k = \frac{\Delta q_k}{\sum_{j \in \mathcal{B}} \Delta q_j} \in [1/|\mathcal{B}|, 1].$$

[To be written. Concepts to cover:]

- Why MW-weighting on σ_p : profit-uncertainty scales with tranche size, so big tranches drive σ_p and decoy micro-tranches at distant prices barely move it.

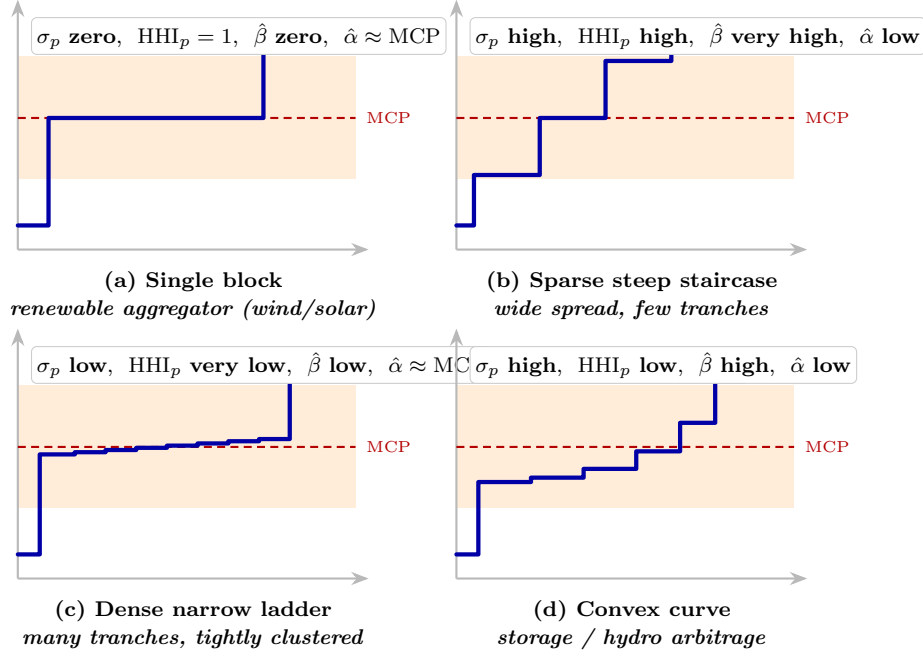


Figure 5: Four stylized in-band bid shapes spanning the $(\sigma_p, \text{HHI}_p, \hat{\beta}, \hat{\alpha})$ space. Each panel shows one bid ladder against a common MCP (red dotted) and bandwidth δ (orange band), with the implied outcome values noted in the corner.

- Why the two-stage construction for $(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$: keeps coefficient interpretation clean; $\hat{\gamma}$ from the quadratic OLS is the curvature orthogonal to the linear trend (Frisch–Waugh).
- Identification of $\hat{\gamma}$ requires $K \geq 3$ tranches; $K = 2$ identifies only $(\hat{\alpha}, \hat{\beta})$.
- σ_p and HHI_p are robust to mechanical changes in tranche-count and price-grid density (both shifted at 2024-06-14 and 2025-03-19); $\hat{\beta}$ and $\hat{\gamma}$ are mechanically sensitive — treat them as supporting, not headline.
- None of the five outcomes depends on price-level seasonality: all are computed relative to the contemporaneous MCP through the bandwidth δ . Figure 5 illustrates them on stylised ladders.

4.4 Aggregate outcomes

Takeaway. System-level outcomes built off the bid stack and the cleared prices: in-band offered energy per (tech, market, day, hour-class); the day-ahead/intraday wedge $\Delta_{d,h}$ and its within-day SD Δ_d^{SD} ; the intraday-auction in-band sell-share by tech; system adjuste cost (per day, EUR M); and five post-MTU15-only within-hour dispersion outcomes $D_{d,h}^\bullet$ that summarise across-quarter shifts in price level, volume, system MCP, ladder spread, and tranche count.

In-band offered energy (per (tech, market, day, hour-class)). $m_{u,d,p} = \sum_{k \in \mathcal{B}_{u,d,p}} \Delta q_k$ aggregated with period length τ_p ($= 1$ pre-MTU15, $1/4$ post-MTU15) to convert MW to MWh:

$$E_{t,m,d,\mathcal{X}}^{\text{band}} = \frac{1}{|\mathcal{X}|} \sum_{h \in \mathcal{X}} \sum_{p \in h} \sum_{u \in t} m_{u,d,p} \cdot \tau_p \quad (\text{MWh per class-hour}).$$

Cross-market wedge (per clock-hour). With $\text{MCP}_{d,h}^m$ the within-market mean of OMIE clearing prices over sub-hour structure on date d :

$$\Delta_{d,h} = \text{MCP}_{d,h}^{\text{DA}} - \text{MCP}_{d,h}^{\text{IDA}}, \quad \Delta_d^{\text{SD}} = \text{sd}_h(\Delta_{d,h}).$$

IDA in-band sell-share (per (tech, day)). Captures production-unit participation between sell-heavy (rent-capture) and buy-heavy (rebalancing):

$$SS_{t,d} = \frac{\sum_{q,k \in \mathcal{B}} q_{t,d,q,k}^{\text{sell}}}{\sum_{q,k \in \mathcal{B}} (q_{t,d,q,k}^{\text{sell}} + q_{t,d,q,k}^{\text{buy}})}.$$

System ajuste cost (per day, EUR M). REE’s adjustment-cost components from ESIOS: up-direction (TR-up, aFRR-up, mFRR-up, Fase I) and down-direction (Fase II, RR-dn). Stacked positive in Figure 6.

Within-hour dispersion outcomes (post-MTU15 only). For each (unit, date, clock-hour) with four quarters $q = 1, \dots, 4$:

$$D_{d,h}^{\bar{p}} = \text{sd}_q(\bar{p}_q), \quad D_{d,h}^m = \text{sd}_q(m_q)/\bar{m}, \quad D_{d,h}^{\text{MCP}} = \text{sd}_q(\text{MCP}_q), \quad D_{d,h}^{\sigma_p} = \text{sd}_q(\sigma_{p,q}), \quad D_{d,h}^{\text{HHI}} = \text{sd}_q(\text{HHI}_{p,q}).$$

All mechanically zero under MTU60; any positive value post-MTU15 is a direct trace of within-hour discrimination.

4.5 Critical / flat hour-class partition

Takeaway. Granular pricing has economic content only where within-hour residual demand actually varies. Measured by 15-minute residual-demand SD, clock-hours partition into critical (morning ramp 5–8h, evening ramp 16–22h; SD 522–1303 MW), flat (overnight 1–3h; SD 336–357 MW), and midday (11–14h; SD 468–502 MW, kept for falsification only). bid-shape DiD uses critical-vs-flat as the within-day identification contrast.

Intuition. Granular pricing only has economic content where within-hour residual demand² actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\text{within}}(h, d) = \text{sd}_q[\text{load} - \text{wind} - \text{solar}]_{q,h,d}$ on 15-min ENTSO-E data (A65 load; A75 PSR codes B16, B18, B19 for wind+solar) and partition clock-hours by median σ_{within} post-2024-06-14:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5, \dots, 8\}$ (morning ramp) and $\mathcal{C}_E = \{16, \dots, 22\}$ (evening ramp); $\sigma_{\text{within}} \in [522, 1303]$ MW.
- **Flat** $\mathcal{F} = \{1, 2, 3\}$: overnight; $\sigma_{\text{within}} \in [336, 357]$ MW.
- **Midday** $\mathcal{M} = \{11, \dots, 14\}$ (solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (Appendix A.2).

Descriptive critical – flat differential. Let $D_{d,h}^\bullet$ stand for any one of the five within-hour dispersion outcomes of §4.4 on date d , clock-hour h . Fit per outcome on the post-MTU15 sample, $h \in \mathcal{C} \cup \mathcal{F}$:

$$D_{d,h}^\bullet = \eta_d + \theta \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon_{d,h}$$

with date fixed effects η_d (distinct from the bandwidth δ). θ is the critical–flat dispersion gap; a descriptive contrast, not a causal estimate.

²Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

Outcome	Subsample	ID15	DA15	crit/flat ratio (IDA, DA)
$D^{\bar{p}}$ (EUR/MWh)	per unit (pooled)	0.39*** (0.05)	1.35*** (0.11)	4.1×, 7.4 ×
D^m (CV)	per unit (pooled)	0.024*** (0.003)	0.033*** (0.002)	4.3×, 4.9×
D^{MCP} (EUR/MWh)	system	4.55*** (0.40)	3.97*** (0.37)	2.8×, 2.6×
D^{σ_p} (EUR/MWh)	CCGT	−0.19 (0.36)	0.61*** (0.12)	0.8×, 6.5 ×
	Hydro	0.48*** (0.08)	1.32*** (0.17)	2.2×, 3.3×
	Hydro pump	0.14*** (0.03)	0.70*** (0.17)	3.3×, 10.9×
	Wind	0.00 (0.01)	0.03** (0.01)	0.9×, 1.9×
$D^{1/\text{HHI}}$	CCGT	0.17 (0.16)	0.17*** (0.02)	1.1×, 14.3 ×
	Hydro	0.05*** (0.01)	0.19*** (0.03)	2.0×, 2.8×
	Hydro pump	0.03*** (0.01)	0.13*** (0.03)	2.6×, 4.3×
	Wind	0.00 (0.00)	0.01 (0.00)	1.0×, 1.3×

Table 3: Post-MTU15 critical–flat within-hour dispersion gap θ , by outcome. Post-only cross-sectional gap. Windows: ID15 post 2025-03-19 to 2025-04-27; DA15 post 2025-10-01 to 2025-12-31. SE clustered by date; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (applies to all subsequent tables). Bid-level rows ($D^{\bar{p}}$, D^m , D^{MCP}) are bandwidth-free; the per-tech D^{σ_p} and $D^{1/\text{HHI}}$ rows use the window-specific p_{90} bandwidth (§4.3): ID15 column $\delta=62$ (post-ID15 IDA); DA15 column $\delta=50$ (post-DA15 DA). $D^{1/\text{HHI}}$ uses the effective-tranche-count framing ($N_{\text{eff}} = 1/\text{HHI}_p$) because the within-hour SD of HHI is mechanically tiny on a $[0, 1]$ scale; the inverse-HHI quantity carries the same content on a more readable scale.

5 Empirical strategy

Takeaway. Two specifications for two economic objects. (i) For **level outcomes** (prices, wedge SD, imbalance penalty, cleared MWh): hourly OLS and BSTS counterfactual on long pre-windows (2022 onwards) with year-by-year renewable interactions; convex-renewable robustness via quadratic and cubic specs. (ii) For **bid-shape outcomes** (σ_p , $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, HHI_p): bid-curve-level DiD with flat hours as the within-day control for critical hours.

5.1 Level variables: hourly OLS with convex-renewable controls and BSTS counterfactual

[To be written. Concepts to cover:]

- **Windows.** Long pre-window (2022-01-01 onwards) to capture full seasonality across years; short post (40 days for ID15, pre-blackout; 92 days for DA15). DA15 windows hold reforzada constant across the cutover by starting on 2025-04-28.
- **Headline OLS spec** (no linear trend, hourly): $y_t \sim \text{post} + \text{wind} + \text{solar} + \text{gas} + \text{wind} \times \mathbf{1}\{\text{year} \geq 2024\} + \text{solar} \times \mathbf{1}\{\text{year} \geq 2024\} + \text{hour FE} + \text{month FE} + \text{DOW FE}$. Newey-West HAC SE lag = 24×7 .
- **Why year-by-renewable interactions are load-bearing.** Spanish solar PV grew +43% between Jan 2024 and Jan 2025; spring-2025 IDA prices clearing roughly flat would be a price drop net of solar’s marginal merit-order push.
- **Convex-renewable robustness.** Replace linear renewables with quadratic (wind², solar², year-interacted on both linear and quadratic terms) and cubic specs. Quadratic spec absorbs the merit-order curvature; cubic spec adds essentially nothing on top.

- **BSTS counterfactual** (Brodersen et al., 2015), EPEX-style (Märkle-Hußet al., 2020): state-space level μ_t (random walk with stochastic slope) + 7-day seasonality + exogenous regressors with spike-and-slab prior. Same regressors as OLS. Treatment effect $\bar{\tau} = \frac{1}{|\text{post}|} \sum_{t \geq T_r} (y_t - \hat{y}_t(0))$.
- **Same-calendar placebo** one year before each cutover guards against recurring seasonal artefacts. Real effect is retained only if the placebo is null or opposite-signed.
- **Per-firm residual-demand slope** $b_{f,t}$ following Ito and Reguant (2016) §III.A: rebuild rivals' aggregate supply $S_{-f}(p)$ from the OMIE offer book, form $Q_f^{\text{res}}(p) = D - S_{-f}(p)$, read the slope at MCP via a quadratic fit on $\text{MCP} \pm 50$ EUR/MWh.
- **Strategic markup proxy.** Cournot first-order condition: $(p - MC)/p = q_f^{\text{strat}}/(b_f \cdot p)$, where q_f^{strat} subtracts the focal firm's own renewable cleared MW (infra-marginal at $p \approx 0$). Pre/post comparison of q_f^{strat}/b_f per firm captures both the residual-demand thickening and the loss of strategic capacity to renewables.

5.2 Bid-shape outcomes: per-curve DiD with critical-vs-flat hours

Takeaway. The bid-shape DiD identifies WITHIN-DAY changes in the bid curve: per (unit, date, period) we summarise the in-band ladder with the quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p)$ and use flat hours as the within-day control for critical hours. The flat-hour partition is the identifying restriction; no time-series extrapolation is needed.

For per-tech bid-shape outcomes $Y \in \{\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p\}$ we use the DiD:

$$Y_{u,d,p[s]} = \alpha_u + \beta \mathbf{1}\{d \geq T_r\} + \xi \mathbf{1}\{h \in \mathcal{C}\} + \theta D_{d,h} + \varepsilon, \quad \text{SEs clustered by date.}$$

[To be written. Concepts to cover:]

- **Why critical vs flat.** Differentiated bidding only pays off where residual demand is steep enough to reward it. In critical hours that condition holds and four quarter-prices unlock real discrimination; in flat hours the curve is shallow and the extra granularity is mostly wasted optionality. The critical–flat differential θ is the granularity-enabled discrimination, not a residual we are netting out.
- **Identification assumption.** Parallel trends in critical and flat hours absent the reform. σ_p and HHI_p satisfy this cleanly for CCGT and Hydro-pump (the headline cells); $\hat{\beta}$ and $\hat{\gamma}$ show pre-trend drift because they are mechanically sensitive to tranche-count and price-grid density (both shifted at 2024-06-14 and 2025-03-19). Treat $\hat{\beta}$ and $\hat{\gamma}$ as supporting, not headline.
- **Threats absorbed by construction.** Bid-shape outcomes are robust to seasonal level shifts (all computed relative to contemporaneous MCP via bandwidth δ). Main residual threat: post-reform shocks that affect critical or flat hours asymmetrically in *shape* (not level).
- **Each reform studied separately** — standard 2×2 DiD, no staggered-treatment complications (de Chaisemartin and D'Haultfoe uille, 2026, ch. 3).

6 Results

[To be written. Headline answer mirroring the slide deck: granularity made the market more competitive without moving aggregate quantities. Three first-order questions:]

- **Q1 Prices.** ID15 cut IDA prices substantially; DA15 was null. Cross-market response appears on the bid *shape*, not on the price level.

- **Q2 Imbalance and ancillary services.** Aggregate imbalance volume unchanged; penalty per MWh dropped by a third at ID15; ancillary-services cost rise is reforzada-driven, not granularity-driven.
- **Q3 Market power, bid shape, quantity migration.** Per-firm residual demand thickened at every reform; under a strategic-markup proxy that nets out infra-marginal own renewables, every Big-4 firm sees a markup squeeze at DA15 DA. Dispatchables grade their critical-hour ladders more finely; renewables compress on the reformed venue.

6.1 Q1 — Prices in each reform and cross-market effects

Takeaway. ID15 cut the granular-leg (IDA) clearing price by -30 to -46 EUR/MWh across linear specs and -23 to -24 EUR/MWh under a convex-renewable spec; the same magnitude appears on the DA cross-market response (anticipation through backward induction). DA15 own-venue and cross-market are null under every spec tried. The pricing-chain primitive — per-firm residual-demand slope b_f (Ito and Reguant, 2016) — thickens at every reform on every Big-4 dominant firm; under the Cournot first-order condition the implied markup squeezes.

[To be written. Concepts to cover:]

- **Headline robustness ladder** (ID15 IDA): OLS daily base -32^{***} ; + year-by-renewable (2024+25 pooled) -39^{***} ; OLS hourly + year-by-renewable -30^{***} ; OLS hourly + quadratic renewables (D) -24^{***} ; OLS hourly + cubic renewables (E) -23^{***} ; BSTS daily -34^* .
- **Cross-market** (ID15 on DA): same magnitude -29 to -47 across specs. Once IDA is granular and elastic, the DA cleanly absorbs the discount through backward-induction arbitrage on the bid book.
- **DA15 null** on both own venue (-9 to $+3$) and cross-market IDA (-10 to $+2$). No spec rescues a level effect.
- **Convex-renewable robustness.** The quadratic spec adds wind², solar² with year-by-renewable interactions on both linear and quadratic terms; absorbs merit-order curvature in renewable price-suppression. Cubic adds essentially nothing (diminishing returns).
- **Renewable capacity vs production check, 2024 vs 2025.** Spring 2025 had $+43\%$ more solar capacity but only $+13\%$ more realised solar (spring 2024 already very sunny); wind essentially flat; gas $+73\%$. Naively predicted spring-2025 price from gas alone would be much higher; the residual -23 to -39 EUR/MWh is the granularity-attributable channel.
- **Other robustness** in Appendix A: 90- and 180-day post-windows, hour-class heterogeneity, per-IDA-session breakdown, 2024 same-calendar placebo (null at ID15, residual signal at DA15).

Reform-side leg	Per-firm residual-demand slope	Equilibrium clearing price
	%-change in b_f (4-firm range)	Effect range across specs (EUR/MWh)
ID15 IDA (granular spot)	$+15$ to $+25\%$	-23 to -46
ID15 DA (cross-market)	$+93$ to $+141\%$	-23 to -47
DA15 DA (granular forward)	$+19$ to $+42\%$	null (-9 to $+3$)
DA15 IDA (cross-market)	$+13$ to $+20\%$	null (-10 to $+2$)

Table 4: Headline match between pricing-chain primitive and equilibrium price. Left: per-firm hourly OLS on $\log b_f$ (no linear trend), with hour FE, month FE, DOW FE, gas, wind, solar, year-by-renewable interactions. Newey-West HAC SE lag 24×7 . Range across the four Big-4 dominant firms (Iberdrola, Endesa, Naturgy, EDP). Right: clearing-price effect range across the linear / quadratic / cubic OLS and BSTS specs (Appendix A.1). The per-firm slope thickens at every reform-side leg; ID15 DA cross-market is the largest single effect (every firm sees b_f roughly double). The price effect is robustly negative at ID15 (both legs) and null at DA15 (both legs).

6.2 Q2 — Imbalance penalty per MWh drops at ID15, aggregate volume unchanged

Takeaway. ID15 cut the imbalance penalty per MWh by a third; DA15 added nothing. Aggregate imbalance volume essentially unchanged at both reforms. Granularity reorganises within-hour timing of imbalance, not its size: the same MWh of residual imbalance is resolved with cheaper balancing reserves.

[To be written. Concepts to cover:]

- **Penalty per MWh of deviation** (the imbalance settlement price) is set ex-post by REE via formula, *not discretionary*: the MW-weighted price of the reserves used to balance the system.
- ID15 cuts it by -33% ; DA15 is flat.
- Aggregate volume essentially unchanged (-0.1% ID15, -4% DA15). Forecast errors are physical (wind/solar/demand stochasticity); granularity reshuffles their within-hour timing, not their size.
- **Mechanism.** Granular spot reveals firms' positions earlier $\Rightarrow$ REE schedules cheaper reserves ahead of time and avoids dipping into expensive ones. Same MWh of residual imbalance, resolved with cheaper reserves.

Outcome	Pre mean	ID15 (March)	DA15 (October)
Penalty per MWh of deviation (EUR/MWh)	3.45	-1.14*** (-33%)	-0.05 (null, flat)
Aggregate imbalance (MWh/day)	1,676	-2 (null, -0.1%)	-67 (null, -4%)

Table 5: BSTS on imbalance penalty and aggregate volume. ID15 pre-window 2024-12-01 to 2025-03-18, post 2025-03-19 to 2025-04-27 (pre-blackout). DA15 pre 2025-04-28 to 2025-09-30, post 2025-10-01 to 2025-12-31. BSTS with seasonality, wind, solar, and gas covariates. Relative effect in parentheses. *** denotes posterior probability > 0.999 that the effect is non-zero. Penalty per MWh is the imbalance settlement price (the MW-weighted price of the reserves used to balance the system).

6.3 Q2 (cost evidence) — ancillary-services costs are dominated by reforzada, not granularity

Takeaway. REE's adjustment-cost lines balloon post-blackout, almost entirely on Fase I up-redispach and almost entirely on CCGT. This is a reforzada effect, mechanistically distinct from MTU15. Pre-blackout (blue zone in Figure 6), real-time channels (TR-up, aFRR-up) shrink slightly across ISP15-win and MTU15-IDA, but our identification is too tight to pin a granularity-attributable component on the EUR cost aggregates.

[To be written. Concepts to cover:]

- The system operator is paying CCGT to produce (expensive) through pre-IDA Fase I redispach and curtailing solar (cheap). Fase I up-redispach costs balloon by EUR ~ 5 M/day post-blackout, almost entirely CCGT voltage support.

- Granularity vs reforzada: the real-time channels shrink at the same time the pre-real-time Fase I rises. The displacement reading (procurement timing shifts upstream) and the efficiency reading (granular auction absorbs intra-hour forecast errors) are observationally equivalent on the EUR aggregates; we do not separate them.
- Independent corroboration of the reforzada cost rise: Brown and Reguant (2026); Daví-Arderius and Graf (2026).

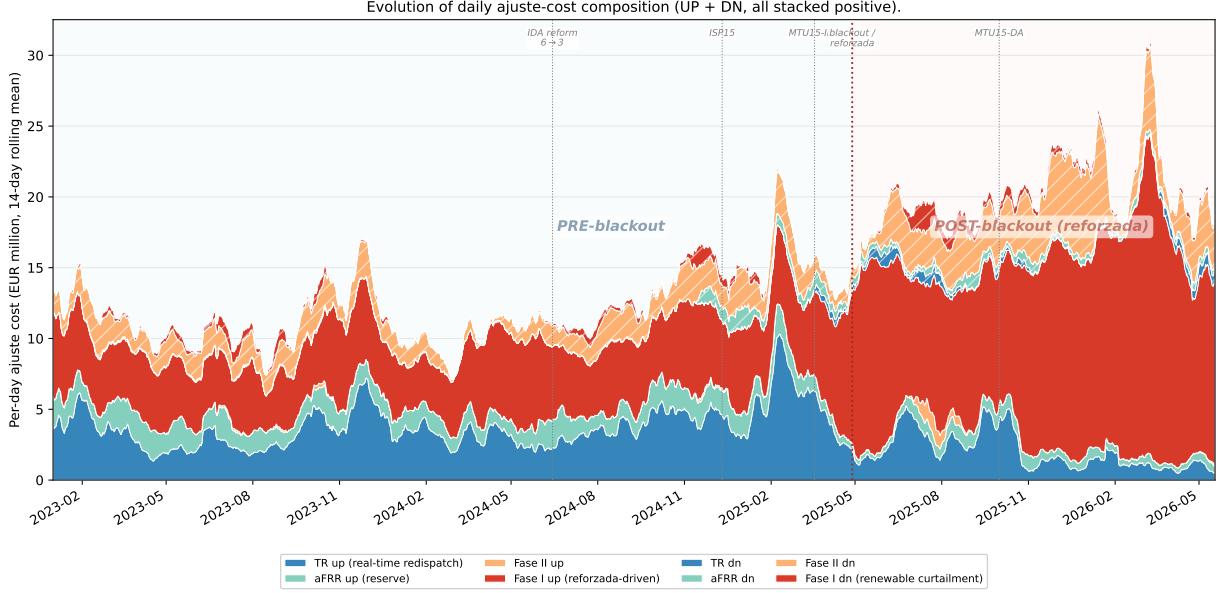


Figure 6: Daily adjustment-cost composition, 2023–2026 (EUR M / day, 14-day rolling mean). Each channel is the daily sum of the pay-as-bid system-cost series REE publishes via ESIOS: *TR up/dn* is real-time technical restrictions under PO 3.7 (cost of energy REE buys/sells to enforce grid security between gate closure and delivery); *aFRR up/dn* is the automatic Frequency Restoration Reserve energy cost; *Fase I up/dn* is the pre-IDA technical-restrictions process applied to PDBF (pay-as-bid of a separate restriction offer per PO 3.2 and PO 14.4 ap. 20.1, distinct from the day-ahead market offer); *Fase II up/dn* is the post-Fase-I rebalance that re-absorbs the up-redispatch into the schedule. All series are EUR sums collected at native 15-min granularity (raw daily totals smoothed only by the 14-day moving average), with down-direction channels taken as absolute values so the chart reads as gross costs REE pays providers in both directions. Up-direction stacked solid (TR-up, aFRR-up, Fase II up, Fase I up); down-direction stacked hatched. Blue background: pre-blackout (granularity reforms shrink the up-direction TR and aFRR channels through the ISP15-win and MTU15-IDA cutovers, visible against the stable 2023 baseline). Red background: post-blackout reforzada (Fase I up-redispatch balloons by EUR ~5 M/day, almost entirely CCGT voltage support; this is the level shift between the 2023 baseline and 2025-Q4 totals).

6.4 Q3 (wedge evidence) — within-hour wedge SD jumps at ID15 and per-IDA-session levels diverge

Takeaway. The within-day SD of the DA–IDA wedge jumps at ID15 because four IDA quarter-prices spread across each hourly DA price (mechanical). At DA15, with both auctions clearing at 15 min, the SD does *not* collapse — the empirical fingerprint of bounded per-product arbitrage capacity. Per-session wedge level: ID15 widens S1 by $\sim +3$ EUR/MWh (robust), narrows S3 by ~ -7 (robust), S2 null; DA15 null across sessions.

[To be written. Concepts to cover:]

- ID15 within-day wedge SD: overall $+1.1^{**}$, critical hours $+2.1^{***}$, morning ramp $+2.3^{***}$, evening ramp $+1.2^*$, midday -1.6^{**} , flat null. Sharpest in the morning ramp.

- DA15 wedge SD: no significant effect across hour classes.
- Per-IDA-session wedge level at ID15: S1 +3.0** (robust across OLS daily, hourly, BSTS), S3 −7.0* (robust); ID15 placebo null. DA15 per-session null with residual signal at the placebo cutover.

6.5 Q3 (quantity-shift evidence) — bid-shape widening in critical hours

Takeaway. Dispatchables grade their critical-hour ladders more finely; renewables compress on the reformed venue. Headline cells from the bid-curve-level DiD (critical vs flat, day-clustered SE): CCGT widens DA σ_p at ID15 by +4.45*** EUR/MWh, with HHI deconcentrating −0.18***; same direction at DA15 DA (+1.05***, −0.17***). Hydro-pump’s DA ladder coarsens at ID15 (σ_p −1.48***): once IDA is granular, storage withholds upstream. Renewables compress: wind, biomass, cogen.

[To be written. Concepts to cover:]

- Headline metrics are σ_p and HHI: robust to mechanical tranche-count and price-grid-density changes. Slope $\hat{\beta}$ and curvature $\hat{\gamma}$ in Appendix — pre-trends drift on those, treat as supporting.
- CCGT DA widens at both reforms: ID15 σ_p +4.45***, HHI −0.18***; DA15 σ_p +1.05***, HHI −0.17***. Marginal-cost units grade their critical-hour ladder into many small price steps.
- Strategic renewables compress on the reformed venue: ID15 DA wind −0.17***, biomass −0.23***; DA15 IDA CCGT cross-market −1.01***.
- Hydro-pump anomaly: ID15 DA σ_p −1.48***. Once IDA is granular, pump storage withholds upstream and the DA ladder coarsens.
- HHI: CCGT DA deconcentrates at both reforms; cogen, wind, biomass IDA HHI rises at ID15 (fewer fatter tranches).
- Robustness: morning-ramp-only DiD reproduces the headline signs; pre-trend figures clean for σ_p and HHI on the headline cells (Appendix).

Outcome (DiD θ)	Tech	ID15		DA15	
		DA	IDA	DA	IDA
σ_p (EUR/MWh)	CCGT	+4.45***	+0.24	+1.05***	−1.01***
	Hydro	−0.35**	+0.24	+0.85***	+0.44***
	Pump-storage	−1.48***	+0.03	+0.13	−0.02
HHI _p	CCGT	−0.18***	−0.04	−0.17***	−0.00
	Hydro	+0.010*	−0.034***	−0.037***	−0.01
	Pump-storage	0.00	−0.02	0.00	+0.01

Slope and curvature in Appendix — $\hat{\beta}$, $\hat{\gamma}$ are mechanically sensitive to tranche-count and price-grid density; treat as supporting.

Table 6: Bid-curve-level DiD on headline bid-shape outcomes. Critical-flat DiD on price dispersion σ_p and bid-stack concentration HHI_p. Tight pre/post windows (reforzada held constant): ID15 pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Bandwidth $\delta = p_{90} - p_{50}$ window-specific. SEs clustered by date. A negative HHI DiD means the bid stack *deconcentrates* in critical hours relative to flat. Full ($\hat{\alpha}, \hat{\beta}, \hat{\gamma}$) structural decomposition, additional technologies (Cogen, Hybrid, Wind, Biomass), morning-ramp-only robustness, and parallel-trends figures in Appendix A.

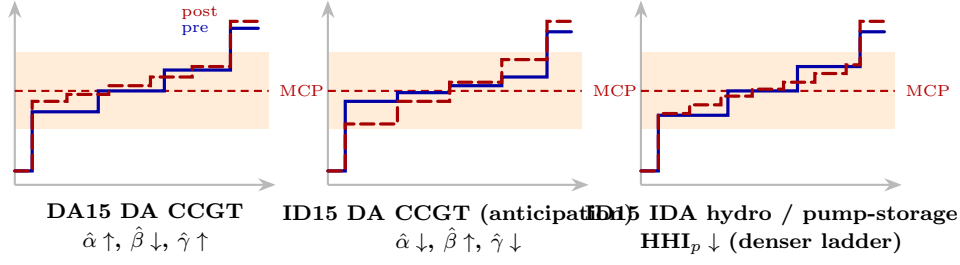


Figure 7: Stylised retained bidding shifts. Three in-band ladder reshapes observed in the data: DA15 day-ahead CCGT (level up, slope flatter, convex bend); ID15 day-ahead CCGT anticipation (level down, steeper, concave bend); ID15 intraday-auction hydro and pump-storage (denser ladder, level and slope unchanged). Blue solid: pre-reform; red dashed: post-reform.

6.6 Synthesis — structural markup squeeze, aggregator vs strategic-bidder split

Takeaway. DA15 delivers a clean Cournot markup squeeze: per-firm residual demand thickens by +24 to +28% on the day-ahead AND the firm’s strategic capacity (total cleared MW minus its own renewable cleared MW) shrinks, so the markup proxy q_f^{strat}/b_f falls −12 to −27% on every Big-4 firm. ID15 also thickens b_f but the markup-proxy signs are compositional (small post window with nuclear out and CCGT heavily called). The two reforms engage two distinct agents: the aggregator (price-taker hedging forecast error) at the spot reform, the strategic bidder (residual monopolist) at the forward reform.

[To be written. Concepts to cover:]

- Strategic markup proxy (Cournot FOC, $q_f^{\text{strat}} = \text{total cleared minus own renewable cleared}$, infra-marginal at $p \approx 0$): DA15 DA −12 to −27% across IB/GE/GN/HC; ID15 markup signs mixed due to compositional shifts in the 40-day window.
- Aggregator footprint at the spot reform (ID15): aggregate imbalance volume unchanged, penalty per MWh drops, granular spot reveals firms’ positions earlier.
- Strategic-bidder footprint at the forward reform (DA15): CCGT day-ahead ladder widening, residual demand thickening for all four firms.
- Pricing chain: b_f thicker $\Rightarrow$ Cournot markup compresses $\Rightarrow$ clearing prices fall. Delivered at ID15 (IDA and cross-market DA); null at DA15.

7 Conclusion

[To be written. Concepts to cover:]

- **Aggregate effects of DA15 are small; ID15 delivered the headline gains.** Significant changes in bidding behaviour at both reforms, but bid-shape signals are mixed across techs. Evidence of rising solar cannibalisation — already a serious profitability issue.
- **Open question:** did market microstructure affect grid stability? Three facts in the open: (i) REE’s post-blackout response is technology-specific (CCGT doubles in restriction dispatch); (ii) realised renewable supply was similar in spring 2024 vs 2025, yet only 2025 blacked out; (iii) nuclear’s auction-plus-bilateral output dropped sharply in the 40 days between ID15 and the blackout (Appendix A.8). The ENTSO-E Expert Panel does not list ID15 as a root cause, but CNMC flags quarter-hourly trading as contributing to abrupt voltage variations.

- **The biggest welfare losses live in the ancillary markets**, not in the wholesale auctions (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026).
- **Urgent need to strengthen network security** to bring down retail prices — not finer time resolution.

CNMC report on the blackout flags quarter-hourly trading among contributors to within-hour voltage variability (CNMC, 2026, p. 44):

The implementation of the 15-minute trading unit in the Spanish intraday market was completed on 18 March 2025, one month before the incident. ... The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.

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A Robustness

A.1 Year-by-year solar coefficient: the renewable response is non-stationary

The BSTS uses long pre-windows (2022 onwards) with $\text{wind}_t \times \mathbf{1}\{\text{year} = y\}$ and $\text{solar}_t \times \mathbf{1}\{\text{year} = y\}$ interactions. The economic reason is that Spanish wind+solar installed capacity grew by a factor of ~ 6 over 2018-2025, and the marginal effect of an additional GWh of renewable output on price is not constant across that period. A flat-coefficient spec mis-fits the cross-year merit-order shift and reports artificially confident treatment effects (§5.1).

Table 7 reports the cumulative solar coefficient (base + year-dummy) by calendar year, extracted from the BSTS posterior of the price spec under both windows (ID15 pre ends 2025-03-18, DA15 pre ends 2025-09-30). **Conditional posterior means** (the coefficient when the spike-and-slab includes the regressor). The wind base coefficient is stable around -0.50 EUR/MWh per GWh; the wind $\times$ year interactions all have inclusion probability below 1.5%. The solar coefficient is the one that varies.

Year	ID15 IDA spec	DA15 IDA spec	DA15 DA spec
2022 (base)	-0.22	-0.23	-0.22
2023	-0.02	-0.23	-0.17
2024	-0.46	-0.47	-0.40
2025 [†]	-0.29 (<i>Jan-Mar only on ID15</i>)	-0.50	-0.49
<i>Implied weighted average for the post-window prediction</i>			
2024+2025 weighted avg [‡]	-0.43	-0.45	-0.44

Table 7: Cumulative solar coefficient (EUR/MWh per GWh of solar) by year, extracted from the BSTS posterior of the corresponding price spec. Cumulative coefficient = base + year-dummy interaction, conditional posterior mean among MCMC draws that include the regressor. [†]The ID15 spec pre-window ends 2025-03-18, so its 2025-by-solar coefficient is fitted on Jan-Mar 2025 only, when Spanish solar production is at its lowest seasonal level; the DA15 spec pre-window extends through 2025-09-30 and includes summer, giving the more representative estimate. [‡]Months-weighted average of the 2024 and 2025 cumulative coefficients (12 months of 2024 + available months of 2025): this is the coefficient one would extrapolate forward if 2024 and 2025 were pooled into a single dummy for the post-window prediction; the per-year BSTS instead extrapolates using the 2025-row coefficient alone.

Reading the table: the DA15 spec (which sees Jan-Sep 2025) tells the cleaner story. Solar’s marginal price-suppressing effect is roughly constant from 2022 to 2023, then nearly doubles from 2023 to 2024 (Spain added ~ 5 GW of solar capacity in 2024 alone), and continues to deepen in 2025 as solar saturates more clock-hours and negative-price events become routine. The ID15 spec’s 2025 coefficient looks deceptively small (Jan-Mar winter sun is weak) but the underlying renewable response in 2025 is the most negative on record. The non-stationarity is exactly what the year interaction is designed to absorb: a flat-coefficient spec would average -0.22 (2022) and -0.50 (2025) into a single value around -0.35 , mis-fitting both ends of the window.

Calibrated-solar robustness: how much of the price drop is solar vs the reform? A natural concern about the headline BSTS counterfactual is that Spanish solar capacity grew rapidly through 2024 and 2025, so the marginal price-suppressing effect of solar is itself non-stationary. The BSTS per-year spike-and-slab attempts to capture this via year-by-solar interactions; the conditional posterior means (Table 7) are -0.46 for 2024 and -0.29 for the Jan-Mar-2025 sample on the ID15 spec, suggesting a strong solar effect that may not be fully credited in the post-window prediction. If the post-window counterfactual under-credits solar, the residual “price drop”

that we attribute to the reform is partly mechanical (it is the solar effect we failed to attribute correctly). We therefore implement a **calibrated-solar counterfactual**: pre-subtract a known solar contribution $\beta_{\text{solar}}^{\text{cal}} \cdot \text{solar}_t \cdot \mathbf{1}\{t \geq 2024\}$ from observed prices, fit the BSTS state-space model on the residual (with the usual base regressors but no year interactions), and add the calibrated solar contribution back to the post-window prediction before computing the treatment effect.

Calibrated solar coefficient $\beta_{\text{solar}}^{\text{cal}}$ (EUR/MWh per GWh)	ID15 IDA			DA15 DA		
	Real	Placebo	Net	Real	Placebo	Net
−0.21 (BSTS unconditional default)	−41	+32	− 73	+30	+43	−13
−0.29 (per-year cond. 2025, ID15 spec)	−38	+36	− 74	+21	+32	−12
−0.40 (per-year cond. 2024, DA15 DA spec)	−30	+36	− 66	+18	+27	−10
− 0.46 (per-year cond. 2024, ID15 spec)	−26	+37	− 63	+9	+23	−13
−0.50 (per-year cond. 2025, DA15 spec)	−23	+37	− 60	+9	+19	−10

Table 8: Calibrated-solar BSTS treatment effects on clearing price. EUR/MWh; −0.46 row uses the user’s preferred “trust the 2024 conditional” substitution. The calibrated approach explicitly subtracts $\beta_{\text{solar}}^{\text{cal}} \cdot \text{solar} \cdot \mathbf{1}\{t \geq 2024\}$ from prices, fits BSTS on the residual (no year interactions), and adds the calibrated contribution back to the post-window prediction. Placebo column uses the same calibration but with the cutover shifted one year earlier (real reform absent in the placebo window). Real, placebo, and net all measured on the price level in EUR/MWh.

The price drop at ID15 IDA survives every solar calibration we tried. Under the most aggressive solar coefficient (−0.50, the DA15-spec conditional 2025 mean) the real-effect point estimate is −23 EUR/MWh; under the user’s preferred “trust the 2024 conditional” value of −0.46 it is −26 EUR/MWh; under the BSTS unconditional default it is −41 EUR/MWh. After 2024-same-calendar placebo-netting (which absorbs whatever cross-season variation exists in the spring window) the net effects are uniformly larger in magnitude (−60 to −74 EUR/MWh) because the placebo windows themselves have positive “effects” under the calibrated specification, reflecting whatever upward residual variation Spring 2024 carried. The DA15 DA real effect is essentially null across all solar calibrations (+9 to +30 EUR/MWh, ≈ 0); the net effect is mildly negative (−10 to −13 EUR/MWh) under every solar calibration but the magnitude is much smaller than at ID15 and the credible intervals always bracket zero.

Bottom line. The ID15 IDA price drop is robustly negative in sign and substantial in magnitude under every solar coefficient we could justify. Solar’s strong 2024 marginal effect (−0.46) does not explain the ID15 price drop — the reform did something to the granular leg’s clearing price on top of what solar alone would predict. The DA15 DA effect is small to null under every calibration. We treat the ID15 drop as a real reform effect and the DA15 drop as not separately identified. The per-firm slope-thickening result (Appendix A.10) remains the cleanly-identified pricing-chain INPUT and is unaffected by this discussion.

OLS cross-check. The wide credible intervals in the calibrated-solar BSTS reflect the state-space level component’s flexibility, which absorbs persistent low-frequency variation that the regression then has no way to pin down. We complement the BSTS with a standard OLS regression on the same daily price panel, including calendar-month and day-of-week fixed effects (the seasonal structure that BSTS encoded via local-level + 7-day seasonal), a time trend (linear or quadratic), gas price, and per-year wind and solar interactions. Standard errors are Newey-West HAC with

7-day bandwidth to account for daily price autocorrelation. The model:

$$p_t = \alpha + \theta \text{POST}_t + \beta_{\text{wind}} \text{wind}_t + \beta_{\text{solar}} \text{solar}_t + \beta_{\text{gas}} \text{gas}_t + \sum_y \gamma_y \text{renew}_t \cdot \mathbb{1}\{\text{year} = y\} + \phi_m + \delta_{\text{dow}} + g(t) + \varepsilon_t.$$

Specification (POST coef θ)	ID15 IDA price		DA15 DA price	
	$\hat{\theta}$	HAC SE	$\hat{\theta}$	HAC SE
Spec 1: base + month FE + DOW FE	−31.7***	8.2	−8.3	5.3
Spec 2: + linear time trend	−23.7***	8.6	+8.3	7.3
Spec 3: + year-by-renewable (per year)	−45.8***	9.6	−8.6	8.1
Spec 4: + year-by-renewable (2024+ pooled)	−38.7***	7.3	−4.6	7.7
Spec 5: + quadratic time trend	−53.1***	6.8	−18.9**	9.2

Table 9: OLS price-effect estimates with full controls. Daily clearing-price regressions with calendar-month and day-of-week fixed effects, gas price, time trend (linear or quadratic), and year-by-renewable interactions. Newey-West HAC standard errors with 7-day bandwidth. POST = 1 on/after the reform cutover. Spec 4 (year-by-renewable with 2024+2025 pooled) implements the user-suggested fix for the Jan-Mar-2025-only-fit problem. ** $p < 0.05$, *** $p < 0.01$. ID15 IDA pre-window 2022-01-01 to 2025-03-18, post 2025-03-19 to 2025-04-27. DA15 DA pre 2022-01-01 to 2025-09-30, post 2025-10-01 to 2025-12-31.

OLS verdict matches the calibrated-solar BSTS verdict. The ID15 IDA price drop is significant at every standard significance level across all five specifications (−24 to −53 EUR/MWh, $p < 0.01$ throughout). The DA15 DA effect ranges from −4.6 (null, $p = 0.55$) to −18.9 (significant under quadratic trend, $p = 0.04$); the bulk of specifications return small null or marginally-significant negative coefficients. OLS with full controls is more precise than the BSTS (smaller standard errors, narrower confidence intervals) but qualitatively delivers the same conclusion: **ID15 has a robust own-venue price drop; DA15 does not.**

This diagnostic also clarifies why several level findings (§6.1) lose significance under the year-interaction spec: forecast uncertainty about what solar’s marginal price impact will be in the post-cutover window is large precisely because the coefficient has been shifting fast in the years immediately preceding the cutover.

Note on bid-shape outcomes in this appendix. The bid-shape DiD robustness tables below report on the headline quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p)$ used in §4.3. The cross-border and offer-type checks lead with σ_p (the robust price-side anchor); for a near-linear curve $\sigma_p \approx |\hat{\beta}| \cdot \sigma_Q$, so the signs on those checks track $\hat{\beta}$ up to estimator noise (and via curvature track $\hat{\gamma}$).

A.2 Pre-trend placebo

Split the pre-window at its midpoint and run the same DiD on the two halves; genuine reform effects should leave that placebo near zero. ID15 own-market cells pass cleanly (sign flips between real and placebo). The DA15 own-market CCGT placebo runs same-sign as the real, so the DA15 CCGT widening is continuation-plus-augmentation rather than a clean jump. Figure 8 is the visual analogue.

	ID15 IDA (own)		ID15 DA (cross)		DA15 DA (own)		DA15 IDA (cross)	
Tech	Hdln	Plac	Hdln	Plac	Hdln	Plac	Hdln	Plac
<i>Intercept α (EUR/MWh; positive = higher curve)</i>								
CCGT	11.8*	4.9	-40.2***	1.8	28.3***	40.2***	6.0**	3.8**
Hydro	87.3	10.2***	0.2	-0.4	19.7	53.3*	9.3***	6.3***
Pump-storage	-3.0	-12.7***	1.1	1.3	-1.8	-4.3	16.1	-4.9
<i>Slope β (EUR/MWh per MW; more negative = steeper)</i>								
CCGT	-0.881*	0.146	0.269***	0.004	-0.150***	-0.206***	0.200	-0.101
Hydro	-1.498	-0.156**	0.154	-1.703***	-1.785***	1.176***	-0.012	-0.022
Pump-storage	0.071	0.248	-0.010	0.015*	0.008	0.046***	-0.254	0.007
<i>Curvature γ (EUR/MWh per MW²; positive = convex)</i>								
CCGT	0.0100	0.0334*	-0.0040***	0.0015***	0.0010***	0.0022***	0.0076	0.0002
Hydro	-0.1088	0.2455	-0.3055	-1.3947	-7.4938***	1.6198	0.0016	-0.0105
Pump-storage	0.0014	0.0005	0.0001	0.0002	0.0001	0.0000	-0.0002	-0.0002

Table 10: Pre-only midpoint placebo for bid-shape DiD, all four (reform, market) cells. Windows: ID15 pre 2024-12-19 to 2025-03-18 (midpoint placebo splits this at $\sim 2025-02-01$); DA15 pre 2025-07-01 to 2025-09-30 (midpoint $\sim 2025-08-15$). “Hdln” reproduces the headline DiD from §6.5 using the full pre/post split; “Plac” fits the same critical/flat DiD on the pre-window split at its midpoint (no reform crosses the midpoint, so “Plac” should be null under parallel trends). Stars: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

σ_p as the rugged visualization. For the parallel-trends figure we lead with σ_p (Figure 8) and report the structural decomposition $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p)$ alongside (Figure 9). σ_p requires only price-side variation, is bounded in $[0, \delta]$, and is well-defined on single-tranche curves; the slope $\hat{\beta}$ needs joint p - q variation and is noisier on sparse curves. For near-linear curves the two are related by $\sigma_p \approx |\hat{\beta}| \sigma_Q$, so the patterns track up to estimator noise.

A.3 Long-pre window with year/season/renewable controls and conditional parallel trends (RA-DiD)

I extend the pre-window to 2022-01-01, add year FE, calendar-month FE, and $\text{crit} \times \text{wind} / \text{crit} \times \text{solar}$ interactions; and as a separate spec run the regression-adjustment DiD of Heckman et al. (1997) under conditional parallel trends. ID15 DA CCGT (anticipation) survives both checks on σ_p and HHI; DA15 DA CCGT survives on σ_p , the HHI sign flips under RA-DiD so the σ_p side is the rugged headline. Side technologies omitted: only cells that survive at least the long-pre spec are reported (Table 11).

Cell	Outcome	Tight (1)	Long-pre + controls (2)	RA-DiD (3)
ID15 DA CCGT (anticipation)	σ_p	+19.42***	+10.78***	+12.71**
	HHI_p	-0.076***	-0.098***	-0.000
DA15 DA CCGT	σ_p	+1.25**	+5.47***	+1.72
	HHI_p	-0.059***	-0.028***	+0.099**

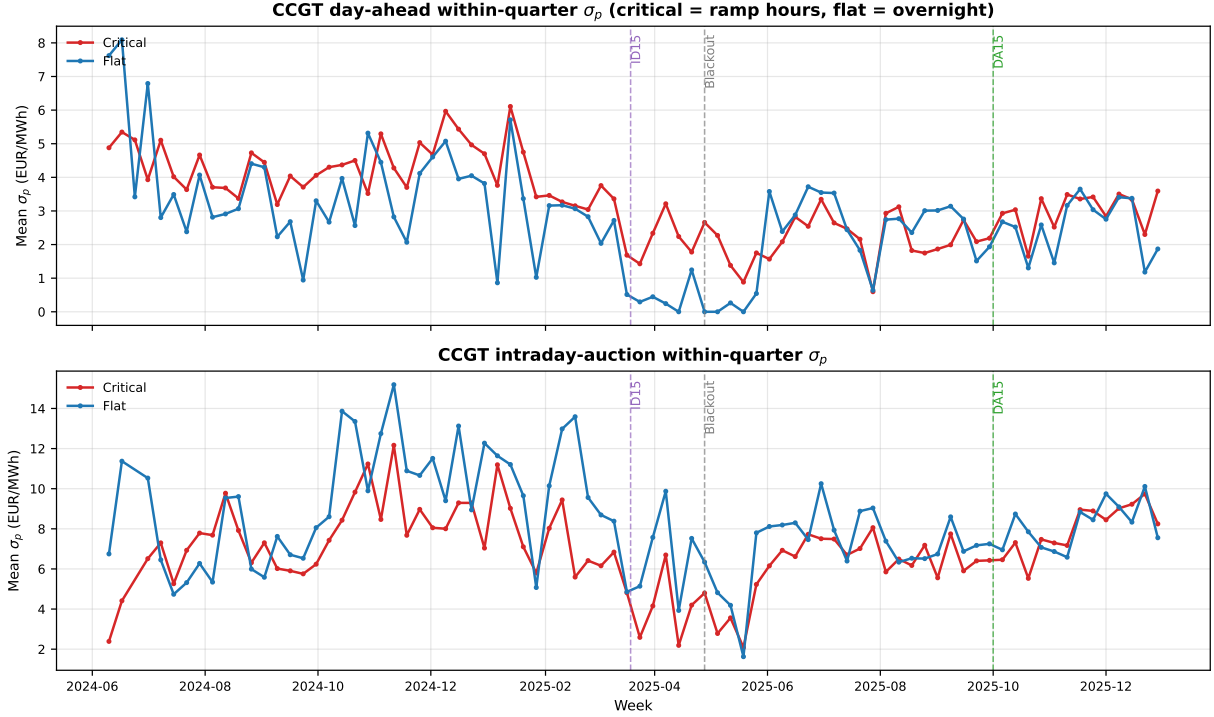


Figure 8: Headline parallel-trends diagnostic: weekly CCGT within-quarter σ_p by hour-class, 2024-06 to 2025-12. *Top:* day-ahead; *Bottom:* intraday-auction (latest-session in-band tranches, $|p_k - \text{MCP}| \leq 50$ EUR/MWh). Red: critical hours; blue: flat hours. Vertical dashes: ID15 (2025-03-19), blackout (2025-04-28), DA15 (2025-10-01). σ_p is the rugged price-side dispersion that visualizes the within-day reform effect without requiring q -side identification (cf. the structural decomposition in Figure 9).

Table 11: bid-shape DiD robustness: tight (1), long-pre with year+month+renewable controls (2), and regression-adjustment DiD per Heckman et al. (1997) under conditional PT (3). Windows match Table 6. *Bandwidth note:* this robustness uses the wider strategic bandwidth $H = 140$ EUR/MWh of the underlying per-curve panel (the bandwidth of Figure 8), not the window-specific p_{90} of Table 6; σ_p scales with band width, so the magnitudes here are larger than in the headline table but the signs and significances on the headline cells are the comparable object. Spec (2) extends pre to 2022-01-01 and adds year FE, month FE, crit $\times$ wind / crit $\times$ solar interactions. Spec (3) regresses per-unit pre $\rightarrow$ post change on average wind and solar on FLAT hours, predicts the counterfactual at critical units' covariates, ATT = realized – predicted; bootstrap SEs 500 reps. Cells reported are those surviving at least spec (2).

Per-firm b_{mt} under the same long-pre control. I run the same long-pre + year + month FE on the per-firm b_{mt} thickening: magnitude shrinks from the tight-window range +13%–+28% to +8%–+15% across the four dominant CCGT-owning firms, all firm cells still significant. Direction robust, magnitude roughly halved.

A.4 Hour-class falsification (midday vs. flat)

I substitute midday hours for critical in the same DiD. ID15 IDA CCGT α flips sign at midday (CCGT does not engage the granularity instrument in solar-saturated hours). DA15 DA CCGT carries the same sign in midday as in critical — the day-ahead reform engaged the whole non-flat zone, not critical alone. Falsification passes.

I add hourly Spain-France cross-border net flow as a control on the ID15 IDA DiD. Per-tech coefficients essentially unchanged. Rules out the contemporaneous XBID/SIDC sibling reform as the source.

A.5 Per-CCGT-firm decomposition

I run the DiD firm by firm on each firm’s own CCGT fleet (unit FE within-firm). Naturgy is the dominant scale-up at DA15 DA (Table 12); Iberdrola flat on level; EDP-HC small same-sign. ID15 IDA spreads more evenly across firms. Sign and firm-ranking are interpretable; magnitudes are not (firm intercepts not netted out).

Firm	α	β	γ	Units
<i>DA15 DA (own-market)</i>				
Naturgy	69.9***	−0.402***	0.0022**	17
Iberdrola	0.0	0.001	0.0000	10
EDP-HC	5.2*	−0.021*	−0.0002	7
<i>ID15 IDA (own-market)</i>				
Naturgy	10.3	−1.197*	0.0109	17
Iberdrola	32.2	−0.253	—	9
Endesa	−2.9	0.109	0.0044*	7
EDP-HC	1.7	−0.211**	−0.0029	6

Table 12: Per-firm CCGT bid-shape DiD, own-market cells of both reforms. Windows match Table 6: ID15 IDA pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 DA pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31 (both post-blackout). One regression per firm on the firm’s own CCGT fleet (unit FE within-firm). Endesa DA15 DA omitted: with only six units and uneven within-fleet activation the coefficients are numerically unstable. Magnitudes are not directly comparable to the pooled headlines; sign and ranking are interpretable.

A.6 Bandwidth δ

I rerun the DiD at $\delta = 100, 140, 200$ EUR/MWh against the headline window-specific p_{90} (Table 13). CCGT β and γ stay sign-stable across all bandwidths (the DA15 DA CCGT slope even steepens monotonically); hydro γ collapses to near zero at $\delta = 200$, exposing its narrow-band character. The headline CCGT findings are robust to the bandwidth choice; hydro is not.

δ	ID15 IDA			DA15 DA		
	CCGT	Hydro	Pump	CCGT	Hydro	Pump
<i>Slope β (EUR/MWh per MW)</i>						
Headline (p_{90})	−0.881*	−1.498	0.071	−0.150***	−1.785***	0.008
$\delta = 100$	−0.254***	−0.781	0.035	−0.388***	−0.636***	0.012
$\delta = 140$	−0.234***	−0.613	0.010	−0.493***	−0.015	0.007
$\delta = 200$	−0.245***	−0.862	−0.007	−0.481***	−0.024	0.001
<i>Curvature γ (EUR/MWh per MW²)</i>						
Headline (p_{90})	0.0100	−0.1088	0.0014	0.0010***	−7.4938***	0.0001
$\delta = 100$	0.0309**	0.1572	0.0008	0.0057***	−0.0591	0.0001***
$\delta = 140$	0.0761	0.1459	0.0006	0.0059***	0.0009	0.0002**
$\delta = 200$	0.0761	0.1541	0.0007	0.0050***	−0.0004	0.0001**

Table 13: Bandwidth robustness for bid-shape DiD. Windows match Table 6: ID15 IDA pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27; DA15 DA pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Per-curve DiD coefficient θ on the slope β and curvature γ at progressively wider bandwidths. Window-specific p_{90} headline values: $\delta = 62$ for ID15 IDA, $\delta = 50$ for DA15 DA. CCGT β and γ are the most robust: DA15 DA CCGT β steepens monotonically with wider bandwidths ($-0.15 \rightarrow -0.49$) and γ stays convex and significant ($+0.001 \rightarrow +0.006$). ID15 IDA CCGT β steepens uniformly; γ is positive throughout but only sometimes significant. Hydro effects are narrow-band artefacts: DA15 DA hydro γ collapses from -7.6 at p_{90} to near zero at $\delta = 200$. α rows omitted (qualitatively similar to β and γ on the CCGT cells).

Per-IDA-session BSTS estimates with the long pre-window cluster around EUR $-45/\text{MWh}$ across IDA1, IDA2, IDA3, confirming session symmetry. A winter-only pre-window inflates estimates to $\sim -80/\text{MWh}$ because the renewable coefficient is identified off too narrow a range. This motivates the long pre with year-by-year renewable interactions in BSTS.

A.7 CCGT offer-type composition check

I split CCGT bids by structural type (simple / MIC / block) and re-run the DA15 DiD within each type. The composition shifts toward more block offers post-reform (block offers are mechanically single-tranche, pushing pooled HHI UP), but the within-type DiD is still negative in MIC (-0.046^{***}) and block (-0.043^{***}), and pooled stays -0.059^{***} . The bid-stack deconcentration is real within-tranche behaviour, not a composition artefact.

A.8 Nuclear PDBF shortfall before the blackout

I run the headline BSTS on nuclear auction-plus-bilateral dispatch (PDBF on DA, PHF on IDA) at both reforms. PHF is the post-IDA programme before real-time, the right object for IDA conduct around the cutover.

Reform window	Outcome	Real	Placebo 2024	Placebo-net
ID15 (Mar 19 – Apr 27, 2025, pre-blackout)	PDBF (DA + bilateral)	-72.8^{***}	-18.9	-53.9
	PHF (post-IDA total)	-54.1^{***}	$+17.3$	-71.4
DA15 (Oct 1 – Dec 31, 2025, reforzada-constant)	PDBF	-6.1	-29.4	$+23.3$
	PHF	-7.8	-31.2	$+23.4$

Table 14: Nuclear PDBF and PHF BSTS, ID15 and DA15. GWh/day. Headline BSTS methodology (long pre-window 2022-01-01 onwards, year-by-year wind and solar interactions); placebos are the same-calendar 2024 windows.

Nuclear PDBF drops -72.8^{***} GWh/day at ID15 with a near-zero placebo (placebo-net -53.9); DA15 placebo-net brackets zero on both PDBF and PHF. Reading: in the 40-day post-ID15 pre-blackout window, inflexible nuclear had limited room above its routine refuelling baseline as DA prices ran near EUR $20/\text{MWh}$ with several near-zero days; by autumn 2025 the price compression reversed and the post-blackout regime held nuclear synchronised for voltage support.

A BSTS on daily-mean α confirms that the bid-shape DiD α finding is a within-day redistribution, not a level shift: the placebo on the same-calendar window absorbs the level rise.

A.9 Demand-side bid shape

I re-run bid-shape DiD symmetrically on buy offers (tranches sorted descending in price) per demand class. At DA15 the buy-side α rises in step with sell-side α in critical: equilibrium re-coordination at a higher reservation level, not a one-sided supply shift.

Demand class	ID15 IDA	ID15 DA	DA15 DA	DA15 IDA
<i>Intercept α (EUR/MWh)</i>				
Retailer	-10.6**	5.4	15.0***	33.1***
DirectCons	0.2	—	-3.1	19.5***
PumpBuy	0.3	-4.9	-2.1*	0.8
<i>Slope β (EUR/MWh per buy MW; on a descending price-ordered curve)</i>				
Retailer	-1.688	-1.624	2.607***	-2.272**
DirectCons	-5.308***	—	-3.746***	1.555***
PumpBuy	-0.012	0.133	0.039***	0.017***
<i>Curvature γ (EUR/MWh per buy MW²)</i>				
Retailer	-6.1746	-0.7050	2.7389	4.9286
DirectCons	-25.4***	—	-5.7976*	-0.0206
PumpBuy	-0.0002	0.0002	0.0008**	0.0005

Table 15: Demand-side bid-shape DiD, critical vs. flat, unit FE. Per-curve OLS on the buy side with tranches sorted descending in price; β measures how price falls as cumulative buy MW expands down the demand ladder. Retailer = competitive supplier (BIG-4 dominated); DirectCons = large direct industrial buyer; PumpBuy = pump-storage in charging mode.

Retailer DA15 IDA α at +33*** tracks the sell-side widening; ID15 IDA retailer runs the opposite direction, corroborating the ID15 IDA price compression in §6.1.

A.10 Residual-demand slope inside the bandwidth: empirical b_{mt}

I replicate Ito and Reguant (2016) Section III.A Method 1 per firm. For each (date, period, market, focal firm) I build the residual demand the firm faces as aggregate demand minus all-other-firms’ supply, sample it on a price grid ± 50 EUR/MWh around MCP, fit a quadratic, and take the slope at MCP. Per-firm means in Table 16.

Firm	ID15 IDA				DA15 DA				DA15 IDA			
	pre	post	Δ	ratio	pre	post	Δ	ratio	pre	post	Δ	ratio
Iberdrola	156	180	+24	1.15	230	285	+55	1.24	149	168	+19	1.13
Endesa	180	214	+34	1.19	255	321	+66	1.26	171	202	+31	1.18
Naturgy	173	212	+39	1.23	266	330	+64	1.24	159	187	+28	1.17
EDP-HC	167	193	+26	1.16	231	295	+64	1.28	157	181	+24	1.15

Table 16: Per-firm residual-demand slope b_{mt} at the market clearing price, Ito and Reguant (2016) Section III.A methodology. Units: MW per EUR/MWh (larger = more elastic residual demand = thicker market for the focal firm). Per (date, period, market, focal firm) the residual demand curve is aggregate demand minus all-other-firms’ supply, sampled on a grid ± 50 EUR/MWh around the period’s MCP, with the slope at MCP extracted from a quadratic fit. ID15 IDA pre = 2024-06-14 to 2025-03-18 (hourly IDA); ID15 IDA post = 2025-03-19 to 2025-04-27 (15-min IDA). DA15 DA pre = 2024-06-14 to 2025-09-30 (hourly DA); DA15 DA post = 2025-10-01 to 2026-03-06 (15-min DA).

Residual demand thickens for every firm in every (reform, market) cell by 13–28% (tight) and 8–15% (long-pre + year + month FE). The Cournot first-order condition then implies markup compression. Per-firm b is the model’s pricing primitive and it moves as predicted.

Per-IDA-session decomposition. I rebuild b_{mt} per IDA session (S1, S2, S3) by joining each session’s MCP to its own bids. Pre-reform the model predicts b thins toward delivery (S1 < S2 < S3); the data confirm it. Post-MTU15-IDA the S2/S3 gap collapses (Table 17).

Grid bandwidth	Pre-MTU15-IDA			Post-MTU15-IDA		
	S1	S2	S3 (closest to delivery)	S1	S2	S3
± 50 EUR/MWh (baseline)	80.2	68.3	59.4	99.9	73.0	72.3
± 70 EUR/MWh	71.7	62.6	55.0	84.0	61.8	60.5
± 90 EUR/MWh	64.2	57.1	50.8	73.4	54.5	53.0

Table 17: Mean per-firm b_{mt} by IDA session and grid bandwidth. Units MW per EUR/MWh (larger = more elastic residual demand). Mean across the four dominant firms (Iberdrola, Endesa, Naturgy, EDP-HC). Pre-window: 2024-06-14 to 2025-03-18 (hourly IDA). Post-window: 2025-03-19 to 2025-04-27 (15-min IDA, pre-blackout). At every bandwidth, three patterns hold: (i) pre-reform residual demand thins monotonically toward delivery ($S1 > S2 > S3$), confirming the sequential-markets prediction that liquidity falls as the gate closes; (ii) post-reform S1 stays the most elastic session (thickens by 14–24%); (iii) the S2-vs-S3 gap collapses post-reform — the granularity reform makes the two later sessions look like the same product.

Three patterns hold at every bandwidth: (i) pre-reform thinning toward delivery ($b_{S1} > b_{S2} > b_{S3}$); (ii) post-reform S1 thickens by 14–24%; (iii) the S2-vs-S3 gradient collapses to essentially zero. The granularity reform makes the two later IDA sessions look like the same product. Industry reporting confirms the underlying liquidity ordering (IDA1 and IDA2 each carry $\sim 37\%$ of intraday liquidity; IDA3 only 12%).

A.11 Per-channel BSTS on REE ajuste-cost decisions

I run BSTS separately on each of the eight daily ajuste-cost channels REE publishes via ESIOS (TR up/dn, aFRR up/dn, Fase I up/dn, Fase II up/dn). MTU15-IDA shrinks TR up and aFRR up cleanly; MTU15-DA is dominated by reforzada (Fase I up rises). Total daily-cost change is roughly $-\text{EUR } 3.9\text{M/day}$ at MTU15-IDA and $+\text{EUR } 5.1\text{M/day}$ at MTU15-DA on the raw BSTS posterior mean. Unlike the price specs we do not report a same-calendar placebo: these are regulatory cost lines reflecting REE’s procurement-and-dispatch decisions, not market outcomes, so a year-ago window is not a valid counterfactual for what REE *would have done* absent the reform. BSTS controls for renewable trend and seasonality but cannot isolate market-mechanism effects from operator-discretion changes.

Channel	MTU15-IDA (pre-blackout post)	MTU15-DA (post-window straddles reforzada)
Fase I up	+1,988	+4,554
Fase I dn	−197**	−123
Fase II up	−24	−143***
Fase II dn	−963*	+1,959**
TR up	−4,182**	−709
TR dn	+390***	−29
aFRR up	−842***	−325
aFRR dn	−44	−132
Sum	$\approx -3,874$	$\approx +5,102$

Table 18: BSTS on daily REE ajuste-cost channels (kEUR/day, raw posterior mean only). Long pre-window 2022-01-01 onwards, year-by-year wind and solar interactions, gas covariate, 7-day seasonal cycle. MTU15-IDA real post: 2025-03-19 to 2025-04-27 (pre-blackout). MTU15-DA real post: 2025-10-01 to 2025-12-31. Stars indicate the 95% credible interval excludes zero. We do *not* report a same-calendar placebo here because these channels are regulatory cost lines reflecting REE’s procurement-and-dispatch decisions, not market clearing outcomes; the year-ago window is not a counterfactual for what REE *would have done* absent the reform (its procurement strategy itself evolved across the post-blackout regime). Granularity-attributable interpretation of these movements is bounded above by the small pre-blackout gap visible in Figure 6; the cleaner reform identification sits on the auction side (bid-shape DiD, wedge SD, per-firm b_{mt}).

Reading: at MTU15-IDA the granularity-correlated channels (TR up, aFRR up) shrink robustly without substitution into Fase I up. At MTU15-DA the reforzada-driven Fase I rise dominates the total. Daví-Arderius and Graf (2026)’s event-study estimate of $\sim$ EUR 1.24B/year for the reforzada cost rise is on the same order of magnitude as the raw MTU15-DA Fase I up posterior here (+EUR 4.6M/day $\approx$ +EUR 1.7B/year); the agreement is reassuring directionally but *not* a placebo-net identification claim (§6.3).

A.12 Battery bid behavior in the IDA — descriptive

The bid-shape DiD bid-shape framework does not transfer to batteries directly because every battery curve we observe in OMIE post-2024 is single-tranche by construction ($\text{HHI}_p = 1$, $\sigma_p = 0$, $\hat{\beta}$ and $\hat{\gamma}$ undefined). What does change at the reforms is participation intensity and bid placement relative to the clearing price. I list all OMIE units with technology *Almacenamiento Compra/Venta* (eleven buy-side and three sell-side units), pull their IDA bids across the same window as the headline analysis, and compare across three regimes (pre-MTU15-IDA / post-MTU15-IDA pre-blackout / post-DA15 onwards).

Regime	Side	Curves	Median $p_{\text{bid}} - \text{MCP}$	In-band share (± 50)	Avg MW/bid
Pre-MTU15-IDA	Charge	8	+38.0	63%	4.4
	Sell	4	−47.6	50%	4.2
Post-MTU15-IDA	Charge	1,115	+0.2	96%	3.3
	Sell	1,017	+0.2	98%	3.9
Post-DA15	Charge	2,184	+27.4	74%	3.3
	Sell	1,598	−10.0	72%	3.1

Table 19: Battery IDA bidding by regime (descriptive). Eleven *Almacenamiento Compra* units and three *Almacenamiento Venta* units in the OMIE master list. Single-tranche curves throughout. “Median $p_{\text{bid}} - \text{MCP}$ ” positive on the charge side means the unit bids ABOVE the clearing price to guarantee charging clears; symmetric below MCP on the sell side. Pre-MTU15-IDA sample is small (only twelve curves total). Post-MTU15-IDA is a near-at-MCP strategy (price-taker). Post-DA15 the bidder pulls the charge bid up and the sell bid down (revealed willingness-to-pay / willingness-to-accept), and overall battery participation more than doubles.

Three observations. *Participation*: battery curves rise from a dozen pre-MTU15-IDA to thousands post-DA15 — the granularity reforms run alongside a build-out of the asset class, and the IDA is the venue where batteries place their volumes. *Strategy at MTU15-IDA cutover*: bids cluster essentially AT the clearing price (96–98% within ± 50 EUR/MWh, median $p_{\text{bid}} - \text{MCP} \approx 0$) — price-taking with both buy and sell curves straddling the realised MCP, consistent with an asset that wants to clear as much volume as available capacity allows. *Strategy at DA15 cutover*: bids spread out, charge bids well above MCP and sell bids well below it (revealed reservation prices). The shift from at-MCP price-taking to spread-around-MCP volumetric bidding is what an actively-arbitraging storage operator looks like.

Battery arbitrage: charge/sell mix and dual-bidding across reforms. I pair the three batteries with both compra and venta unit codes (Iberdrola *AFIBG*, EDP Soto *ASRIHC*, Batalha *SMB*) and ask two questions: how does the charge-versus-sell mix shift across reforms, and how often does the same battery bid BOTH sides in the same period (a bracketing-arbitrage signal). Pre-ID15: only Iberdrola is active, and only on nine periods total. Post-ID15: Iberdrola cycles nearly evenly (53% charge / 47% sell of its unit-periods), Batalha is overwhelmingly charging (88% charge). Post-DA15: Iberdrola shifts toward charging-heavy (69% charge), EDP Soto enters with a charge-leaning mix (62% charge), and *Batalha* starts placing bids on both sides in 18% of its periods (32 out of 175) — a bracket where the charge bid lands well above MCP and the sell bid well below, so that the position clears in either direction as the realised MCP settles in between. The bracketing strategy is consistent with active two-sided arbitrage. Sample is tiny (three battery pairs) and we cannot identify a treatment effect; we report the descriptive evolution.

A.13 Per-session IDA bid-shape decomposition: CCGT bidding intensity is IDA2-concentrated pre-reform and homogenizes post-reform

I split the pooled IDA bid-shape DiD per IDA session. Pre-reform CCGT bidding intensity concentrates in IDA2 (σ_p and $\hat{\beta}$ roughly $2.5\times$ IDA1). Post-reform the three sessions converge in level. Tight descriptive, not the same identification as the pooled DiD; see Table caveat.

Outcome	Pre-MTU15-IDA			Post-MTU15-IDA (pre-blackout)			Post-DA15 (incl. re-reforzada)		
	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3
σ_p (EUR/MWh)	1.74	4.08	0.34	4.03	5.77	2.47	3.64	4.79	2.07
$\hat{\beta}$ (EUR/MWh per MW)	0.125	0.335	0.161	0.122	0.135	0.123	0.099	0.125	0.098
HHI_p	0.25	0.50	0.60	0.21	0.33	0.33	0.22	0.25	0.33
n_{curves}	7,647	11,398	10,515	8,962	10,254	7,208	56,484	60,147	43,998

Table 20: Per-session IDA CCGT bid-shape, critical evening hours 16–22. Three windows: pre-MTU15-IDA 2024-06-14 to 2025-03-18 (post-IDA-reform 3-session at MTU60); post-MTU15-IDA 2025-03-19 to 2025-04-27 (40 days, pre-blackout); post-DA15 2025-10-01 to 2026-04-27 (post-blackout, post-DA15, includes re-reforzada from Nov 2025). Medians across in-band per-curve fits at bandwidth $\delta = 150$ EUR/MWh. IDA3 covers hours 13–24, so it has no flat-hour reference; critical evening hours intersect all three sessions for cross-session comparison. The collapse of pre-reform IDA2 dominance ($\text{HHI}_p = 0.50$, $\hat{\beta} = 0.335$) into post-reform session-homogeneity is the per-session companion to the pooled critical/flat DiD reported in Table 6. *Caveat:* this is a descriptive comparison across three calendar windows with different seasonal mixes; we report it as suggestive of a session-order collapse, not as an identified treatment effect. The same long-pre robustness applied to the pooled DiD in §A.3 has not been applied per-session due to small per-session samples.

The figure version separates the four windows visually.

A.14 Post-DA15 IDA activity rise and the Fase I displacement channel

I compute pre-DA15 vs post-DA15 IDA bidding activity per session (block orders excluded). CCGT supply MW/day in IDA3 falls 15%; demand-side MW/day rises +15%/+10%/+29% across IDA1/IDA2/IDA3. The reading is a Fase I displacement channel: re-reforzada (§2) shifts CCGT dispatch from real-time into pre-IDA Fase I, leaving residual imbalance for the demand side to absorb in late IDAs.

Side / outcome	IDA1		IDA2		IDA3	
	MW/day	tranches/curve	MW/day	tranches/curve	MW/day	tranches/curve
CCGT supply	+11.6%	−3.8%	+0.5%	−1.9%	− 15.0%	−18.3%
All supply	−5.7%	+16.7%	−10.2%	+6.9%	−17.9%	+3.8%
Demand	+ 14.7%	+14.9%	+ 10.0%	+12.0%	+ 29.4%	+9.7%

Table 21: IDA bidding activity, post-DA15 vs. pre-DA15 (reforzada-constant window). Pre-DA15: 2025-04-28 to 2025-09-30 (post-blackout, post-ID15, pre-DA15); post-DA15: 2025-10-01 to 2025-12-31. “MW/day” is the sum of in-band offered quantities per session-day; “tranches/curve” is the mean over (date, unit, period) curves. Block orders excluded. CCGT IDA3 supply MW/day falls 15% while demand-side IDA3 MW/day rises 29% — consistent with re-reforzada Fase I displacement: CCGT is dispatched ahead of IDA close via Fase I, leaving residual imbalance for the demand side to absorb in late IDAs.

Composes with the BSTS reading of Table 18: the TR-up shrink across the MTU15-DA cutover is procurement-timing substitution into pre-IDA Fase I, not a market-mechanism cleanup. Industry source verbatim confirms (§6.3).

A.15 Strategic markup proxy: q_f^{strat}/b_f per firm

Cournot first-order condition: $(p - MC)/p = q_f^{\text{strat}}/(b_f \cdot p)$, where q_f^{strat} is the focal firm’s strategic cleared MW (total cleared minus own renewable cleared MW: wind, solar PV, run-of-river, CSP, biomass-renewable, RE Tarifa CUR). Renewables clear at marginal cost ≈ 0 regardless of bidding choices, so they are infra-marginal and do not enter the markup calculation. Reported as the period mean of q_f^{strat}/b_f per (firm, window). Bigger = more pricing power per strategic MW.

	%Δ b_f (residual demand)				%Δ markup _f			
	IB	GE	GN	HC	IB	GE	GN	HC
ID15 IDA	+15	+19	+23	+16	−12	+152 [‡]	−13	+62 [‡]
ID15 DA	+ 116	+ 93	+ 100	+ 100	−19	+29 [‡]	− 38	+23 [‡]
DA15 DA	+ 24	+ 26	+ 24	+ 28	− 27	− 26	−12	−18
DA15 IDA	+13	+18	+17	+15	+7	+10	−15	+40

Table 22: Per-firm residual-demand slope and strategic markup proxy, pre vs post. b_f recovered as in Ito and Reguant (2016) (rebuild rivals’ supply from OMIE offer book, quadratic fit at MCP±50 EUR/MWh, period mean). Markup proxy = q_f^{strat}/b_f with q_f^{strat} from focal-firm cumulative sells at MCP minus renewable-unit subset. [‡]: ID15 markup signs are compositional, not behavioural — the 40-day post window saw nuclear out and CCGT heavily called, so GE and HC’s strategic dispatch jumped +80 to +170% from a small base, inflating the markup numerator. **DA15 DA is the cleanest market-power result**: all four dominant firms see b_f thicken (+24 to +28%) AND the markup proxy fall (−12 to −27%).

A.16 Convex-renewable robustness on the price OLS

Hourly OLS on the ID15 IDA price, no linear trend, Newey-West HAC lag 24×7 . Pre window 2022-01-01 onwards; post = 40 days for ID15 (pre-blackout), 92 days for DA15. **D**: quadratic spec adds wind², solar² with year-by-renewable interactions on both linear and quadratic terms. **E**: cubic spec adds wind³, solar³ with year-by-renewable interactions on all three orders. The quadratic spec absorbs merit-order curvature in renewable price-suppression; the cubic adds essentially nothing (diminishing returns).

	ID15 (IDA reform)		DA15 (DA reform)	
	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS hourly — + year-by-renewable (2024+25 pooled)	−30***	−29***	+3	+2
D + quadratic renew × year	−24***	−24***	−2	−3
E + cubic renew × year	−24***	−23***	−2	−2
Newey-West HAC SE (D)	7.00	7.10	5.36	5.31
Newey-West HAC SE (E)	6.85	6.95	5.37	5.32

Table 23: Convex-renewable robustness: quadratic and cubic specs. The quadratic D spec absorbs the merit-order curvature in renewable price-suppression. The ID15 own-venue coefficient shrinks from −30 (linear year-by-renew) to −24 (D), $p=0.001$. The cubic E spec adds essentially nothing (additional shrinkage < 1 EUR/MWh, SE unchanged). The ID15 floor under convex-renewable controls is ≈ -23 EUR/MWh. DA15 stays null under both convex specs.

A.17 Renewable capacity vs realised production, 2024 vs 2025

(a) **Installed capacity at year-start** (ENTSO-E, MW). Solar PV expanded +43%; wind essentially flat.

	Jan 2024	Jan 2025	Δ MW	Δ %
Solar PV (B16)	29,047	41,433	+12,386	+42.6%
Wind (B18+B19)	30,932	32,451	+1,519	+4.9%
Solar + Wind	59,979	73,885	+13,906	+23.2%
Coal (B05)	1,820	1,258	−562	−30.9%
Nuclear (B14), Gas (B04)	no change	no change	—	—

(b) **Realised production in the matched ID15 window** (Mar 19 – Apr 27).

	Spring 2024	Spring 2025	Δ	Δ %
Wind (GWh/d)	167.5	161.4	−6.1	−4%
Solar (GWh/d)	122.2	138.4	+16.2	+13%
Gas (EUR/MWh)	431	746	+315	+73%
IDA price (EUR/MWh, raw mean)	10.9	25.1	+14.2	—

Reading. The +43% solar-capacity expansion delivered only a +13% generation increment in the matched window — spring 2024 was already very sunny, and the bulk of the 2025 build-out came online later. Wind output is flat. Gas, the dispatchable margin’s marginal cost, was 73% higher in 2025 and on its own would have pushed prices *up*. The headline OLS isolates the residual (−23 to −39 EUR/MWh) after subtracting these channels — the reform did real work.

B Continuous-market response at the three structural reforms

BSTS on ES-leg continuous-market series under headline BSTS. ID15 is the only structural cutover of the three: trade count jumps +24,269/day placebo-net ($p < 0.001$) as participants take up the four 15-min contracts per clock-hour, and energy turnover drops −12.36 GWh/day placebo-net ($p < 0.001$) as smaller contracts and partial migration toward the now-finer IDA auctions thin

total volume. The continuous-market volume-weighted price is not separately identifiable from the renewable-driven price drop once year-interactions control the wind/solar coefficient (real -43.6^* , placebo-net -18.3 ns). ISP15 is a methodological null (settlement-only reform, contract grid unchanged); DA15 leaves the continuous-market contract grid alone (already 15-min post-ID15) and no outcome survives the year-interaction control. The continuous-market reading is consistent with the auction-side reading: granularity reforms shift the contract grid and the participation pattern, but the price-level component cannot be separated from the renewable-coefficient drift.

C Market-structure context

C.1 The two-tier CCGT bid curve and the day-ahead/intraday contrast

Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost and a high *parked* tier above the clearing-price ceiling. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier with negligible withholding in the intraday auction, after Fase I has cleared. A unit's marginal cost does not change between two auctions hours apart. The parked tier is a placeholder rather than a Fase I bid: Fase I up-redispach is settled through a separate restriction offer submitted to REE.³ MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier; the CCGT auction-cleared scale-up in §6.1 is the competing tier expanding into the four quarters of each peak hour.

C.2 Market concentration migrated from wholesale to ancillary services

Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

C.3 The reforzada cost increase concentrates on CCGT

The post-blackout REE intervention is technology-specific. *Fase I* (*Resolución de restricciones técnicas por garantía de suministro*) is the pre-IDA technical-restrictions process REE runs on the PDBF programme: REE buys up-redispach and sells down-redispach to clear transmission-security constraints, settled pay-as-bid of a *separate restriction offer* the unit submits to REE (Procedimiento de Operación PO 3.2; settlement PO 14.4 ap. 20.1). The Fase I restriction offer is a distinct instrument from the unit's day-ahead market offer. *Fase II* (*Resolución de restricciones por balance*) is the post-Fase-I rebalance that absorbs the up-redispach back into the schedule through symmetric down-redispach elsewhere (PO 3.2 §6); it has been operationally near-zero since 2024. Decomposing weekly dispatch into its components — the day-ahead-cleared base (PDBC), bilateral executions, and the post-DA top-up routed through IDA and REE real-time redispach — the CCGT panel of Figure 12 shows the day-ahead-cleared layer (blue) collapsing post-blackout while the IDA+REE-RT layer (red) balloons. Other technologies are largely flat across the same window.

This per-technology pattern is the cleanest evidence that the reforzada cost increase is a separate policy effect from the granularity reforms: a granularity reform that touched all bid-side

³Procedure PO 3.2, settlement PO 14.4 ap. 20.1; see also CNMC (2025c,a).

firms equally would not produce a CCGT-only intervention pattern. The day-ahead CCGT cleared scale-up of §6.1 (DA15 only, critical hours) sits alongside this REE-intervention story: both are CCGT-side responses to the operational environment, but the granularity-driven response is competitive-clearing (and consumer-favourable) while the reforzada-driven response is REE-paid redispatch.

C.4 Representative bid curves by technology

Figures 13–14 show real day-ahead bid step functions for one representative unit per technology, at an evening-ramp critical-hour quarter (Figure 13) and at a flat overnight hour on a windy night (Figure 14). The visual contrast supports the §5.2 exclusion of solar from bid-shape DiD and the §6.6 reading of wind’s small bid-shape DiD coefficients: dispatchables structure their competing tier inside the in-band region; renewables submit single or near-single blocks far below the band, so a within-band σ_p or HHI_p outcome is mechanically zero or one. Wind aggregators do occasionally submit multi-tranche micro-ladders below the band — the WMVD088 panel in Figure 14 shows 15 tranches over the narrow $[-20, +1.5]$ EUR/MWh range, consistent with the operational (imbalance-bill) rather than strategic incentive that drives wind’s bidding (§6.6) — but these micro-ladders sit outside the in-band region and so do not enter the bid-shape DiD outcomes.

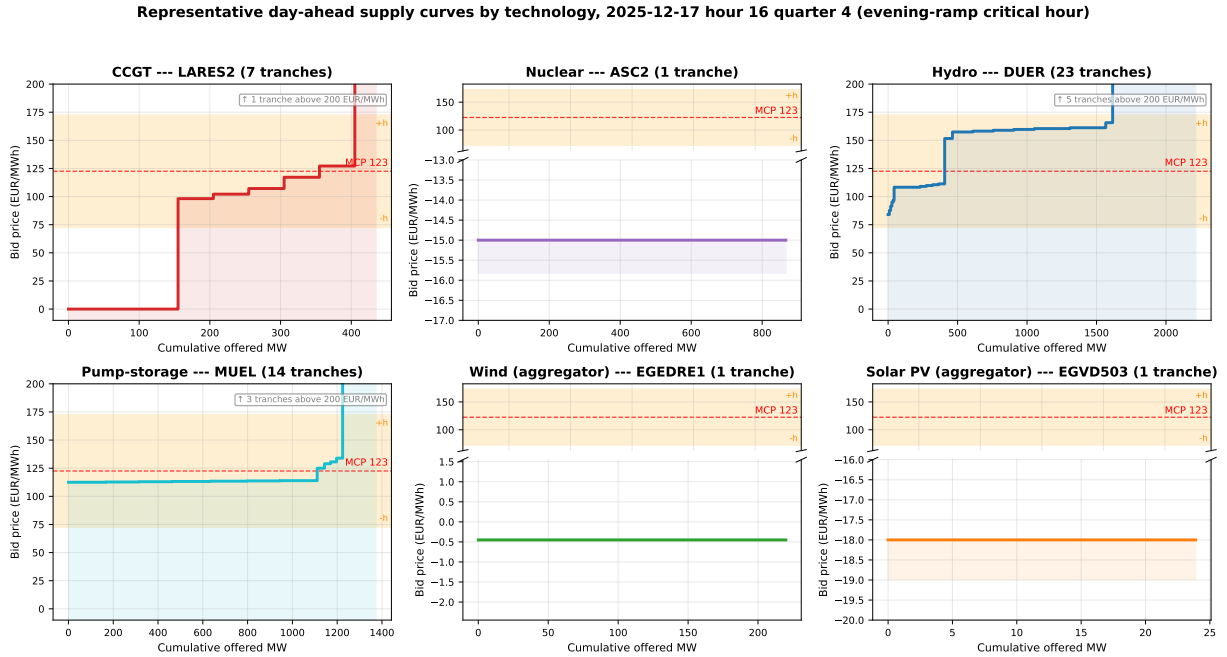


Figure 13: Representative day-ahead supply curves by technology, 2025-12-17 hour 17 quarter 4 (post-MTU15-DA, evening-ramp critical hour). Red dashed: MCP; orange band: in-band region $[\text{MCP} \pm \delta]$ with $\delta = 50$ (§4.3). Panels with bid ranges far below the band use a broken y-axis for readability.

Representative day-ahead supply curves by technology, 2025-12-04 hour 00 quarter 4 (flat overnight hour on a windy night)

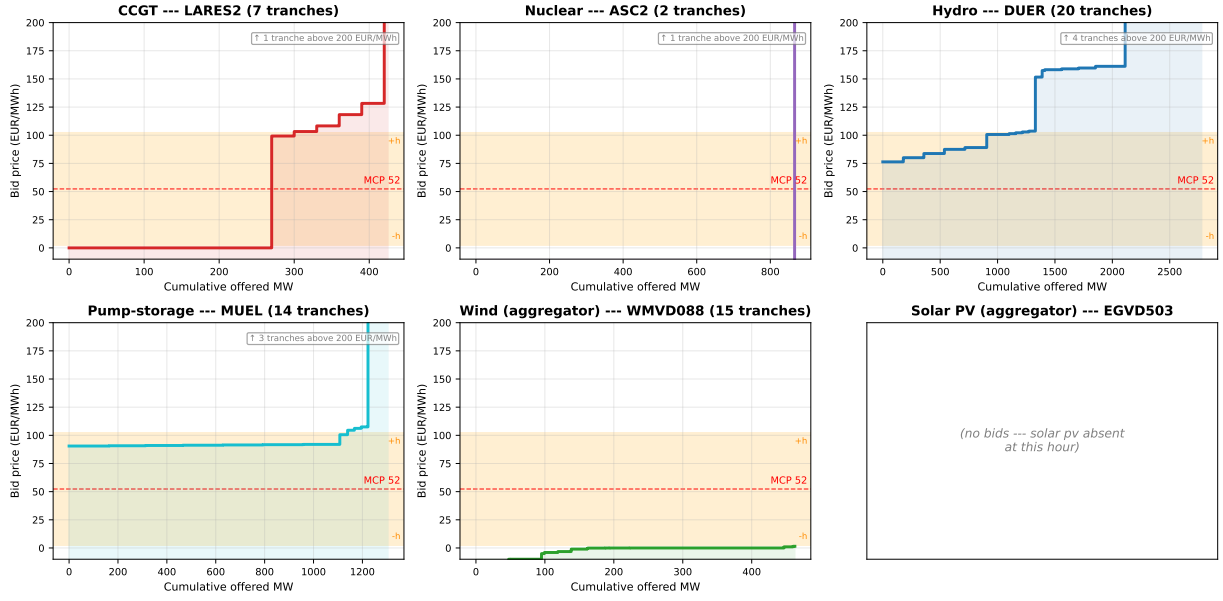


Figure 14: Representative day-ahead supply curves by technology, 2025-12-04 hour 01 quarter 4 (flat overnight hour on a windy night). Same colour scheme and broken y-axis convention as Figure 13. The wind panel (WMVD088) shows a 15-tranche aggregator micro-ladder concentrated in $[-20, +1.5]$ EUR/MWh, well below the band.

Buy vs sell IDA bid shapes (pre-ID15). Figure 15 shows representative IDA bid curves with the sell side (blue, ascending) and the buy side (red, descending) overlaid for one unit per tech in the pre-ID15 window, when buy-side activity was largest (§6.4). The shapes differ sharply across techs: CCGT and pump-storage submit dense sell ladders with a thin one-tranche buy book (sellers with a small rebuy option); hydro and renewable aggregators write multi-tranche curves on *both* sides (forecast-driven rebalancing that can clear in either direction); nuclear barely participates on the buy side (the entire 9-month pre-ID15 window contains a single nuclear (date, session, period) cell with both sides). The renewable-aggregator panels are the clearest illustration of the buy-side rebalancing role of IDA: the buy and sell curves intersect near the realised MCP, and the aggregator clears on whichever side its quarter-level forecast falls.

Representative IDA sell vs buy bid curves by technology (pre-ID15)

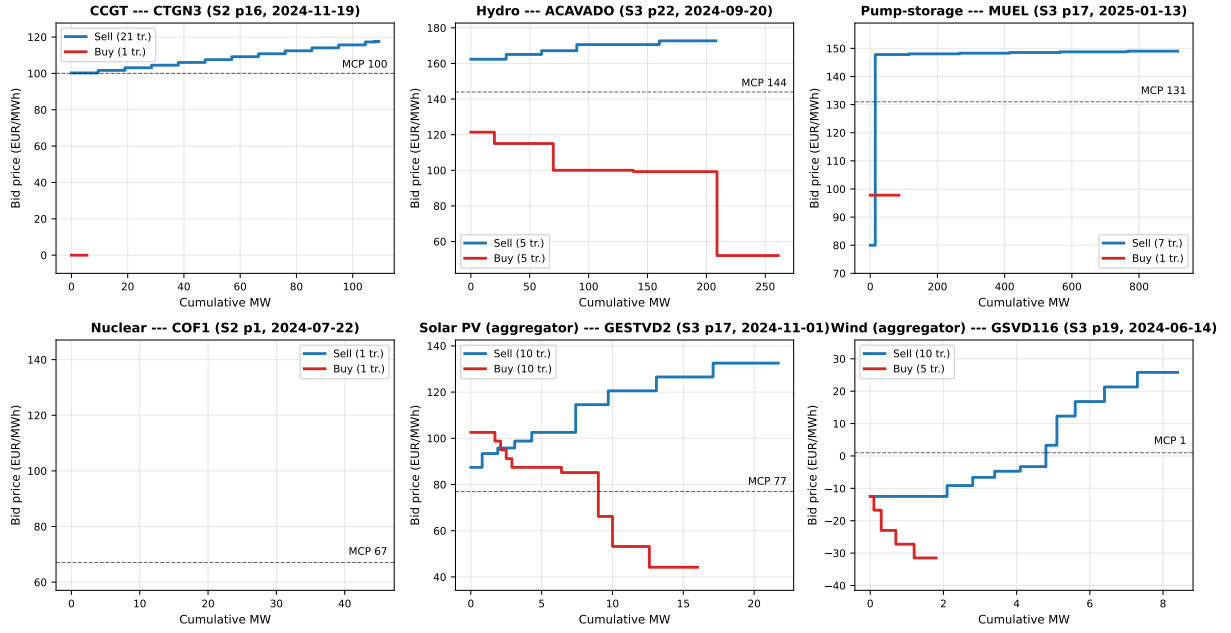


Figure 15: Representative IDA sell vs buy bid curves by technology, pre-ID15. One representative (unit, date, session, period) per tech. Blue: sell curve (supply, ascending in price). Red: buy curve (demand, descending in price). Dashed: MCP. Pre-ID15 window 2024-06-14 to 2025-03-18.

D Theoretical derivations

D.1 Setup details and information arrival

This subsection records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (forward gate): $\bar{A}$, the profile $\{\zeta_t\}$, and all distributions are common knowledge; each forward product clears against its realised clearing state v_{1t} , screened by the stage-1 ladder. Between the gates the within-hour update ω_t realises: $\omega_t = \zeta_t + \mu_t$ when the forward is hourly, $\omega_t = \mu_t$ when it is per-quarter. Stage 2 (spot gate): each spot product clears against ω_t and its own clearing state v_{2t} , screened by the stage-2 ladder. Stage 3 (balancing): the unpriceable noise ε_t realises and the settlement prices residual deviations at the penalty γ .

Residual demand (Ito and Reguant (2016) form). The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery quarter t , with all quantities in MWh:

$$q_{1t} = \bar{A} + \zeta_t \mathbb{1}\{Q^F = Q\} + v_{1t} - b_{1t}p_{1t} - s_t - S_{1t}^{op}, \quad q_{2t} = b_{2t}(p_{1(t)} - p_{2t}) + \omega_t + s_t - S_{2t}^{op}, \quad (7)$$

where $p_{1(t)}$ is the forward price covering quarter t (one number per clock-hour when $Q^F = 1$), S_{mt}^{op} is the aggregator's offered MWh in market m at quarter t (uniform S_m^{op}/Q when market m is hourly), and b_{mt} is market thinness with per-quarter asymmetry allowed (when market m is hourly, $b_{mt} \equiv b_m$ and $p_{mt} \equiv p_m$).

Hourly residual demand by specialisation. When market m is hourly, its product clears against the within-hour *average* of the per-quarter objects in (7): the profile and every redistributive component drop out by $\sum_t \zeta_t = \sum_t \mu_{mt} = 0$, the screened state is the hour-level $\theta_m = Q^{-1} \sum_t v_{mt}$, and the spot's traded update is the hour-level $\bar{\omega} = Q^{-1} \sum_t \omega_t$. The within-hour variation that this averaging suppresses does not vanish from the model: it loads onto the balancing residual (§D.7) and shows up in the system imbalance cost C_{sys}^I in the welfare derivations (§D.8).

Tiling lemma: uniform amplification under aggregation. The body of §3.1 assumes the screened state is uniform at each granularity ($\theta_m \sim U[-\Delta_m, \Delta_m]$ for hourly products, $v_{mt} \sim U[-\rho_\chi \Delta_m, +\rho_\chi \Delta_m]$ for per-quarter products). The following lemma shows the uniform family is exactly closed under the aggregation $v_{mt} = \theta_m + \mu_{mt}$, with ρ pinned by the support of the redistributive component.

Lemma 2 (Tiling). *Let $\theta_m \sim U[-\Delta_m, \Delta_m]$ and let μ_{mt} be supported on $\{-\Delta_m, +\Delta_m\}$, taking each value with probability $1/2$ in each quarter (with $\sum_t \mu_{mt} = 0$ enforced across quarters). Then $v_{mt} = \theta_m + \mu_{mt} \sim U[-2\Delta_m, +2\Delta_m]$ exactly: $\rho_\chi = 2$. More generally, supporting μ_{mt} on the integer multiples $\{-(2j-1)\Delta_m, +(2j-1)\Delta_m\}$ for $j = 1, \dots, J$ delivers $v_{mt} \sim U[-(2J)\Delta_m, +(2J)\Delta_m]$ exactly, i.e. $\rho_\chi = 2J$, an arbitrary even integer.*

Proof. For the $J = 1$ case: $\theta_m + \Delta_m \sim U[0, 2\Delta_m]$ when $\mu_{mt} = +\Delta_m$, and $\theta_m - \Delta_m \sim U[-2\Delta_m, 0]$ when $\mu_{mt} = -\Delta_m$. Each sub-distribution has density $1/(2\Delta_m)$ on its support; the equiprobable mixture has density $1/(4\Delta_m)$ on $[-2\Delta_m, 2\Delta_m]$, which is $U[-2\Delta_m, 2\Delta_m]$. The two sub-supports tile $[-2\Delta_m, 2\Delta_m]$ without overlap or gap. The $J > 1$ case is the same tiling argument with J pairs of sub-supports. $\square$

The body carries a general ρ_χ rather than the even-integer values delivered by the tiling; we treat ρ_χ as a primitive of the hour class, pinned by the support of the redistributive component the venue's per-quarter object carries. The same construction with a smaller support delivers the news-only value $\rho_\chi^\mu \leq \rho_\chi$ used in §3.6(iii).

D.2 Operational aggregator: closed-form gain from per-period scheduling

We derive the aggregator's expected hedging gain $G(\Delta_\eta)$ from per-period (rather than hourly) spot scheduling in closed form under uniform shocks. We then explain why the aggregator's *individual* forward-vs-spot allocation is not pinned down by an interior first-order condition under standard penalty specifications, and why the empirical migration onto the granular leg (Result 3) is therefore an equilibrium statement about G being positive, not an aggregator-level Δ^* identification.

D.2.1 Problem statement and primitives

The aggregator allocates total hourly expected production $\bar{Y}_h$ between forward and spot. Forecast-error shock $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$, i.i.d. across t , with variance $\sigma_\eta^2 = \Delta_\eta^2/3$. We set the structural per-subperiod expected production constant ($\bar{y}_t = \bar{Y}_h/Q$) for the closed-form derivation; the structural variation $\sigma_a^2 = \text{Var}(\bar{y}_t - \bar{Y}_h/Q)$ adds linearly to G and is handled in the closing remark of §D.2.2. Imbalance penalty is $\gamma|\mathcal{I}_t|$ per period (linear in the magnitude of the gap).

D.2.2 Expected penalty under each regime

Granular spot ($Q^S = Q$). At Stage 2 the aggregator observes η_t and sets the per-subperiod allocation $S_{2t}^{op*} = \bar{y}_t + \eta_t - S_1^{op}/Q$. The realised per-period imbalance is 0 (modulo post-Stage-2

noise ε_t , which is regime-invariant and we drop for clarity):

$$\mathbb{E}[\mathcal{C}_{\text{gran}}] = 0.$$

Hourly spot ($Q^S = 1$). The aggregator commits $S_1^{\text{op}} + S_2^{\text{op}} = \bar{Y}_h$ before η_t resolves; the per-subperiod schedule is therefore uniform at $\bar{Y}_h/Q$. With $\bar{y}_t = \bar{Y}_h/Q$ the per-period imbalance is exactly η_t . For $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$:

$$\mathbb{E}|\eta_t| = \frac{\Delta_\eta}{2}, \quad \mathbb{E}[\mathcal{C}_{\text{hour}}] = \gamma Q \cdot \frac{\Delta_\eta}{2}.$$

Closed-form hedging gain. The aggregator's expected hourly saving from per-period (granular) spot scheduling relative to hourly spot scheduling is

$$G(\Delta_\eta) = \mathbb{E}[\mathcal{C}_{\text{hour}}] - \mathbb{E}[\mathcal{C}_{\text{gran}}] = \frac{\gamma Q \Delta_\eta}{2}. \quad (8)$$

Comparative statics are immediate:

$$\frac{\partial G}{\partial \Delta_\eta} = \frac{\gamma Q}{2} > 0, \quad \frac{\partial G}{\partial \sigma_\eta^2} = \frac{\gamma Q}{2\sqrt{3}\sigma_\eta} > 0. \quad (9)$$

The gain is strictly positive and strictly increasing in the forecast-error variance. *Closing remark (structural variation).* When $\bar{y}_t \neq \bar{Y}_h/Q$, the hourly regime's uniform schedule additionally misses the deterministic profile, adding $\gamma \sum_t |\bar{y}_t - \bar{Y}_h/Q|$ to its bill; granular scheduling removes this term too, so it enters G additively and we omit it from the display.

D.2.3 Why Δ^* is not pinned down by an individual FOC

A natural follow-up is to ask whether G pins down the aggregator's allocation S_1^{op} via a first-order condition. It does not. The aggregator's expected profit under either regime is

$$\Pi(S_1^{\text{op}}) = (p_1 - p_2) S_1^{\text{op}} + p_2 \bar{Y}_h - \mathbb{E}[\mathcal{C}],$$

where the expected imbalance cost $\mathbb{E}[\mathcal{C}]$ is independent of S_1^{op} once we impose the no-average-bias condition $S_1^{\text{op}} + S_2^{\text{op}} = \bar{Y}_h$. The FOC reduces to $p_1 - p_2 = 0$, a corner solution rather than an interior Δ^* . This is a structural feature of the price-taker setup: the imbalance cost depends on the resolution of information (whether the aggregator can react to η_t), not on how forward and spot allocations split.

The migration onto the granular leg is therefore an *equilibrium* statement: across aggregators, $G(\Delta_\eta) > 0$ generates a positive premium for participation on the granular spot; equilibrium prices adjust so that the wedge $p_1 - p_2$ partially incorporates this premium; and the empirical aggregate shift in supply onto the granular leg (Result 3, sell-share rise of nuclear and wind at the March reform) is the equilibrium counterpart. We claim only the direction of the shift via $G > 0$; we do not derive a level prediction for Δ^* at the individual aggregator's optimum.

D.3 Strategic bidder: the optimal n -tranche ladder along the monopoly locus

We prove the bid-ladder propositions of §3.2 in turn: the continuous benchmark (Lemma 1), the optimal n -tranche ladder (Proposition 1), the closed-form bid-shape statistics (Corollary 1), level neutrality (Remark 1), the optimal tranche depth under a complexity cost (Proposition 2), the granularity–variation interaction (Corollary 2), and the stage-1 forward locus slope (Corollary 3).

Fix one delivery product. Residual demand is $q = A - bp$ with $A = \bar{a} + v$ and $v \sim U[-\Delta, \Delta]$; the bidder has constant marginal cost c . The auction is uniform-price; the bidder submits a non-decreasing step function $S(p) = \sum_k \Delta q_k \cdot \mathbf{1}\{p \geq p_k\}$ and the clearing pair (p^*, q^*) solves $S(p^*) = A - bp^*$ on the relevant segment.

D.3.1 Continuous benchmark (proof of Lemma 1)

In state A , the full-information monopoly point maximises $(p - c)(A - bp)$ over p ; the FOC gives $p^m(A) = (A + bc)/(2b)$ and $q^m(A) = (A - bc)/2$. Eliminating A , the monopoly locus is $\{(q, p) : q = b(p - c)\}$, a line of slope b through $(p, q) = (c, 0)$. Define $S^*(p) = b(p - c)$. For any state A ,

$$S^*(p^m(A)) = b\left(\frac{A + bc}{2b} - c\right) = \frac{A - bc}{2} = q^m(A),$$

so $(p^m(A), q^m(A))$ is on S^* in every state. *Uniqueness on the contested range*: a non-decreasing schedule that clears at the monopoly point in every state must contain the locus, which is strictly increasing; a monotone schedule containing a strictly increasing curve coincides with it on its range. Expected profit at S^* is $\mathbb{E}[(p^m(A) - c)q^m(A)] = \mathbb{E}[(A - bc)^2]/(4b)$, the unconditional monopoly value. $\square$

D.3.2 Optimal n -tranche ladder (proof of Proposition 1)

Clearing geometry under a step offer. Index the n competing tranches $1, \dots, n$ by ascending price $p_1 < \dots < p_n$, let q_0 be the inframarginal base priced at the floor, and let $Q_k = q_0 + \sum_{j \leq k} \Delta q_j$ be cumulative quantity ($Q_0 = q_0$). Three clearing regimes are possible per state A , alternating riser–horizontal–riser:

- (V_0) *Base riser.* For $A \in [\bar{a} - \Delta, Q_0 + bp_1]$: quantity is Q_0 and the realised demand sets the price, $p = (A - Q_0)/b$ between the floor and p_1 .
- (H_k) *Horizontal segment at p_k .* For $A \in [Q_{k-1} + bp_k, Q_k + bp_k]$: clearing is $(p_k, A - bp_k)$ — the marginal tranche is partially filled.
- (V_k) *Riser at Q_k .* For $A \in [Q_k + bp_k, Q_k + bp_{k+1}]$ (the top riser V_n closing at $\bar{a} + \Delta$): clearing is $((A - Q_k)/b, Q_k)$ — the realised demand sets the price.

This partitions the state space into $2n + 1$ full cells: one base riser, n horizontals of length Δq_k (in A -units), and n risers of length $b(p_{k+1} - p_k)$. Profit is continuous at every switch: both adjacent regimes deliver the same price–quantity pair at the boundary state.

Each instrument serves an interval of states and loses a squared distance. Profit in state A is a quadratic hill around the monopoly point. On a horizontal segment at price p , completing the square gives $\pi(p; A) = \pi^m(A) - b(p - p^m(A))^2$; on a riser at quantity q (the base included), $\pi(q; A) = \pi^m(A) - (q - q^m(A))^2/b$. Each instrument is therefore exactly right in one state — the step at p in the state whose monopoly price is p , the riser at q in the state whose monopoly quantity is q ; call that state the instrument's *target* $\hat{A}$. Because p^m and q^m are linear in A with slopes $1/(2b)$ and $1/2$, both losses reduce to the same expression,

$$\pi^m(A) - \pi = \frac{(A - \hat{A})^2}{4b},$$

so the bidder's problem is to place $2n + 1$ targets so that whichever state realises, it lands close to a target.

Equal intervals are optimal. An instrument serving an interval of length L does best with its target at the interval's midpoint, since the loss is symmetric in the distance; the mean squared distance of a uniform draw from the midpoint is $L^2/12$ (the variance of a uniform), and the interval is reached with probability $L/(2\Delta)$. Its contribution to the expected loss is therefore at least

$$\frac{L^2}{12} \cdot \frac{1}{4b} \cdot \frac{L}{2\Delta} = \frac{L^3}{96b\Delta},$$

and the total expected loss is at least $\sum_j L_j^3 / (96 b \Delta)$ over the $2n + 1$ intervals with $\sum_j L_j = 2\Delta$. Since x^3 is convex, moving length from a longer interval to a shorter one always lowers the sum, so the minimum splits the state range into *equal* intervals of width $w \equiv 2\Delta / (2n + 1)$.

The clearing rule places the boundaries halfway by itself. The bidder chooses only the targets; the interval boundaries then follow from the clearing geometry, and they land exactly halfway between adjacent targets. The switch from the step at p_k to the riser at Q_k occurs at the state $A^* = Q_k + b p_k$; substituting $p_k = p^m(\hat{A}_{H_k})$ and $Q_k = q^m(\hat{A}_{V_k})$,

$$A^* = \frac{\hat{A}_{H_k} + bc}{2} + \frac{\hat{A}_{V_k} - bc}{2} = \frac{\hat{A}_{H_k} + \hat{A}_{V_k}}{2},$$

and likewise at every other switch. Placing the targets at the equal-grid midpoints $m_j = \bar{a} - \Delta + (j - \frac{1}{2})w$, $j = 1, \dots, 2n + 1$, therefore reproduces exactly the equal-interval partition with every target at its interval's midpoint: the lower bound is attained.

Closed forms and feasibility. The alternation assigns V_0 to interval 1, H_k to interval $2k$, V_k to interval $2k+1$, so

$$q_0 = q^m(\bar{a} - \Delta + \frac{w}{2}), \quad p_k = p^m(m_{2k}) = p^m(\bar{a} - \Delta + (2k - \frac{1}{2})w), \quad Q_k = q^m(m_{2k+1}) :$$

every competing tranche carries $\Delta q_k = Q_k - Q_{k-1} = (m_{2k+1} - m_{2k-1})/2 = w$ MW (the first measured from the base) and the price grid is even with step $p_{k+1} - p_k = (m_{2k+2} - m_{2k})/(2b) = w/b$. Feasibility under Assumption 1: the base $q_0 = q^m(\bar{a} - \Delta) + w/4 \geq w/4 > 0$, and the lowest realised price (in V_0 at $A = \bar{a} - \Delta$) is $p^m(\bar{a} - \Delta - w/2) \geq c$ iff $q^m(\bar{a} - \Delta) \geq w/4$ — the stated condition.

Expected profit. With equal intervals the mean squared distance to the target is $w^2/12$ everywhere, so the total expected loss is $w^2/(48b) = \Delta^2/(12b(2n+1)^2)$ and

$$\mathbb{E}[\pi_n] = \frac{\mathbb{E}[(A - bc)^2]}{4b} - \frac{\Delta^2}{12b(2n+1)^2}. \quad \square$$

D.3.3 Bid-shape statistics (proof of Corollary 1)

The competing tranches carry equal MW w , so $w_k \equiv \Delta q_k / \sum_j \Delta q_j = 1/n$. Hence

$$\text{HHI}_p = \sum_{k=1}^n w_k^2 = \sum_{k=1}^n \frac{1}{n^2} = \frac{1}{n}.$$

The tranche prices form an arithmetic progression $p_k = p^m(\bar{a}) + (k - (n+1)/2)(w/b)$ centred on $p^m(\bar{a})$. The MW-weighted mean price is $\bar{p} = (1/n) \sum_k p_k = p^m(\bar{a})$ by symmetry, and the MW-weighted variance is

$$\sigma_p^2 = \frac{1}{n} \sum_{k=1}^n (p_k - \bar{p})^2 = \frac{1}{n} \left(\frac{w}{b}\right)^2 \sum_{k=1}^n \left(k - \frac{n+1}{2}\right)^2 = \frac{1}{n} \left(\frac{w}{b}\right)^2 \cdot \frac{n(n^2-1)}{12} = \left(\frac{w}{b}\right)^2 \frac{n^2-1}{12}.$$

Substituting $w = 2\Delta / (2n + 1)$,

$$\sigma_p = \frac{2\Delta}{(2n+1)b} \sqrt{\frac{n^2-1}{12}}.$$

As $n \rightarrow \infty$, $\sigma_p \rightarrow \Delta / (2b\sqrt{3}) = \sigma_v / (2b)$ since $\sigma_v = \text{sd}_v(v) = \Delta / \sqrt{3}$ for $v \sim U[-\Delta, \Delta]$.

For $\hat{\beta}$, the cumulative quantity at tranche k is $Q_k = q_0 + k w$ and the price is $p_k = p^m(\bar{a}) + (k - (n+1)/2)(w/b)$. Both are linear in k , so the OLS slope of p_k on Q_k is exactly the ratio of slopes:

$$\hat{\beta} = \frac{p_{k+1} - p_k}{Q_{k+1} - Q_k} = \frac{w/b}{w} = \frac{1}{b}.$$

For $\hat{\gamma}$: p_k is an exact affine function of Q_k , so the least-squares quadratic fit reproduces the affine relation with zero residual and zero curvature coefficient. Hence $\hat{\gamma} = 0$. $\square$

D.3.4 Level neutrality (proof of Remark 1)

Take expectation of the clearing price over states. Each cell contributes its midpoint-monopoly price weighted by cell length: horizontal cells contribute $p_k = p^m(m_{2k})$ over length w ; riser cells contribute the average realised price over the riser, $\mathbb{E}[(A - Q_k)/b \mid V_k] = (m_{2k+1} - q^m(m_{2k+1}))/b = p^m(m_{2k+1})$, using $A - b p^m(A) - q^m(A) \equiv 0$; the base riser contributes $p^m(m_1)$ by the same demand-sets-price computation. The resulting weighted average over all $2n+1$ cells (each of length w) is the simple average of the $2n+1$ midpoint-monopoly prices, which — because p^m is affine and the midpoints m_j are symmetric around $\bar{a}$ — equals $p^m(\bar{a})$. So $\mathbb{E}[\text{clearing price}] = p^m(\bar{a})$, independent of n . $\square$

Distributional consequence. The step structure reshapes the variance and the per-state surplus split, but never the mean clearing price. Bid-shape changes (regarding the ladder) are therefore *level-neutral* on price; mean price changes come only from changes in residual demand (the slope b or the venue supply S^{op}), which is the channel underpinning the markup identity (6) in the body.

D.3.5 Tranche budget (proof of Proposition 2)

With a complexity cost $\kappa > 0$ per competing tranche, the objective is

$$\mathbb{E}[\pi_n] - \kappa n = \frac{\mathbb{E}[(A - bc)^2]}{4b} - \frac{\Delta^2}{12b(2n+1)^2} - \kappa n.$$

Treating n as continuous, the FOC in n is

$$\frac{d}{dn} \left[-\frac{\Delta^2}{12b(2n+1)^2} \right] = \kappa \iff \frac{\Delta^2}{3b(2n+1)^3} = \kappa \iff 2n^* + 1 = \left(\frac{\Delta^2}{3b\kappa} \right)^{1/3}.$$

The integer optimum is one of the two adjacent integers; either way $n^* + \frac{1}{2} = \frac{1}{2} \Delta^{2/3} (3b\kappa)^{-1/3}$ at the continuous optimum. Joint signature with Corollary 1: a wider screened range Δ raises σ_p and lowers $\text{HHI}_p = 1/n^*$ — they move together. $\square$

D.3.6 Granularity $\times$ within-hour variation (proof of Corollary 2)

(i) Under the continuous benchmark the clearing price is $p^m(A) = (A + bc)/(2b)$, affine in A with slope $1/(2b)$; over $A \in [\bar{a} - \Delta, \bar{a} + \Delta]$ its range is $2\Delta/(2b) = \Delta/b$. Substituting $(\rho_{\mathcal{X}}\Delta, b')$ gives $\rho_{\mathcal{X}}\Delta/b'$.

(ii) The span $\frac{2(n-1)\Delta}{(2n+1)b}$ and the dispersion $\sigma_p = \frac{2\Delta}{(2n+1)b} \sqrt{(n^2-1)/12}$ of §D.3.3 are linear in Δ at fixed n , so replacing (Δ, b) by $(\rho_{\mathcal{X}}\Delta, b')$ multiplies both by $\rho_{\mathcal{X}} b/b'$; $\text{HHI}_p = 1/n$ involves neither Δ nor b .

(iii) Substituting $(\rho_{\mathcal{X}}\Delta, b')$ into Proposition 2: $2n^* + 1 = ((\rho_{\mathcal{X}}\Delta)^2/(3b'\kappa))^{1/3} = \rho_{\mathcal{X}}^{2/3} (\Delta^2/(3b'\kappa))^{1/3}$.

(iv) The flat-hour change is $\sigma_p(\Delta; b') - \sigma_p(\Delta; b)$ — the pure thinness shift, since $\rho_{\text{flat}} = 1$ leaves the screen at Δ . Subtracting it from the critical-hour change $\rho_{\mathcal{X}}\sigma_p(\Delta; b') - \sigma_p(\Delta; b)$ leaves

$$(\rho_{\mathcal{X}} - 1)\sigma_p(\Delta; b') = (\rho_{\mathcal{X}} - 1) \frac{2\Delta}{(2n+1)b'} \sqrt{\frac{n^2-1}{12}},$$

zero iff $\rho_{\mathcal{X}} = 1$ and strictly increasing in $\rho_{\mathcal{X}}$. $\square$

D.3.7 Continuation slope and forward locus (proof of Corollary 3)

Apply backward induction at the spot stage. With Q spot products $t = 1, \dots, Q$ each of slope b_{2t} , the per-quarter strategic best response is $p_{2t}^*(p_1) = (p_1 + c)/2$ at the symmetric solution (omitting arbitrage and aggregator terms which are p_1 -independent for the slope calculation). The per-quarter cleared quantity is $q_{2t}^*(p_1) = b_{2t}(p_1 - p_{2t}^*) = b_{2t}(p_1 - c)/2$. Spot profit at the best response is

$$\pi_{2t}^*(p_1) = (p_{2t}^* - c) q_{2t}^* = \frac{p_1 - c}{2} \cdot \frac{b_{2t}(p_1 - c)}{2} = \frac{b_{2t}(p_1 - c)^2}{4}.$$

Summing across spot products, the stage-1 total profit in state v at forward intercept $A_1 \equiv \bar{A} + v$ is

$$\Pi_1(p_1, v) = (p_1 - c)(A_1 - b_1 p_1) + \frac{(p_1 - c)^2}{4} \sum_t b_{2t} = (p_1 - c)(A_1 - b_1 p_1) + \frac{B_2(p_1 - c)^2}{4},$$

with $B_2 \equiv \sum_t b_{2t}$. The stage-1 FOC:

$$\frac{\partial \Pi_1}{\partial p_1} = (A_1 - b_1 p_1) + (p_1 - c)(-b_1) + \frac{B_2(p_1 - c)}{2} = 0 \iff A_1 - 2b_1 p_1 + b_1 c + \frac{B_2}{2}(p_1 - c) = 0.$$

Solving for p_1^* :

$$p_1^*(v) = \frac{A_1 + c(b_1 - B_2/2)}{2b_1 - B_2/2} = \frac{\bar{A} + v + c(b_1 - B_2/2)}{2b_1 - B_2/2}.$$

The stage-1 monopoly-locus slope in v is therefore

$$\boxed{\frac{dp_1^*}{dv} = \frac{1}{2b_1 - B_2/2}}$$

exactly as claimed in the body (4). Assumption 2 ($B_2 < 2b_1$) ensures the denominator is positive, so the locus is increasing in v and the stage-1 problem is concave at the interior optimum.

Price range. Over the screened states $v \in [-\Delta_1, \Delta_1]$ the stage-1 clearing-price range under the continuous benchmark is exactly

$$2\Delta_1 \cdot \frac{dp_1^*}{dv} = \frac{2\Delta_1}{2b_1 - B_2/2},$$

so the forward tier's price span widens one-for-one with the locus slope.

Measured ladder slope $\hat{\beta}_1$. The cleared quantity at the monopoly point in state v is $q_1^*(v) = A_1 - b_1 p_1^*(v)$. Substituting,

$$q_1^*(v) = \frac{(b_1 - B_2/2)(\bar{A} + v - b_1 c)}{2b_1 - B_2/2}, \quad \frac{dq_1^*}{dv} = \frac{b_1 - B_2/2}{2b_1 - B_2/2}.$$

Assumption 2 also makes $dq_1^*/dv > 0$: the locus is increasing in *both* coordinates, hence implementable by a non-decreasing ladder. Building the step offer along this locus as in §D.3.2, tranche prices and cumulative quantities both advance linearly with the state — at the rates dp_1^*/dv and dq_1^*/dv — so both are evenly spaced sequences and the OLS slope of p_k on Q_k is the ratio of the two rates, independent of the grid width:

$$\hat{\beta}_1 = \frac{dp_1^*/dv}{dq_1^*/dv} = \frac{1/(2b_1 - B_2/2)}{(b_1 - B_2/2)/(2b_1 - B_2/2)} = \frac{1}{b_1 - B_2/2}.$$

Spot-reform jump. At the asymmetric state $(1, Q)$, the spot venue becomes per-quarter, so B_2 jumps from a single hourly slope b_2 to the sum $\sum_t b_{2t}$ of Q per-quarter slopes. Empirically the per-quarter spot slopes are of the same order as the hourly one (Table 17), so the continuation coefficient B_2 jumps from b_2 to roughly $Q \cdot b_2$ — a large factor. The denominators $2b_1 - B_2/2$ and $b_1 - B_2/2$ shrink correspondingly (within the SOC): the stage-1 price range $2\Delta_1/(2b_1 - B_2/2)$ widens and the measured forward ladder slope $\hat{\beta}_1 = 1/(b_1 - B_2/2)$ steepens, both in exact closed form. This is the model’s cross-market anticipation channel of §3.5(M4); the proof is complete. $\square$

D.4 (1, 1) equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§D.4.1), then the four arbitrage regimes at Stage 1 (§D.4.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it. *Convention:* throughout this subsection we report the stage-1 problem in the Ito and Reguant form, suppressing the stage-1 continuation term; Corollary 3 restores it and shows it tilts the stage-1 locus (slope $1/(2b_1 - b_2/2)$ here, since $B_2 = b_2$ at this state) without altering the regime classification or the wedge formulas, which depend on s and the spot best response only.

D.4.1 Stage 2 best response

The strategic bidder’s Stage 2 problem, taking p_1 , q_1 and s as given:

$$\max_{p_2} (p_2 - c)q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) = \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2b_2}. \quad (10)$$

D.4.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (10) into the strategic bidder’s total profit and using $q_1 = \bar{A} - b_1p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) = (p_1 - c)(\bar{A} - b_1p_1 - s - S_1^{op}) + (p_2^*(p_1, s) - c)q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

(R1) No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1c)/(2b_1)$. Combining with (10) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

(R2) Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (10) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} = \frac{\bar{A} - \bar{Y}_h + b_1c}{2b_1}, \quad p_2^{*,F} = p_1^{*,F}. \quad (11)$$

Note that $s > 0$ *reduces* the strategic bidder’s total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

(R3) Limited arbitrage ($s \leq K_a$ binding). If K_a is below the full-arbitrage level, $s = K_a$ binds and $p_1 - p_2 > 0$. From (10), $p_1 - p_2 = (p_1 - c)/2 - (K_a - S_2^{op})/(2b_2)$; the wedge depends on K_a and on S_2^{op} (Ito and Reguant Result 3). The smaller K_a , the larger the residual wedge.

(R4) Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (10):

$$\pi^A(s) = s \left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2 b_2} \right].$$

FOC in s :

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2 b_2} = 0 \iff s^{*,S}(p_1) = \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (10),

$$p_2^{*,S} = \frac{p_1 + c}{2} + \frac{p_1 - c}{4} - \frac{S_2^{op}}{4 b_2} = \frac{3p_1 + c}{4} - \frac{S_2^{op}}{4 b_2}, \quad p_1 - p_2^{*,S} = \frac{p_1 - c}{4} + \frac{S_2^{op}}{4 b_2}.$$

With the aggregator's spot leg netted into the spot intercept ($S_2^{op} = 0$ in the display — the $S_2^{op}/(4b_2)$ term is a level shift common across regimes), this is the boxed body benchmark (1): $p_1 - p_2 = (p_1 - c)/4$ (Ito and Reguant Result 4).

D.5 (1, Q) equilibrium

Forward hourly, spot per-period. Per-period spot best response, then within-block dispersion under full and strategic arbitrage, then the within-hour wedge SD.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand in (7) (suppressing the update term, restored in §D.5.3) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^*(p_1, s_t) = \frac{p_1 + c}{2} + \frac{s_t - S_{2t}^{op}}{2 b_{2t}}. \quad (12)$$

D.5.1 Full arbitrage: block parity and within-block dispersion

Block parity requires $p_1 = (1/Q) \sum_t p_{2t}$. Substituting (12),

$$p_1 = \frac{p_1 + c}{2} + \frac{1}{2Q} \sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}} \iff p_1 - c = \frac{1}{Q} \sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}}.$$

Setting $S_{2t}^{op} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the thinness mechanism, and $Q = 2$ for concreteness,

$$p_1 - c = \frac{\bar{s}}{2} \left(\frac{1}{b_{21}} + \frac{1}{b_{22}} \right) = \frac{\bar{s}}{2} \cdot \frac{b_{21} + b_{22}}{b_{21} b_{22}} \iff \bar{s} = \frac{2(p_1 - c) b_{21} b_{22}}{b_{21} + b_{22}}.$$

The within-block dispersion follows from differencing p_{2t}^* across t :

$$p_{21} - p_{22} = \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}} \right) = \frac{\bar{s}}{2} \cdot \frac{b_{22} - b_{21}}{b_{21} b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} = (p_1 - c) \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \quad (\text{full arbitrage}). \quad (13)$$

D.5.2 Strategic arbitrage: block wedge and attenuated dispersion

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q) \sum_t p_{2t} = \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (1) across the Q matched spot products. Repeating the algebra of §D.5.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S = \frac{(p_1 - c) b_{21} b_{22}}{b_{21} + b_{22}} = \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} = \frac{p_1 - c}{2} \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \quad (\text{strategic arbitrage}). \quad (14)$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives.

D.5.3 Wedge SD in the asymmetric state

Update channel: $\sigma_\omega/(2b_2)$ (derivation of body eq. 3). Decompose the spot residual demand at the per-quarter level using the within-hour state of §3.1: $A_{2t} = \bar{A} + \zeta_t + v_{2t}$. While the forward is hourly, the update $\omega_t = \zeta_t + \mu_t$ realises between the gates and becomes the spot-side traded object (plus an hour-level component common across t , which is absorbed into the mean and does not affect dispersions). The strategic spot best response (12) acquires an ω_t term:

$$p_{2t}^*(p_1, s_t, \omega_t) = \frac{p_1 + c}{2} + \frac{\omega_t + s_t - S_{2t}^{op}}{2 b_{2t}}.$$

With p_1 constant across t (hourly forward), the per-subperiod wedge $w_t = p_1 - p_{2t}^*$ inherits the cross-period variation in p_{2t}^* . Taking variance across t at uniform $b_{2t} \equiv b_2$, s_t , S_{2t}^{op} (so the only t -varying object is ω_t):

$$\text{sd}_t(p_{2t}^*) = \frac{\text{sd}_t(\omega_t)}{2 b_2} = \frac{\sigma_\omega}{2 b_2}, \quad \sigma_\omega^2 = \sigma_\zeta^2 + \sigma_\mu^2,$$

and the within-hour wedge SD equals $\text{sd}_t(p_{2t}^*)$ exactly because p_1 contributes no cross-period variation. This is the main-body (3). Largest in ramp hours where σ_ζ is large; mechanically zero before the reform because the hourly spot averages ω_t to its hour-level component $\bar{\omega}$. (Arbitrage on the hourly forward is a block position, $s_t \equiv \bar{s}$: it disciplines the mean wedge but cannot move the dispersion of the granular leg.)

Thinness-asymmetry channel (block-parity arbitrage). The $\sigma_\omega/(2b_2)$ channel above holds at uniform b_{2t} . With per-subperiod thinness b_{2t} varying across quarters, an additional dispersion term emerges through the block-parity arbitrage of §D.5.1–§D.5.2. For two matched products ($Q = 2$) the population SD contribution from thinness asymmetry alone (setting $\omega_t = 0$) is

$$\text{sd}_t^{F,\text{thin}}(w_t) = \frac{(p_1 - c) |b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \quad \text{sd}_t^{S,\text{thin}}(w_t) = \frac{(p_1 - c) |b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \quad (15)$$

the strategic value exactly half of the full-arbitrage value. Under limited arbitrage with per-product capacity $s_t = K_a/Q$ binding, the thinness-asymmetry SD becomes

$$\text{sd}_t^{L,\text{thin}}(w_t) = \frac{K_a |b_{22} - b_{21}|}{8 b_{21} b_{22}} \quad (Q = 2, s_t = K_a/2), \quad (16)$$

which carries K_a rather than $(p_1 - c)$. The total wedge SD in the asymmetric state is the sum (in quadrature) of the update channel $\sigma_\omega/(2b_2)$ and the thinness-asymmetry channel above; the update channel dominates empirically because σ_ζ is large in ramp hours.

D.6 (Q, Q) equilibrium

Both markets per-period. Per-product forward best response, three arbitrage regimes, Cournot cleared-MWh scale-up, and the stage-1 value of the granular continuation.

The strategic per-period forward best response is the continuation-adjusted locus of Corollary 3 with the matched single spot product ($B_2 = b_{2t}$); arbitrage and aggregator legs shift the intercept only and are omitted from the display:

$$p_{1t}^*(v) = \frac{\bar{A} + \zeta_t + v + c(b_{1t} - b_{2t}/2)}{2b_{1t} - b_{2t}/2}, \quad \lambda_1 \equiv \frac{1}{2b_{1t} - b_{2t}/2}. \quad (17)$$

The ramp ζ_t and the screened state v both enter at the locus slope λ_1 , so the across-quarter dispersion of expected forward prices is $\text{sd}_t(\mathbb{E} p_{1t}^*) = \lambda_1 \sigma_\zeta$, and the per-product forward screen is the ρ_X -amplified $v \sim U[-\rho_X \Delta_1, \rho_X \Delta_1]$.

D.6.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S_{2t}^{op}$ (same structure as the (1, 1) full-arbitrage condition, but per product). The wedge SD is zero by construction.

D.6.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage — the (R4) derivation of §D.4 applied product by product — gives $p_{2t} = (3p_{1t} + c)/4$ (aggregator leg netted). The per-product wedge is $(p_{1t} - c)/4$. Cross-period SD:

$$\text{sd}_t(p_{1t} - p_{2t}) = \frac{1}{4} \text{sd}_t(p_{1t}^*) = \frac{\lambda_1 \sigma_\zeta}{4},$$

using $\text{sd}_t(\mathbb{E} p_{1t}^*) = \lambda_1 \sigma_\zeta$ from (17) — the unlimited-strategic benchmark of the body.

D.6.3 Limited arbitrage at the symmetric state (empirical regime)

With Q matched products per clock-hour the arbitrageur's capacity K_a is spread over Q positions, so the effective per-product capacity is K_a/Q . When this capacity binds — the empirically relevant case in electricity (see §3.7) — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_a/Q$ in the strategic Stage 2 FOC (12), with the matched per-product forward price p_{1t} in place of the hourly p_1 :

$$p_{2t}^* = \frac{p_{1t}^* + c}{2} + \frac{K_a/Q - S_{2t}^{op}}{2b_{2t}}.$$

The per-product wedge is

$$p_{1t}^* - p_{2t}^* = \frac{p_{1t}^* - c}{2} - \frac{K_a/Q - S_{2t}^{op}}{2b_{2t}}.$$

Wedge SD: $\lambda_1 \sigma_\zeta / 2$. At (Q, Q) the per-quarter forward absorbs the ramp upstream: by (17) the forward price across quarters has SD $\text{sd}_t(p_{1t}^*) = \lambda_1 \sigma_\zeta$. The capacity term $(K_a/Q - S_{2t}^{op})/(2b_{2t})$ is

common across quarters when b_{2t} and S_{2t}^{op} are, so only the first term disperses, feeding the wedge at half the rate:

$$\boxed{\text{sd}_t(p_{1t}^* - p_{2t}^*) = \frac{1}{2} \text{sd}_t(p_{1t}^*) = \frac{\lambda_1 \sigma_\zeta}{2}.}$$

This is the main-body (5). Under unlimited strategic arbitrage the wedge SD halves further to $\lambda_1 \sigma_\zeta / 4$ (the doubling under limited per-product arbitrage is the empirical-asymmetric-persistence channel). If per-period thinness b_{2t} also varies across t the wedge inherits an additional thinness-asymmetry component on top of $\lambda_1 \sigma_\zeta / 2$; the empirically observed sustaining of the within-hour wedge SD at granular-state magnitudes (§6.4) is consistent with this combined effect.

Per-product capacity binding. For the limited regime to bind, K_a/Q must be smaller than the strategic-arbitrage equilibrium position $s_t^{\text{strat}} = b_{2t}(p_{1t} - c)/2$ (aggregator spot leg netted into the intercept). Binding therefore requires $K_a \leq Q \cdot \bar{s}^{\text{strat}}$ where $\bar{s}^{\text{strat}}$ is the average per-product strategic position. Empirically, hourly arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

D.6.4 Cournot cleared-MWh under per-period thinness

To isolate the thinness channel, shut the continuation and arbitrage channels in both states ($B_2 = 0, s_t = 0$) — a channel isolation, not an approximation: both states are evaluated under the same convention. The per-period cleared monopoly quantity is then $q_{1t}^* = (\bar{A} + \zeta_t - b_{1t}c - S_{1t}^{op})/2$. Summing across t and using $\sum_t \zeta_t = 0$:

$$\mathbb{E}\left[\sum_t q_{1t}^*\right] = \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S_{1t}^{op}}{2}.$$

The hourly counterpart is $Q \cdot q_1^*|_{(1,1)} = Q \cdot (\bar{A} - b_1c - S_1^{op})/2$. Comparing at equal aggregator totals ($\sum_t S_{1t}^{op} = S_1^{op}$):

$$\mathbb{E}\left[\sum_{t=1}^Q q_{1t}^*\right]\Big|_{(Q,Q)} - Q \cdot q_1^*|_{(1,1)} = \frac{c}{2} \left(Q b_1 - \sum_{t=1}^Q b_{1t}\right). \quad (18)$$

Positive whenever $\sum_{t=1}^Q b_{1t} < Q b_1$, as expected (each 15-min product is thinner than the hourly aggregate it replaces). This is the Cournot scale-up of main-body §3.6(iv).

D.6.5 Stage-1 value of the granular continuation

Maximised monopoly profit is convex in the demand intercept ($\pi^m(A) = (A - bc)^2/(4b)$), so a bidder that can re-optimize per quarter values the update as a mean-preserving spread: expected spot profit exceeds spot profit at the mean update by exactly $\mathbb{E}[\mu_t^2]/(4b_{2t}) = \Delta_\mu^2/(12b_{2t})$ per product under the uniform update — an exact equality, because the profit is quadratic. The term raises the stage-1 *value* of holding a granular continuation (part of why the granular state is privately valuable to the strategic bidder), but it does not involve p_1 , so it leaves the stage-1 first-order condition unchanged: the forward-ladder response to the spot reform operates entirely through the continuation slope of Corollary 3, not through an option-value tilt.

D.7 Balancing discrepancy

Per quarter, the balancing residual collects whatever no auction carried and no agent could re-schedule:

$$\Delta_t = (\text{within-hour demand objects unpriced by any auction}) + (\text{unhedged production error}) + \varepsilon_t. \quad (19)$$

State (1, 1). Both auctions hourly: the spot trades only the hour-level update $\bar{\omega}$, so the ramp, the redistributive news, and the aggregator's forecast error all land in balancing: $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$.

State (1, Q). The spot becomes per-period: it absorbs the full update $\omega_t = \zeta_t + \mu_t$, and the aggregator re-schedules η_t per quarter (§D.2): $\Delta_t = \varepsilon_t$.

State (Q, Q). The per-quarter forward carries the ramp ζ_t , the spot carries the news μ_t , the aggregator hedges η_t : $\Delta_t = \varepsilon_t$.

State	ramp ζ_t	redistributive μ_t	forecast error η_t	residual Δ_t
(1, 1)	balancing	balancing	balancing	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$
(1, Q)	spot	spot	aggregator (hedged)	ε_t
(Q, Q)	forward	spot	aggregator (hedged)	ε_t

The rows for (1, Q) and (Q, Q) are identical in the residual column: the forward reform relocates rows between the two auctions but produces no further imbalance gain — the body's §3.6(v). These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

D.8 Welfare derivations

We collect closed-form welfare components by state, consistent with the balancing ledger of §D.7 and with the channels listed in the body §3.8.

D.8.1 Components

Per clock-hour:

$$W = CS + \pi^{SB} + \pi^{Agg,net} + \pi^{Arb} - C_{sys}^I. \quad (20)$$

For linear residual demand $q = \bar{A} - bp$, consumer surplus at p^* is $CS = (\bar{A} - bp^*)^2 / (2b)$. Gross surplus from consuming q at state A (willingness to pay minus production cost at c ; balancing premia accounted separately) is

$$S(q; A) = \int_0^q \frac{A - x}{b} dx - cq = \frac{q(A - bc)}{b} - \frac{q^2}{2b}.$$

D.8.2 Imbalance cost decomposition (closed form, quadratic penalty)

Under the quadratic penalty $\gamma \mathbb{E}[\Delta_t^2]$ per quarter and the ledger of §D.7, with ζ_t deterministic ($Q^{-1} \sum_t \zeta_t^2 = \sigma_\zeta^2$) and μ_t, η_t mean-zero and independent:

	(1, 1)	(1, Q)	(Q, Q)
$\gamma \mathbb{E}[(\mathcal{I}^{Agg})^2]$ (in $\pi^{Agg,net}$)	$\gamma Q \sigma_\eta^2$	0	0
C_{sys}^I (system side: $\zeta + \mu$)	$\gamma Q (\sigma_\zeta^2 + \sigma_\mu^2)$	0	0
Total imbalance cost	$\gamma Q (\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2)$	0	0

(21)

omitting the unpriceable $\gamma Q \sigma_\varepsilon^2$, common to all three states. The two channels are disjoint (η_t enters only $\pi^{Agg, net}$; ζ_t and μ_t only C_{sys}^I). For uniform shocks, $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$. The whole imbalance saving arrives at the spot reform; the (Q, Q) column equals the $(1, Q)$ column, the model's DA15 imbalance null.

D.8.3 Allocative cost of pricing the state per quarter (discrimination term)

Pricing the within-hour state per quarter is not free: it is third-degree price discrimination by a residual monopolist. Fix one clock-hour, the within-hour object ω_t with $\sum_t \omega_t = 0$, and the margin b at which it is priced; write $\bar{q} = q^m(\bar{A})$ and $A_t = \bar{A} + \omega_t$, so $A_t - bc = 2\bar{q} + \omega_t$.

Regime U (one price for the hour). The monopolist sets $\bar{p} = p^m(\bar{A})$; quarter- t consumption realises on the demand curve, $q_t^U = A_t - b\bar{p} = \bar{q} + \omega_t$; the contracted profile is flat at $\bar{q}$ and the balancing layer supplies the residual ω_t . Substituting into S :

$$S(q_t^U; A_t) = \frac{(\bar{q} + \omega_t)(3\bar{q} + \omega_t)}{2b}.$$

Regime D (per-quarter prices). The monopolist prices each quarter at $p^m(A_t)$; consumption is $q_t^D = q^m(A_t) = \bar{q} + \omega_t/2$ — monopoly pass-through one half — and there is no balancing residual:

$$S(q_t^D; A_t) = \frac{3(\bar{q} + \omega_t/2)^2}{2b}.$$

Difference. Summing over t and using $\sum_t \omega_t = 0$,

$$\sum_t S^D - \sum_t S^U = \frac{1}{2b} \sum_t \left[3\bar{q}^2 + 3\bar{q}\omega_t + \frac{3}{4}\omega_t^2 - 3\bar{q}^2 - 4\bar{q}\omega_t - \omega_t^2 \right] = -\frac{\sum_t \omega_t^2}{8b}:$$

total output is unchanged and output is misallocated across quarters, at expected cost $Q \mathbb{E}[\omega^2]/(8b)$. Against it stands the avoided quadratic balancing penalty $\gamma Q \mathbb{E}[\omega^2]$, so the net system-side gain from pricing ω at margin b is

$$Q \mathbb{E}[\omega^2] \left(\gamma - \frac{1}{8b} \right), \quad \text{positive iff } \gamma > \frac{1}{8b},$$

a threshold independent of the distribution of ω . At the spot reform the relevant margin is b_2 ; balancing premia in Spanish electricity markets exceed wholesale margins by a factor of 2–10, far above $1/(8b_2)$.

D.8.4 Markup channel

From the Cournot identity (6), $\text{markup}_m = q_m/b_{mt}$: at given cleared quantity, $\partial \text{markup}/\partial b_{mt} = -q_m/b_{mt}^2$, so per-firm residual-demand thickening compresses the markup — and, by level neutrality (Remark 1), this is the *only* channel through which the reform moves mean prices; ladder geometry contributes nothing to the mean.

D.8.5 Discreteness channel

The optimal-ladder expected profit shortfall relative to the continuous benchmark is $\Delta^2/(12b(2n^* + 1)^2)$ (Proposition 1); the deeper endogenous depth at amplified screens (Corollary 2(iii)) reduces it. By Remark 1 the mean clearing price is unchanged by the ladder, so within a product this channel is mostly a $CS \leftrightarrow \pi^{SB}$ transfer, net of tranche costs κn^* ; the efficiency content is the discreteness loss itself.

D.8.6 Less strategic withholding under limited per-product arbitrage

The spot deadweight-loss triangle at markup $M = p_2 - c$ and slope b_2 is $b_2 M^2/2$. Under unlimited strategic arbitrage, $p_2 - c = 3(p_1 - c)/4$ (§D.4, R4): $\text{DWL} = 9 b_2 (p_1 - c)^2/32$. Under binding per-product capacity, the wedge is $(p_1 - c)/2$, so $p_2 - c = (p_1 - c)/2$ and $\text{DWL} = b_2 (p_1 - c)^2/8 = 4 b_2 (p_1 - c)^2/32$. The per-product saving from the regime change is

$$\frac{5 b_2 (p_1 - c)^2}{32},$$

per Ito and Reguant Result 2 logic — a welfare gain that accompanies the wedge-SD non-collapse of §3.6(ii), not a cost of it.

D.8.7 Net welfare deltas (closed form)

Combining the contractibility channels (writing $a_1 \equiv \bar{A} - b_1 c$, aggregator and arbitrage legs netted into $\bar{A}$):

$$\Delta W^{\text{spot}} = \gamma Q \sigma_\eta^2 + Q (\sigma_\zeta^2 + \sigma_\mu^2) \left(\gamma - \frac{1}{8 b_2} \right), \quad (22)$$

$$\Delta W^{\text{forward}} = \frac{Q \sigma_\zeta^2}{8} \left(\frac{1}{b_2} - \frac{1}{b_1} \right) + \frac{c a_1}{4 b_1} \left(Q b_1 - \sum_t b_{1t} \right). \quad (23)$$

The spot-reform delta is the aggregator’s hedging saving plus the net value of pricing the update at the spot margin: positive whenever $\gamma > 1/(8b_2)$, and dominated by the imbalance saving at observed balancing premia. The forward-reform delta is a *relocation*: the ramp’s discrimination cost moves from the spot margin b_2 to the (thicker) forward margin b_1 — the first term, positive when $b_1 > b_2$ and zero when the margins coincide — plus the thinness scale-up of (18) valued at the pre-existing forward markup, carrying the sign of $Q b_1 - \sum_t b_{1t}$. Both forward-reform terms are second-order relative to (22): no imbalance term appears at the forward reform (the (Q, Q) column of (21) equals the $(1, Q)$ column), which is the model’s account of the small DA15 aggregate footprint. The markup-compression and withholding channels add on top wherever b_f thickens or the arbitrage regime tightens, with the signs derived above; the bid-ladder rent channel is mostly distributional.

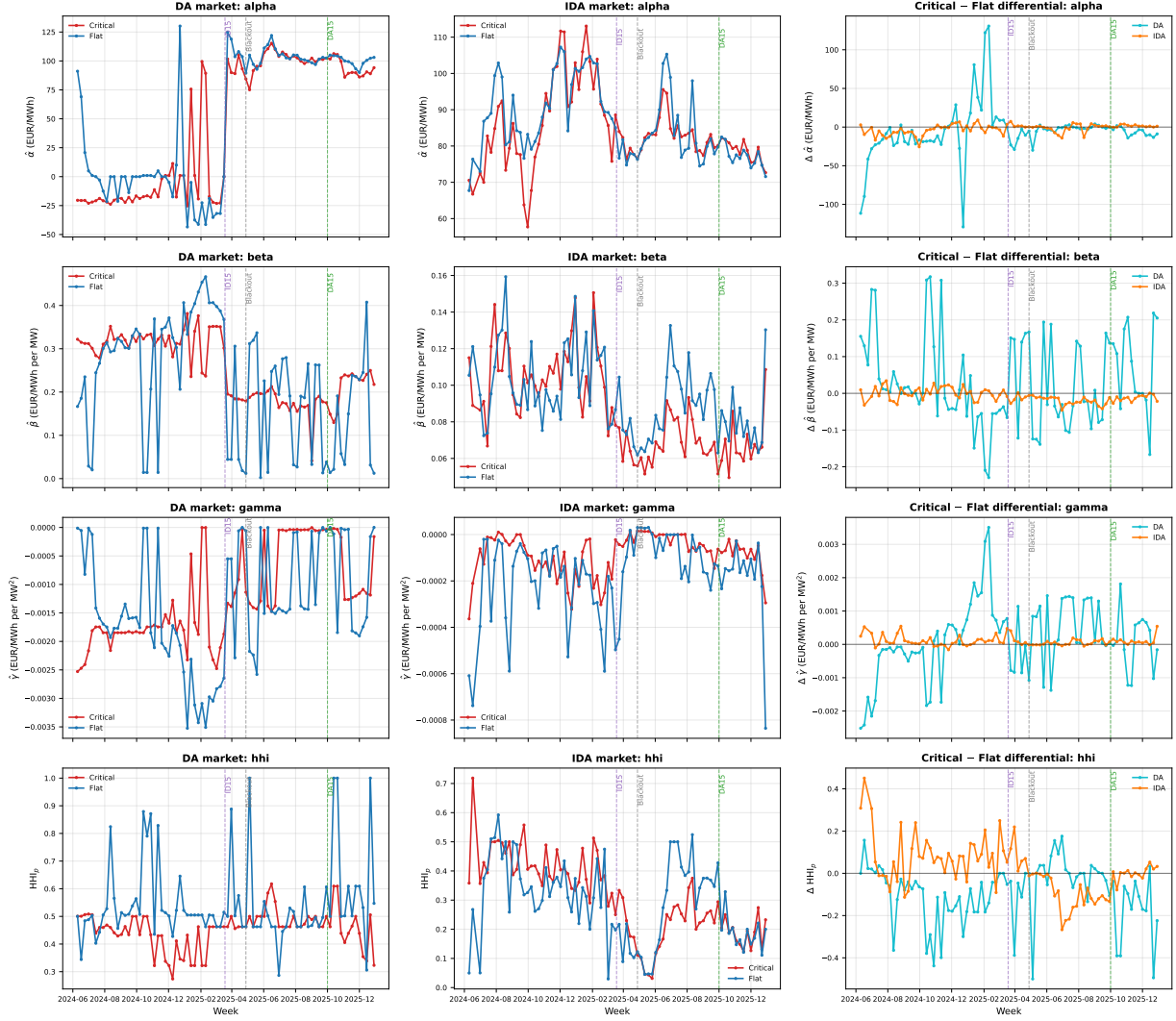


Figure 9: Structural decomposition: weekly CCGT per-curve $\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \text{HHI}_p$ by hour-class, 2024-06 to 2025-12. *Left column:* DA market levels by hour-class. *Middle column:* IDA market levels. *Right column:* critical – flat differential by market (cyan DA, orange IDA), the visual analogue of the bid-shape DiD coefficient. In-band per-curve OLS quadratic fit at $\delta = 150$ EUR/MWh; per-cell aggregation uses the median across curves with $n_{\text{tranche}} \geq 4$ and $\Delta q \geq 20$ MW for the $\hat{\beta}$ and $\hat{\gamma}$ rows (joint p - q identification floor); $\hat{\alpha}$ and HHI_p have no such floor and use all curves. Vertical dashes mark the three reform dates. The HHI_p row carries the cleanest visual signal (DA critical sits sustainably below flat post-DA15, the visual signature of bid-stack deconcentration); $\hat{\alpha}$ is dominated by the spring-2025 shared negative-price shock; $\hat{\beta}$ and $\hat{\gamma}$ rows are noisier per the identification argument in the paragraph above and should be read in conjunction with the σ_p panel of Figure 8.

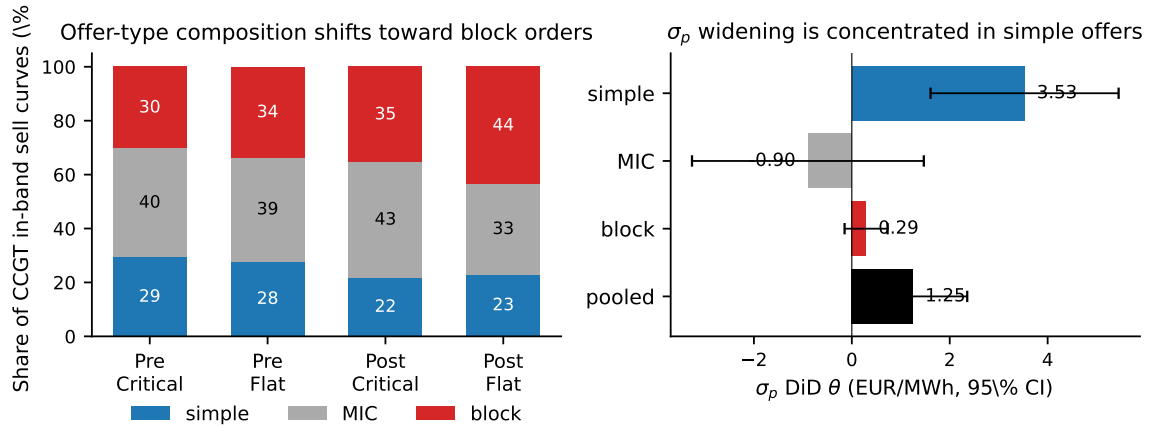


Figure 10: CCGT offer-type composition shift and within-type bid-shape DiD. DA15 windows: pre 2025-07-01 to 2025-09-30; post 2025-10-01 to 2025-12-31. *Left:* composition of CCGT in-band sell curves, pre vs post DA15, critical vs flat. *Right:* bid-shape DiD by offer type with 95% CI; the panel is plotted on the σ_p outcome, and the HHL_p DiD by offer type carries the same sign pattern with negative coefficients in every type (simple -0.007 n.s., MIC -0.046^{***} , block -0.043^{***} , pooled -0.059^{***}).

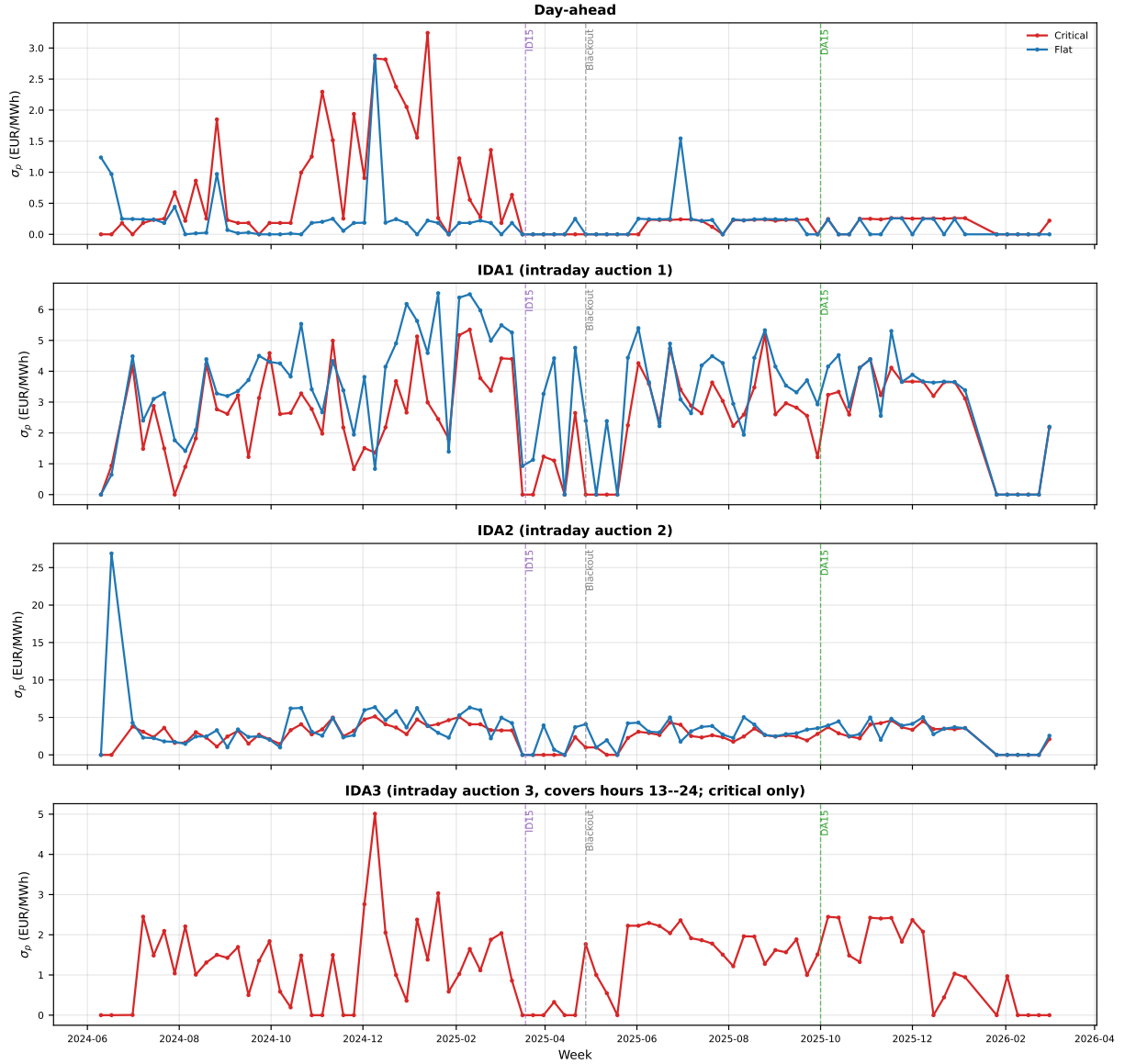


Figure 11: Per-session parallel-trends diagnostic: weekly CCGT σ_p by hour-class, 2024-06 to 2026-05. Stacked panels: day-ahead, then each of the three IDA sessions. Red: critical hours (5–8, 16–22). Blue: flat hours (1–3). IDA3 covers only hours 13–24, so its panel reports critical only. Bandwidth $|p_k - \text{MCP}| \leq 50$ EUR/MWh. Vertical dashes mark ID15 (2025-03-19), blackout (2025-04-28), and DA15 (2025-10-01). The IDA2 panel carries the most-active CCGT bid-shape pre-MTU15-IDA, and the three IDA panels visibly converge in level post-reform — the per-session story behind the pooled bid-shape DiD.

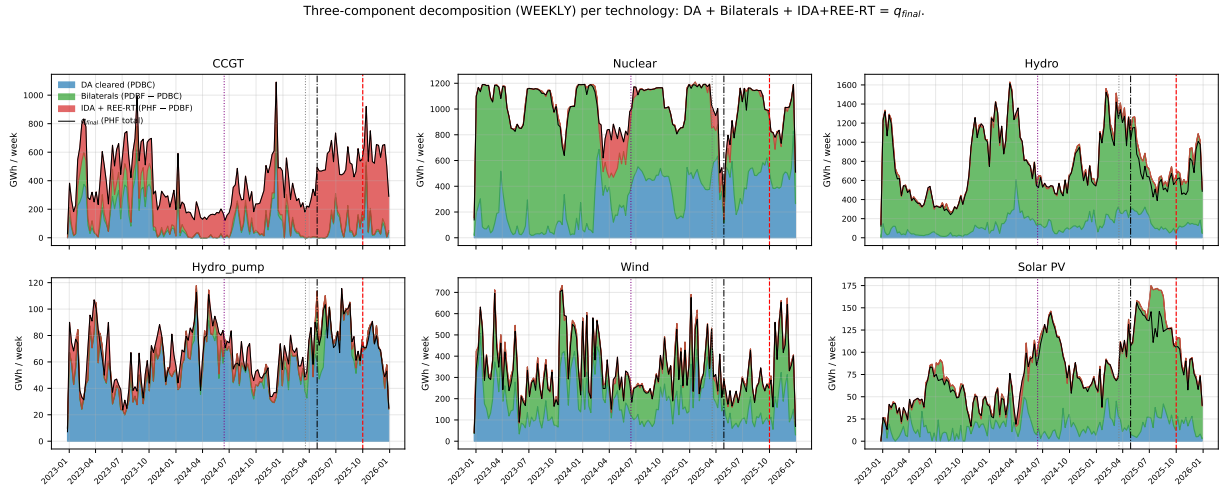


Figure 12: Weekly three-component decomposition of dispatch, by technology. PDBC (blue, day-ahead-cleared) + bilateral executions (PDBF – PDBC, green) + post-DA REE intervention layer (PHF – PDBF, red) = final post-IDA dispatch (black, equivalent to PHF). The red layer aggregates everything that happens to the schedule between the day-ahead programme (PDBF) and the post-IDA programme (PHF): *Fase I* pre-IDA up-redispatch and *Fase II* rebalance under PO 3.2 (paid pay-as-bid via the restriction-offer channel of PO 14.4 ap. 20.1, distinct from the day-ahead market offer), plus the net of all IDA clearings and REE’s post-IDA real-time technical-restrictions step. Within the red layer, *Fase I* dominates the post-blackout growth (Figure 6 and §6.3). Vertical guides: MTU15-IDA, April 2025 blackout, MTU15-DA.